

40. USB 2.0 FS Host/Function Module (USBb)

40.1 Overview

This MCU incorporates a USB 2.0 FS host/function module (USB0).

The USB module is a USB controller that is equipped to operate as a host controller or function controller. The module supports full-speed and low-speed (only for the host controller) transfer as defined in Universal Serial Bus (USB) Specification 2.0. The module has an internal USB transceiver and supports all of the transfer types defined in USB Specification 2.0.

The USB has buffer memory for data transfer, providing a maximum of 10 pipes. Pipes 1 to 9 can be assigned any endpoint number based on peripheral devices used for communication or based on the user system.

Table 40.1 shows the specifications of the USB.

Table 40.1 USB Specifications

Item	Specifications
Features	• USB Device Controller (UDC) and transceiver for USB 2.0 are incorporated. • Host controller, function controller, and On-The-Go (OTG) are supported (one channel) • The host controller and the function controller can be switched by software. • Self-power mode or bus power mode can be selected. When the host controller is selected: • Full-speed transfer (12 Mbps) and low-speed transfer (1.5 Mbps) are supported • Automatic scheduling for SOF and packet transmissions • Programmable intervals for isochronous and interrupt transfers • Multiple peripheral devices can be connected for communication via a one-stage hub. When the function controller is selected: • Full-speed transfer (12 Mbps) is supported*1 • Control transfer stage control function • Device state control function • Auto response function for SET_ADDRESS request • SOF interpolation function
Communication data transfer type	• Control transfer • Bulk transfer • Interrupt transfer • Isochronous transfer
Pipe configuration	• Buffer memory for USB communication is provided. • Up to 10 pipes can be selected (including the default control pipe). • Pipes 1 to 9 can be assigned any endpoint number. Transfer conditions that can be set for each pipe: • Pipe 0: Control transfer, 64-byte single buffer • Pipes 1 and 2: 64-byte double buffer can be specified for bulk transfer and 256-byte double buffer for isochronous transfer • Pipes 3 to 5: Bulk transfer, 64-byte double buffer • Pipes 6 to 9: Interrupt transfer, 64-byte single buffer
Others	• Reception ending function using transaction count • Function that changes the BRDY interrupt event notification timing (BFRE) • Function that automatically clears the buffer memory after the data for the pipe specified at the DnFIFO (n = 0, 1) port has been read (DCLRM) • NAK setting function for response PID generated by end of transfer (SHTNAK) • On-chip pull-up and pull-down resistors of D+/D-
Low power consumption function	Module-stop state can be set.

Note 1. When the function controller is selected, low-speed transfer (1.5 Mbps) is not supported.

Figure 40.1 shows a block diagram of the USB.

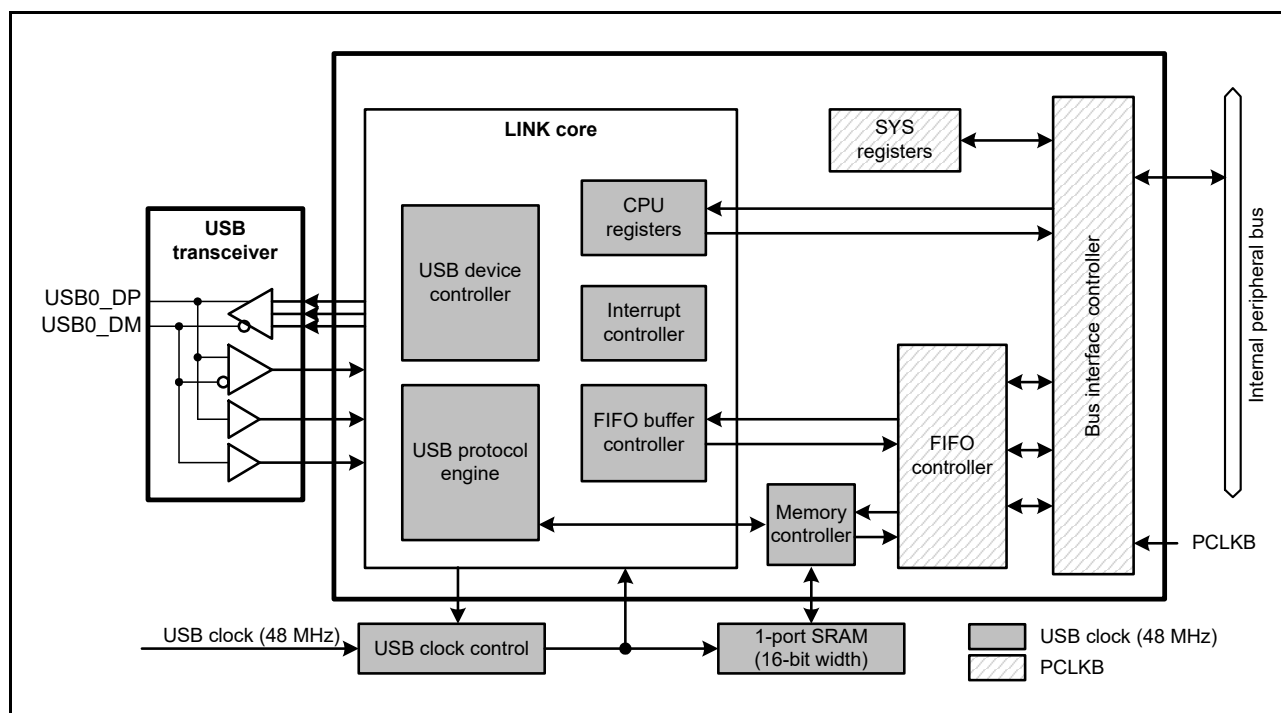


Figure 40.1 USB Block Diagram

Table 40.2 lists the I/O pins of the USB.

Table 40.2 USB Pin Configuration

Port	Pin Name	I/O	Function
USB0	USB0_DP	I/O	D+ I/O pin of the USB on-chip transceiver This pin should be connected to the D+ pin of the USB bus.
	USB0_DM	I/O	D- I/O pin of the USB on-chip transceiver This pin should be connected to the D- pin of the USB bus.
	USB0_VBUS	Input	USB cable connection monitor pin This pin should be connected to VBUS of the USB bus. The VBUS pin status (connected or disconnected) can be detected when the USB module is operating as a function controller.
	USB0_EXICEN	Output	Low-power control signal for external power supply (OTG) chip
	USB0_VBUSEN	Output	VBUS (5 V) supply enable signal for external power supply chip
	USB0_OVRCURA USB0_OVRCURB	Input	External overcurrent detection signals should be connected to these pins. VBUS comparator signals should be connected to these pins when the OTG power supply chip is connected.
	USB0_ID	Input	MicroAB connector ID input signal should be connected to this pin during operation in OTG mode.
Common	VCC_USB	Input	USB power supply pin
	VSS_USB	Input	USB ground pin

40.2 Register Descriptions

40.2.1 System Configuration Control Register (SYSCFG)

Address(es): USB0.SYSCFG 000A 0000h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	SCKE	—	—	—	DCFM	DRPD	DPRPU	—	—	—	USBE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	USBE	USB Operation Enable	0: USB operation is disabled. 1: USB operation is enabled.	R/W
b3 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	DPRPU	D+ Line Resistor Control	0: Pulling up the line is disabled. 1: Pulling up the line is enabled.	R/W
b5	DRPD	D+/D– Line Resistor Control	0: Pulling down the lines is disabled. 1: Pulling down the lines is enabled.	R/W
b6	DCFM	Controller Function Select	0: Function controller is selected. 1: Host controller is selected.	R/W
b9 to b7	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b10	SCKE	USB Clock Enable*1	0: Stops supplying the clock signal to the USB. 1: Enables supplying the clock signal to the USB.	R/W
b15 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. After writing 1 to the SCKE bit, read it and confirm it is set to 1.

USBE Bit (USB Operation Enable)

The USBE bit enables or disables operation of the USB.

Modifying the USBE bit from 1 to 0 initializes the register bits listed in Table 40.3.

This bit should be modified while the SCKE bit is 1.

When the host controller is selected, this bit should be set to 1 after setting the DRPD bit to 1, eliminating SYSSTS0.LNST[1:0] flag chattering, and confirming that the USB bus state is stabilized.

Table 40.3 Registers Initialized by Writing 0 to the SYSCFG.USBE Bit

Selected Function	Register	Bit	Remarks
Function controller	SYSSTS0	LNST[1:0]	The value is retained when the host controller is selected.
	DVSTCTR0	RHST[2:0]	
	INTSTS0	DVSQ[2:0]	The value is retained when the host controller is selected.
	USBADDR	USBADDR[6:0]	The value is retained when the host controller is selected.
	USBREQ	BREQUEST[7:0], BMREQUESTTYPE[7:0]	The value is retained when the host controller is selected.
	USBVAL	—	The value is retained when the host controller is selected.
	USBINDX	—	The value is retained when the host controller is selected.
	USBLENG	—	The value is retained when the host controller is selected.
Host controller	DVSTCTR0	RHST[2:0]	
	FRMNUM	FRNM[10:0]	The value is retained when the function controller is selected.

DPRPU Bit (D+ Line Resistor Control)

The DPRPU bit enables or disables pulling up the D+ line when the function controller is selected.

When the DPRPU bit is set to 1 while the function controller is selected, the bit forces a pull-up of the D+ line to notify the USB host of connection. Modifying the DPRPU bit from 1 to 0 allows the USB to release the D+ line, thus notifying the USB host of disconnection.

This bit should be set to 1 if the function controller is selected, and should be set to 0 if the host controller is selected.

DRPD Bit (D+/D– Line Resistor Control)

The DRPD bit enables or disables pulling down D+ and D– lines when the host controller is selected.

This bit should be set to 1 if the host controller is selected, and should be set to 0 if the function controller is selected.

DCFM Bit (Controller Function Select)

The DCFM bit selects the function of the USB.

This bit should be modified when the DPRPU and DRPD bits are all 0.

SCKE Bit (USB Clock Enable)

The SCKE bit stops or enables supplying 48-MHz clock signals to the USB.

When this bit is 0, only SYSCFG can be read from and written to; the other registers related to the USB cannot be read from or written to.

40.2.2 System Configuration Status Register 0 (SYSSTS0)

Address(es): USB0.SYSSTS0 000A 0004h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
OVCMON[1:0]	—	—	—	—	—	—	—	—	HTACT	SOFEA	—	—	IDMON	LNST[1:0]	
Value after reset:	0 ^{*1}	0 ^{*1}	0	0	0	0	0	0	0	0	0	0	0	0 ^{*1}	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	LNST[1:0]	USB Data Line Status Monitor Flag	Refer to Table 40.4.	R
b2	IDMON	ID Input Pin Monitor Flag	0: USB0_ID pin is low 1: USB0_ID pin is high	R
b4, b3	—	Reserved	These bits are read as 0 and cannot be modified.	R
b5	SOFEA	SOF Active Monitor Flag When the Host Controller is Selected	Indicates the SOF output status. 0: SOF output is stopped. 1: SOF output is operating.	R
b6	HTACT	USB Host Sequencer Status Monitor Flag	0: Host sequencer of the USB is completely stopped. 1: Host sequencer of the USB is not completely stopped.	R
b13 to b7	—	Reserved	These bits are read as 0 and cannot be modified.	R
b15, b14	OVCMON[1:0]	OVRCURA/OVRCURB Input Pin Monitor Flag	The OVCMON[1] flag indicates the status of the USB0_OVRCURA pin. The OVCMON[0] flag indicates the status of the USB0_OVRCURB pin.	R

Note 1. Depends on the status of the USB0_OVRCURA/USB0_OVRCURB and USB0_ID pins.

LNST[1:0] Flags (USB Data Line Status Monitor Flag)

The LNST[1:0] flags indicate the state of the USB data lines (D+ and D– lines). Refer to Table 40.4.

The LNST[1:0] flags should be read after the connection processing (SYSCFG.DPRPU bit = 1) when the function controller is selected; whereas after enabling pull-down of the lines (SYSCFG.DRPD bit = 1) when the host controller is selected.

SOFEA Flag (SOF Active Monitor Flag When the Host Controller is Selected)

When the USB module is suspended while host mode is used, this bit can be used to confirm whether the last SOF has been output after DVSTCTR0.UACT is set to 0.

Confirm that both the HTACT and SOFEA flags are 0 before stopping the USB module while SYSCFG.USBE is 0 and setting the SYSCFG.SCKE bit to 0 to stop the clock during host mode communication.

HTACT Flag (USB Host Sequencer Status Monitor Flag)

The HTACT flag is 0 when the host sequencer of the USB is completely stopped.

Confirm that the HTACT flag is 0 before setting this controller to the USB suspended state while DVSTCTR0.UACT bit is 0 and setting the SCKE bit to 0 to stop the clock during host mode communication.

OVCMON[1:0] Flags (OVRCURA/OVRCURB Input Pin Monitor Flag)

The OVCMON[1:0] flags indicate the status of overcurrent from an external power supply chip.

Table 40.4 Status of USB Data Bus Lines (D+ Line, D– Line)

LNST[1:0] Flags	During Low-Speed Operation (Only in Host Controller Operation)	During Full-Speed Operation
00b	SE0	SE0
01b	K-State	J-State
10b	J-State	K-State
11b	SE1	SE1

40.2.3 Device State Control Register 0 (DVSTCTR0)

Address(es): USB0.DVSTCTR0 000A 0008h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	HNPBTOA	EXICEN	VBUSEN	WKUP	RWUPE	USBRST	RESUME	UACT	—	RHST[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	RHST[2:0]	USB Bus Reset Status Flag	- When the host controller is selected b2 b0 0 0 0: Communication speed not determined (powered state or no connection) 1 x x: USB bus reset in progress 0 0 1: Low-speed connection 0 1 0: Full-speed connection - When the function controller is selected b2 b0 0 0 0: Communication speed not determined 0 0 1: USB bus reset in progress 0 1 0: USB bus reset in progress or full-speed connection	R
b3	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b4	UACT	USB Bus Enable	0: Downstream port is disabled (SOF transmission is disabled). 1: Downstream port is enabled (SOF transmission is enabled).	R/W
b5	RESUME	Resume Output	0: Resume signal is not output. 1: Resume signal is output.	R/W
b6	USBRST	USB Bus Reset Output	0: USB bus reset signal is not output. 1: USB bus reset signal is output.	R/W
b7	RWUPE	Wakeup Detection Enable	0: Downstream port wakeup is disabled. 1: Downstream port wakeup is enabled.	R/W
b8	WKUP	Wakeup Output	0: Remote wakeup signal is not output. 1: Remote wakeup signal is output.	R/W
b9	VBUSEN	VBUSEN Output Pin Control	0: USB0_VBUSEN pin outputs low. 1: USB0_VBUSEN pin outputs high.	R/W
b10	EXICEN	EXICEN Output Pin Control	0: USB0_EXICEN pin outputs low. 1: USB0_EXICEN pin outputs high.	R/W
b11	HNPBTOA	Host Negotiation Protocol (HNP) Control	This bit is used when switching from device B to device A while in OTG mode. If the HNPBTOA bit is 1, the internal function control keeps the suspended state until the HNP processing ends even though SYSCFG.DPRPU = 0 or SYSCFG.DCFM = 1.	R/W
b15 to b12	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

x: Don't care

RHST[2:0] Flags (USB Bus Reset Status Flag)

The RHST[2:0] flags indicate the status of the USB bus reset.

When the host controller is selected, the RHST[2:0] flags indicate 100b after the USBRST bit has been set to 1 by software.

The USB fixes the value of the RHST[2:0] flags when 0 is written to the USBRST bit by software and the USB completes SE0 driving.

When the function controller is selected, the RHST[2:0] flags indicate 010b (connection while DPRPU = 1) when the USB detects the USB bus reset, and a DVST interrupt is generated.

UACT Bit (USB Bus Enable)

The UACT bit enables operation of the USB bus (controls the SOF packet transmission to the USB bus) when the host controller is selected.

With this bit set to 1, the USB puts the USB port to the USB-bus enabled state and performs SOF output and data transmission and reception.

This module starts outputting SOF packets within one frame after 1 has been written to the UACT bit by software.

With this bit set to 0, the USB enters the idle state after outputting SOF packets.

The USB sets the UACT bit to 0 on any of the following conditions.

- A DTCH interrupt is detected during communication (while UACT = 1).
- An EOFERR interrupt is detected during communication (while UACT = 1).

Writing 1 to this bit should be done at the end of the USB reset processing (writing 0 to the USBRST bit) or at the end of the resume processing from the suspended state (writing 0 to the RESUME bit).

This bit should be set to 0 if the function controller is selected.

RESUME Bit (Resume Output)

The RESUME bit controls the resume signal output when the host controller is selected.

Setting the RESUME bit to 1 allows the USB to drive the port to the K-state and output the resume signal.

The USB sets the RESUME bit to 1 on detecting the remote wakeup signal while RWUPE is 1 in the USB suspended state.

The USB continues outputting K-state while the RESUME bit = 1 (until the RESUME bit is set to 0 by software). The RESUME bit should be 1 (= resume period) for the time defined by USB Specification 2.0.

This bit should be set to 1 in the suspended state.

Write 1 to the UACT bit simultaneously with the end of the resume processing (writing 0 to the RESUME bit).

This bit should be set to 0 if the function controller is selected.

USBRST Bit (USB Bus Reset Output)

The USBRST bit controls the USB bus reset signal output when the host controller is selected.

When the host controller is selected, setting this bit to 1 allows the USB to drive SE0 of the USB port to reset the USB bus.

The USB continues outputting SE0 while USBRST = 1 (until the USBRST bit is set to 1 by software). The USBRST bit should be 1 (= USB bus reset period) for the time defined by USB Specification 2.0.

Writing 1 to this bit during communication (the UACT bit = 1) or during the resume processing (the RESUME bit = 1) prevents the USB from starting the USB bus reset processing until both the UACT and RESUME bits become 0.

Write 1 to the UACT bit simultaneously with the end of the USB bus reset processing (writing 0 to the USBRST bit).

This bit should be set to 0 if the function controller is selected.

RWUPE Bit (Wakeup Detection Enable)

The RWUPE bit enables or disables the downstream port peripheral device to use the remote wakeup function (resume signal output) when the host controller is selected.

With this bit set to 1, on detecting the remote wakeup signal, the USB detects the resume signal (K-state for 2.5 μ s) from the downstream port device and performs the resume processing (drives the port to the K-state).

With this bit set to 0, the USB ignores the detected remote wakeup signal (K-state) from the peripheral device connected to the USB port.

While the RWUPE bit is 1, the internal clock should not be stopped even in the suspended state (SYSCFG.SCKE bit should be set to 1).

This bit should be set to 0 if the function controller is selected.

WKUP Bit (Wakeup Output)

The WKUP bit enables or disables outputting the remote wakeup signal (resume signal) to the USB bus when the function controller is selected.

The USB controls the output time of a remote wakeup signal. When this bit is set to 1, the USB sets this bit to 0 after outputting the 10-ms K-state.

According to USB Specification 2.0, the USB bus idle state must be kept for 5 ms or longer before a remote wakeup signal is sent. If the USB writes 1 to this bit right after detection of the suspended state, the K-state will be output after 2 ms.

Do not write 1 to this bit, unless the device state is in the suspended state (INTSTS0.DVSQ[2:0] flags = 1xxb) and the USB host enables the remote wakeup signal. When this bit is set to 1, the internal clock must not be stopped even in the suspended state (write 1 to this bit while the SYSCFG.SCKE bit = 1).

This bit should be set to 0 if the host controller is selected.

HNPBTOA Bit (Host Negotiation Protocol (HNP) Control)

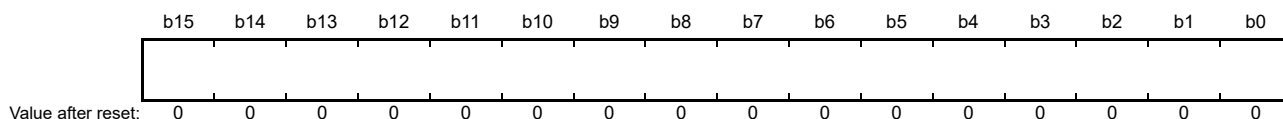
The HNPBTOA bit is used when switching from device B to device A while in OTG mode. If the HNPBTOA bit is 1, the internal function control keeps the suspended state until the HNP processing ends even though the SYSCFG.DPRPU bit = 0 or SYSCFG.DCFM = 1 is set. Even if the falling edge of the D+ signal is detected at this time, no resume (RESM) interrupt is generated.

After this bit is set to 1, write 0 to this bit by software to terminate the HNP processing when connection to the host (pull-up on the target side) or timeout of the HNP processing is detected.

40.2.4 CFIFO Port Register (CFIFO), D0FIFO Port Register (D0FIFO), D1FIFO Port Register (D1FIFO)

(1) When the MBW bit is 1

Address(es): USB0.CFIFO 000A 0014h, USB0.D0FIFO 000A 0018h, USB0.D1FIFO 000A 001Ch



Bit	Symbol	Bit Name	Description	R/W
b15 to b0	—	FIFO Port	This port is used for reading receive data from the FIFO buffer and writing transmit data to the FIFO buffer.	R/W

(2) When the MBW bit is 0

Address(es): USB0.CFIFO.L 000A 0014h, USB0.D0FIFO.L 000A 0018h, USB0.D1FIFO.L 000A 001Ch



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	FIFO Port	This port is used for reading receive data from the FIFO buffer and writing transmit data to the FIFO buffer.	R/W

There are three FIFO ports: CFIFO, D0FIFO, and D1FIFO. Each FIFO port is configured of a port register (CFIFO, D0FIFO, or D1FIFO) that handles reading of data from the FIFO buffer memory and writing of data to the FIFO buffer memory, a port select register (CFIFOSEL, D0FIFOSEL, or D1FIFOSEL) that is used to select the pipe assigned to the FIFO port, and a port control register (CFIFOCTR, D0FIFOCTR, or D1FIFOCTR).

Each FIFO port has the following precautions.

- The FIFO buffer for DCP (control transfer) should be accessed through the CFIFO port.
- Accessing a FIFO buffer using DMA/DTC transfer should be performed through the D0FIFO or D1FIFO port.
- The D0FIFO and D1FIFO ports can also be accessed by the CPU.
- When using functions specific to a FIFO port, the pipe number (selected pipe) specified by the CURPIPE[3:0] bits in the port select register cannot be changed (when the DMA/DTC transfer function is used, etc.).
- The same pipe should not be assigned to two or more FIFO ports.
- There are two FIFO buffer states: the access right for a FIFO buffer can be on the CPU side or on the Serial Interface Engine (SIE) side. When the access right for a FIFO buffer is on the SIE side, the FIFO buffer cannot be accessed from the CPU.

FIFO Port Bit

Accessing the FIFO port bits allows reading the received data from the FIFO buffer or writing the transmit data to the FIFO buffer.

Each FIFO port register can be accessed only while the FRDY flag in each FIFO port control register (CFIFOCTR, D0FIFOCTR, or D1FIFOCTR) is 1.

The valid bits in a FIFO port register depend on the settings of the corresponding MBW bit of the FIFO port select register (CFIFOSEL, D0FIFOSEL, or D1FIFOSEL).

When the MBW bit is 1 (16-bit width), the data arrangement may differ from the data arrangement on the RAM depending on the value of the MDE.MDE[2:0] bits and the setting of the BIGEND bit (CFIFOSEL.BIGEND, D0FIFOSEL.BIGEND, or D1FIFOSEL.BIGEND). Table 40.5 lists the endian operation in 16-bit access. Note that if the total number of transmit data bytes is odd, access the lower byte in bytes when writing the last data.

When the MBW bit is 0 (8-bit width), access the lower byte in bytes.

Table 40.5 Endian Operation in 16-Bit Access

MDE.MDE[2:0] bits	CFIFOSEL.BIGEND Bit D0FIFOSEL.BIGEND Bit D1FIFOSEL.BIGEND Bit	Bits 15 to 8	Bits 7 to 0	Remarks
000b (big endian)	0 (little endian)	Data in address N + 1	Data in address N	Bytes reversed
	1 (big endian)	Data in address N	Data in address N + 1	
111b (little endian)	0 (little endian)	Data in address N + 1	Data in address N	
	1 (big endian)	Data in address N	Data in address N + 1	Bytes reversed

40.2.5 CFIFO Port Select Register (CFIFOSEL), D0FIFO Port Select Register (D0FIFOSEL), D1FIFO Port Select Register (D1FIFOSEL)

- CFIFOSEL

Address(es): USB0.CFIFOSEL 000A 0020h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
RCNT	REW	—	—	—	MBW	—	BIGEND	—	—	ISEL	—	CURPIPE[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	CURPIPE [3:0]	CFIFO Port Access Pipe Specification	b3 b0 0 0 0 0: DCP (Default control pipe) 0 0 0 1: Pipe 1 0 0 1 0: Pipe 2 0 0 1 1: Pipe 3 0 1 0 0: Pipe 4 0 1 0 1: Pipe 5 0 1 1 0: Pipe 6 0 1 1 1: Pipe 7 1 0 0 0: Pipe 8 1 0 0 1: Pipe 9 Settings other than above are prohibited.	R/W
b4	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b5	ISEL	CFIFO Port Access Direction When DCP is Selected	0: Reading from the buffer memory is selected. 1: Writing to the buffer memory is selected.	R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	BIGEND	CFIFO Port Endian Control	0: Little endian 1: Big endian	R/W
b9	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b10	MBW	CFIFO Port Access Bit Width	0: 8-bit width 1: 16-bit width	R/W
b13 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b14	REW	Buffer Pointer Rewind	0: The buffer pointer is not rewind. 1: The buffer pointer is rewind.	R/W*1
b15	RCNT	Read Count Mode	0: The CFIFOCTR.DTLN[8:0] flags are set to 000h when all of the receive data has been read from the CFIFO. (In double buffer mode, the DTLN[8:0] flag value is set to 000h when all the data has been read from only a single plane.) 1: The CFIFOCTR.DTLN[8:0] flags are decremented each time the receive data is read from the CFIFO.	R/W

Note 1. Only 0 can be read.

The same pipe should not be specified by the CURPIPE[3:0] bits in the CFIFOSEL, D0FIFOSEL, and D1FIFOSEL registers. When the CURPIPE[3:0] bits in the D0FIFOSEL and D1FIFOSEL registers are set to 0000b, no pipe is selected.

The pipe number should not be changed while the DMA/DTC transfer is enabled.

CURPIPE[3:0] Bits (CFIFO Port Access Pipe Specification)

The CURPIPE[3:0] bits specify the pipe number using which data is read or written through the CFIFO port.

After writing to these bits, read these bits to check that the written value agrees with the read value before proceeding to the next process.

Do not set the same pipe number to the CURPIPE[3:0] bits in the CFIFOSEL, D0FIFOSEL, and D1FIFOSEL registers. Even if an attempt is made to modify the setting of these bits during access to the FIFO buffer, the current access setting is retained until the access is completed. Then, the modification becomes effective, thus enabling continuous access.

ISEL Bit (CFIFO Port Access Direction When DCP is Selected)

After writing to the ISEL bit with the DCP being a selected pipe, read this bit to check that the written value agrees with the read value before proceeding to the next process.

Set this bit and the CURPIPE[3:0] bits simultaneously.

MBW Bit (CFIFO Port Access Bit Width)

The MBW bit specifies the bit width for accessing the CFIFO port.

When the selected pipe is in the receiving direction, once reading data is started after setting this bit, this bit should not be modified until all the data has been read.

When the selected pipe is in the receiving direction, set the CURPIPE[3:0] bits to a different value once, and then set these bits and the MBW bit simultaneously. For the procedure for modifying the CURPIPE[3:0] bits, follow the description of these bits.

When the selected pipe is in the transmitting direction, the bit width cannot be changed from 8-bit width to 16-bit width while data is being written to the buffer memory.

An odd number of bytes can also be written through byte-access control even when 16-bit width is selected.

REW Bit (Buffer Pointer Rewind)

The REW bit specifies whether or not to rewind the buffer pointer.

When the selected pipe is in the receiving direction, setting the REW bit to 1 while the FIFO buffer is being read allows re-reading the FIFO buffer from the first data (in double buffer mode, re-reading the currently-read FIFO buffer plane from the first data is allowed).

Do not set the REW bit to 1 simultaneously with modifying the CURPIPE[3:0] bits. Before setting the REW bit to 1, be sure to check that the FRDY flag is 1.

To re-write to the FIFO buffer again from the first data for the pipe in the transmitting direction, use the BCLR bit.

- D0FIFOSEL, D1FIFOSEL

Address(es): USB0.D0FIFOSEL 000A 0028h, USB0.D1FIFOSEL 000A 002Ch

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
RCNT	REW	DCLRM	DREQE	—	MBW	—	BIGEND	—	—	—	—	CURPIPE[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	CURPIPE [3:0]	FIFO Port Access Pipe Select	b3 b0 0 0 0 0: DCP (Default control pipe) 0 0 0 1: Pipe 1 0 0 1 0: Pipe 2 0 0 1 1: Pipe 3 0 1 0 0: Pipe 4 0 1 0 1: Pipe 5 0 1 1 0: Pipe 6 0 1 1 1: Pipe 7 1 0 0 0: Pipe 8 1 0 0 1: Pipe 9 Settings other than above are prohibited.	R/W
b7 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	BIGEND	FIFO Port Endian Control	0: Little endian 1: Big endian	R/W
b9	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b10	MBW	FIFO Port Access Bit Width	0: 8-bit width 1: 16-bit width	R/W
b11	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b12	DREQE	DMA/DTC Transfer Request Enable	0: DMA/DTC transfer request is disabled. 1: DMA/DTC transfer request is enabled.	R/W
b13	DCLRM	Automatic FIFO Buffer Memory Clear Mode after Selected Pipe is Read	0: Automatic buffer clear mode is disabled. 1: Automatic buffer clear mode is enabled.	R/W
b14	REW	Buffer Pointer Rewind	0: The buffer pointer is not rewind. 1: The buffer pointer is rewind.	R/W*1
b15	RCNT	Read Count Mode	0: The DnFIFOCTR.DTLN[8:0] flags are set to 000h when all of the receive data has been read from the DnFIFO. (In double buffer mode, the DTLN[8:0] flag value is set to 000h when all the data has been read from only a single plane.) 1: The DnFIFOCTR.DTLN[8:0] flags are decremented each time the receive data is read from the DnFIFO. (n = 0, 1)	R/W

Note 1. Only 0 can be read.

The same pipe should not be specified by the CURPIPE[3:0] bits in the CFIFOSEL, D0FIFOSEL, and D1FIFOSEL registers. When the CURPIPE[3:0] bits in the D0FIFOSEL and D1FIFOSEL registers are set to 0000b, no pipe is selected.

The pipe number should not be changed while the DMA/DTC transfer is enabled.

CURPIPE[3:0] Bits (FIFO Port Access Pipe Select)

The CURPIPE[3:0] bits specify the pipe number using which data is read or written through the D0FIFO port or D1FIFO port.

After writing to these bits, read these bits to check that the written value agrees with the read value before proceeding to the next process.

Do not set the same pipe number to the CURPIPE[3:0] bits in the CFIFOSEL, D0FIFOSEL, and D1FIFOSEL registers. Even if an attempt is made to modify the setting of these bits during access to the FIFO buffer, the current access setting is retained until the access is completed. Then, the modification becomes effective, thus enabling continuous access.

MBW Bit (FIFO Port Access Bit Width)

The MBW bit specifies the bit width for accessing the D0FIFO port or D1FIFO port.

When the selected pipe is in the receiving direction, once reading data is started after setting this bit, this bit should not be modified until all the data has been read.

When the selected pipe is in the receiving direction, set the CURPIPE[3:0] bits to a different value once, and then set these bits and the MBW bit simultaneously. For the procedure for modifying the CURPIPE[3:0] bits, follow the description of these bits.

When the selected pipe is in the transmitting direction, the bit width cannot be changed from 8-bit width to 16-bit width while data is being written to the buffer memory.

An odd number of bytes can also be written through byte-access control even when 16-bit width is selected.

DREQE Bit (DMA/DTC Transfer Request Enable)

The DREQE bit enables or disables the DMA/DTC transfer request to be issued.

Before setting the DREQE bit to 1 to enable the DMA/DTC transfer request to be issued, set the CURPIPE[3:0] bits.

When modifying the setting of the CURPIPE[3:0] bits, set this bit to 0 first.

DCLRM Bit (Automatic FIFO Buffer Memory Clear Mode after Selected Pipe is Read)

The DCLRM bit enables or disables the buffer memory to be cleared automatically after data in the selected pipe has been read.

With this bit set to 1, the USB sets the BCLR bit to 1 for the FIFO buffer of the selected pipe on receiving a zero-length packet while the FIFO buffer assigned to the selected pipe is empty, or on receiving a short packet and reading the data while the PIPECFG.BFRE bit is 1.

When using the USB with the SOFCFG.BRDYM bit set to 1, set this bit to 0.

REW Bit (Buffer Pointer Rewind)

The REW bit specifies whether or not to rewind the buffer pointer.

When the selected pipe is in the receiving direction, setting the REW bit to 1 while the FIFO buffer is being read allows re-reading the FIFO buffer from the first data (in double buffer mode, re-reading the currently-read FIFO buffer plane from the first data is allowed).

Do not set the REW bit to 1 simultaneously with modifying the CURPIPE[3:0] bits. Before setting the REW bit to 1, be sure to check that the FRDY flag is 1.

To re-write to the FIFO buffer again from the first data for the pipe in the transmitting direction, use the BCLR bit.

RCNT Bit (Read Count Mode)

The RCNT bit specifies the read mode for the value in the DnFIFOCTR.DTLN[8:0] flags.

When accessing DnFIFO with the PIPECFG.BFRE bit set to 1, set the RCNT bit to 0.

40.2.6 CFIFO Port Control Register (CFIFOCTR), D0FIFO Port Control Register (D0FIFOCTR), D1FIFO Port Control Register (D1FIFOCTR)

Address(es): USB0.CFIFOCTR 000A 0022h, USB0.D0FIFOCTR 000A 002Ah, USB0.D1FIFOCTR 000A 002Eh

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
BVAL	BCLR	FRDY	—	—	—	—	DTLN[8:0]								
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b8 to b0	DTLN[8:0]	Receive Data Length Flag	Indicate the length of the receive data. These bits indicate different values depending on the setting of the RCNT bit in the port select register. For details, refer to the description on the DTLN[8:0] flags shown below.	R
b12 to b9	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b13	FRDY	FIFO Port Ready Flag	0: FIFO port access is disabled. 1: FIFO port access is enabled.	R
b14	BCLR	CPU Buffer Clear	0: Does not operate. 1: Clears the buffer memory on the CPU side.	R/W ^{*1}
b15	BVAL	Buffer Memory Valid	0: Invalid 1: Writing ended	R/W

Note 1. Only 0 can be read.

Registers CFIFOCTR, D0FIFOCTR, and D1FIFOCTR correspond to CFIFO, D0FIFO, and D1FIFO, respectively.

DTLN[8:0] Flags (Receive Data Length Flag)

The DTLN[8:0] flags indicate the length of the receive data.

While the FIFO buffer is being read, the DTLN[8:0] flags indicate different values depending on the DnFIFOSEL.RCNT bit (n = 0, 1) value as described below.

- RCNT = 0

The USB sets the DTLN[8:0] flags to indicate the length of the receive data until the CPU or DMAC/DTC has read all the received data from a single FIFO buffer plane.

While the PIPECFG.BFRE bit = 1, these bits retain the length of the receive data until the BCLR bit is set to 1 even after all the data has been read.

- RCNT = 1

The USB decrements the value indicated by the DTLN[8:0] flags each time data is read from the FIFO buffer. (The value is decremented by one when the MBW bit is 0, and by two when the MBW bit is 1.)

The USB sets these bits to 0 when all the data has been read from one FIFO buffer plane. However, in double buffer mode, if data has been received in one FIFO buffer plane before all the data has been read from the other plane, the USB sets these bits to indicate the length of the receive data in the former plane when all the data has been read from the latter plane.

FRDY Flag (FIFO Port Ready Flag)

The FRDY flag indicates whether the FIFO port can be accessed by the CPU or DMAC/DTC.

In the following cases, the USB sets the FRDY flag to 1 but data cannot be read via the FIFO port because there is no data to be read. In these cases, set the BCLR bit to 1 to clear the FIFO buffer, and enable transmission and reception of the next data.

- A zero-length packet is received when the FIFO buffer assigned to the selected pipe is empty.
- A short packet is received and the data is completely read while the PIPECFG.BFRE bit = 1.

BCLR Bit (CPU Buffer Clear)

The BCLR bit should be set to 1 to clear the FIFO buffer on the CPU side for the selected pipe.

When double buffer mode is set for the FIFO buffer assigned to the selected pipe, the USB clears only the FIFO buffer on the CPU side even when both planes are read-enabled.

When the selected pipe is the DCP, setting the BCLR bit to 1 allows the USB to clear the FIFO buffer regardless of whether the FIFO buffer is on the CPU side or SIE side. To clear the buffer on the SIE side, set the DCPCTR.PID[1:0] bits for the DCP to 00b (NAK) before setting the BCLR bit to 1.

When the selected pipe is in the transmitting direction, if 1 is written to the BVAL bit and the BCLR bit simultaneously, the USB clears the data that has been written before it, enabling transmission of a zero-length packet.

When the selected pipe is not the DCP, writing 1 to the BCLR bit should be done while the FRDY flag in the FIFO port control register is 1 (set by the USB).

BVAL Bit (Buffer Memory Valid)

The BVAL bit should be set to 1 when data has been completely written to the FIFO buffer on the CPU side for the pipe selected using the CURPIPE[3:0] bits (selected pipe).

When the selected pipe is in the transmitting direction, set the BVAL bit to 1 in the following cases. Then, the USB switches the FIFO buffer from the CPU side to the SIE side, thus enabling transmission.

- To transmit a short packet, set the BVAL bit to 1 after data has been written.
- To transmit a zero-length packet, set the BVAL bit to 1 before data is written to the FIFO buffer.

When data of the maximum packet size has been written for the pipe in continuous transfer mode, the USB sets the BVAL bit to 1 and switches the FIFO buffer from the CPU side to the SIE side, thus enabling transmission.

Writing 1 to the BVAL bit should be done while the FRDY flag is 1 (set by the USB).

When the selected pipe is in the receiving direction, do not set the BVAL bit to 1.

40.2.7 Interrupt Enable Register 0 (INTENB0)

Address(es): USB0.INTENB0 000A 0030h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	VBSE	RSME	SOFE	DVSE	CTRE	BEMPE	NRDYE	BRDYE	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	BRDYE	Buffer Ready Interrupt Enable	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b9	NRDYE	Buffer Not Ready Response Interrupt Enable	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b10	BEMPE	Buffer Empty Interrupt Enable	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b11	CTRE	Control Transfer Stage Transition Interrupt Enable*1	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b12	DVSE	Device State Transition Interrupt Enable*1	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b13	SOFE	Frame Number Update Interrupt Enable	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b14	RSME	Resume Interrupt Enable*1	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b15	VBSE	VBUS Interrupt Enable	0: Interrupt output disabled 1: Interrupt output enabled	R/W

Note 1. The RSME, DVSE, and CTRE bits can be set to 1 only when the function controller is selected. Set these bits to 0 when the host controller is selected.

On detecting the interrupt corresponding to the bit in the INTENB0 register to which 1 has been set by software, the USB generates the USB interrupt request.

The USB sets 1 to each status bit in the INTSTS0 register when a detection condition of the corresponding interrupt source has been satisfied regardless of the setting in the INTENB0 register (regardless of whether the interrupt output is enabled or disabled).

While the status bit in the INTSTS0 register corresponding to the interrupt source indicates 1, the USB generates the USB interrupt request when the corresponding interrupt enable bit in the INTENB0 register is modified from 0 to 1 by software.

40.2.8 Interrupt Enable Register 1 (INTENB1)

Address(es): USB0.INTENB1 000A 0032h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	OVRCRE	BCHGE	—	DTCHE	ATTCH E	—	—	—	—	EOFERRE	SIGNE	SACKE	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	SACKE	Setup Transaction Normal Response Interrupt Enable	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b5	SIGNE	Setup Transaction Error Interrupt Enable	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b6	EOFERRE	EOF Error Detection Interrupt Enable	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b10 to b7	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b11	ATTCH E	Connection Detection Interrupt Enable	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b12	DTCHE	Disconnection Detection Interrupt Enable	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b13	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b14	BCHGE	USB Bus Change Interrupt Enable	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b15	OVRCRE	Overcurrent Input Change Interrupt Enable	0: Interrupt output disabled 1: Interrupt output enabled	R/W

Note: The bits in the INTENB1 register can be set to 1 only when the host controller is selected. Set these bits to 0 when the function controller is selected.

The INTENB1 register specifies the interrupt masks when the host controller is selected, and for the setup transaction. On detecting the interrupt corresponding to the bit in the INTENB1 register to which 1 has been set by software, the USB generates the USB interrupt request.

The USB sets 1 to each status bit in the INTSTS1 register when a detection condition of the corresponding interrupt source has been satisfied regardless of the setting in the INTENB1 register (regardless of whether the interrupt output is enabled or disabled).

While the status bit in the INTSTS1 register corresponding to the interrupt source indicates 1, the USB generates the USB interrupt request when the corresponding interrupt enable bit in the INTENB1 register is modified from 0 to 1 by software.

When the function controller is selected, the interrupts should not be enabled.

40.2.9 BRDY Interrupt Enable Register (BRDYENB)

Address(es): USB0.BRDYENB 000A 0036h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	PIPE9B RDYE	PIPE8B RDYE	PIPE7B RDYE	PIPE6B RDYE	PIPE5B RDYE	PIPE4B RDYE	PIPE3B RDYE	PIPE2B RDYE	PIPE1B RDYE	PIPE0B RDYE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	PIPE0BRDYE	BRDY Interrupt Enable for Pipe 0	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b1	PIPE1BRDYE	BRDY Interrupt Enable for Pipe 1	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b2	PIPE2BRDYE	BRDY Interrupt Enable for Pipe 2	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b3	PIPE3BRDYE	BRDY Interrupt Enable for Pipe 3	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b4	PIPE4BRDYE	BRDY Interrupt Enable for Pipe 4	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b5	PIPE5BRDYE	BRDY Interrupt Enable for Pipe 5	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b6	PIPE6BRDYE	BRDY Interrupt Enable for Pipe 6	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b7	PIPE7BRDYE	BRDY Interrupt Enable for Pipe 7	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b8	PIPE8BRDYE	BRDY Interrupt Enable for Pipe 8	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b9	PIPE9BRDYE	BRDY Interrupt Enable for Pipe 9	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b15 to b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The BRDYENB register enables or disables the INTSTS0.BRDY flag to be set to 1 when the BRDY interrupt is detected for each pipe.

On detecting the BRDY interrupt for the pipe corresponding to the bit in the BRDYENB register to which 1 has been set by software, the USB sets 1 to the corresponding BRDYSTS.PIPEnBRDY flag ($n = 0$ to 9) and the INTSTS0.BRDY flag. If INTENB0.BRDYE = 1 at this time, the USB generates the BRDY interrupt request.

While at least one PIPEnBRDY flag indicates 1, the USB generates the BRDY interrupt request when the corresponding interrupt enable bit in the BRDYENB register is modified from 0 to 1 by software.

40.2.10 NRDY Interrupt Enable Register (NRDYENB)

Address(es): USB0.NRDYENB 000A 0038h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	PIPE9NRDYE	PIPE8NRDYE	PIPE7NRDYE	PIPE6NRDYE	PIPE5NRDYE	PIPE4NRDYE	PIPE3NRDYE	PIPE2NRDYE	PIPE1NRDYE	PIPE0NRDYE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	PIPE0NRDYE	NRDY Interrupt Enable for Pipe 0	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b1	PIPE1NRDYE	NRDY Interrupt Enable for Pipe 1	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b2	PIPE2NRDYE	NRDY Interrupt Enable for Pipe 2	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b3	PIPE3NRDYE	NRDY Interrupt Enable for Pipe 3	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b4	PIPE4NRDYE	NRDY Interrupt Enable for Pipe 4	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b5	PIPE5NRDYE	NRDY Interrupt Enable for Pipe 5	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b6	PIPE6NRDYE	NRDY Interrupt Enable for Pipe 6	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b7	PIPE7NRDYE	NRDY Interrupt Enable for Pipe 7	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b8	PIPE8NRDYE	NRDY Interrupt Enable for Pipe 8	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b9	PIPE9NRDYE	NRDY Interrupt Enable for Pipe 9	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b15 to b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The NRDYENB register enables or disables the INTSTS0.NRDY flag to be set to 1 when the NRDY interrupt is detected for each pipe.

On detecting the NRDY interrupt for the pipe corresponding to the bit in the NRDYENB register to which 1 has been set by software, the USB sets 1 to the corresponding NRDYSTS.PIPEnNRDYE flag (n = 0 to 9) and the INTSTS0.NRDY flag. If INTENB0.NRDYE = 1 at this time, the USB generates the NRDY interrupt request.

While at least one PIPEnNRDYE flag indicates 1, the USB generates the NRDY interrupt request when the corresponding interrupt enable bit in the NRDYENB register is modified from 0 to 1 by software.

40.2.11 BEMP Interrupt Enable Register (BEMPENB)

Address(es): USB0.BEMPENB 000A 003Ah

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	PIPE9B EMPE	PIPE8B EMPE	PIPE7B EMPE	PIPE6B EMPE	PIPE5B EMPE	PIPE4B EMPE	PIPE3B EMPE	PIPE2B EMPE	PIPE1B EMPE	PIPE0B EMPE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	PIPE0BEMPE	BEMP Interrupt Enable for Pipe 0	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b1	PIPE1BEMPE	BEMP Interrupt Enable for Pipe 1	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b2	PIPE2BEMPE	BEMP Interrupt Enable for Pipe 2	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b3	PIPE3BEMPE	BEMP Interrupt Enable for Pipe 3	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b4	PIPE4BEMPE	BEMP Interrupt Enable for Pipe 4	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b5	PIPE5BEMPE	BEMP Interrupt Enable for Pipe 5	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b6	PIPE6BEMPE	BEMP Interrupt Enable for Pipe 6	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b7	PIPE7BEMPE	BEMP Interrupt Enable for Pipe 7	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b8	PIPE8BEMPE	BEMP Interrupt Enable for Pipe 8	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b9	PIPE9BEMPE	BEMP Interrupt Enable for Pipe 9	0: Interrupt output disabled 1: Interrupt output enabled	R/W
b15 to b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The BEMPENB register enables or disables the INTSTS0.BEMP flag to be set to 1 when the BEMP interrupt is detected for each pipe.

On detecting the BEMP interrupt for the pipe corresponding to the bit in the BEMPENB register to which 1 has been set by software, the USB sets 1 to the corresponding BEMPSTS.PIPEnBEMP flag ($n = 0$ to 9) and the INTSTS0.BEMP flag. If INTENB0.BEMPE = 1 at this time, the USB generates the BEMP interrupt request.

While at least one PIPEnBEMP flag in the BEMPSTS register indicates 1, the USB generates the BEMP interrupt request when the corresponding interrupt enable bit in the BEMPENB register is modified from 0 to 1 by software.

40.2.12 SOF Output Configuration Register (SOFCFG)

Address(es): USB0.SOFCFG 000A 003Ch

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	TRNENSEL	—	BRDYM	—	EDGESTS	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	EDGESTS	Edge Interrupt Output Status Monitor Flag*1	Indicates 1 when the edge interrupt output signal is in the middle of the edge processing.	R
b5	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b6	BRDYM	BRDY Interrupt Status Clear Timing	0: Software clears the status. 1: The USB clears the status when data has been read from the FIFO buffer or data has been written to the FIFO buffer.	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b8	TRNENSEL	Transaction-Enabled Time Select*1	0: For non-low-speed communication 1: For low-speed communication	R/W
b15 to b9	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Confirm that this bit is 0 before stopping the clock supply to the USB module.

EDGESTS Flag (Edge Interrupt Output Status Monitor Flag)

The EDGESTS flag indicates 1 when the edge interrupt output signal is in the middle of the edge processing.

Confirm that this flag is 0 before stopping the clock supply to the USB.

BRDYM Bit (BRDY Interrupt Status Clear Timing)

The BRDYM bit specifies the timing for clearing the BRDY interrupt status for each pipe.

TRNENSEL Bit (Transaction-Enabled Time Select)

The TRNENSEL bit selects, for full-speed or low-speed communication, the transaction-enabled time in which the USB issues tokens in a frame via the port.

Set the TRNENSEL bit to 1 when a low-speed device is connected.

The TRNENSEL bit is valid only when the host controller is selected.

This bit should be set to 0 if the function controller is selected.

40.2.13 Interrupt Status Register 0 (INTSTS0)

Address(es): USB0.INTSTS0 000A 0040h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
VBINT	RESM	SOFR	DVST	CTRT	BEMP	NRDY	BRDY	VBSTS	DVSQ[2:0]		VALID	CTSQ[2:0]			
Value after reset:	0	0	0	0/1*1	0	0	0	0	0*2	0*3	0*3	0/1*3	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	CTSQ[2:0]	Control Transfer Stage Flag	b2 b0 0 0 0: Idle or setup stage 0 0 1: Control read data stage 0 1 0: Control read status stage 0 1 1: Control write data stage 1 0 0: Control write status stage 1 0 1: Control write (no data) status stage 1 1 0: Control transfer sequence error	R
b3	VALID	USB Request Reception Flag	0: Setup packet is not received. 1: Setup packet is received.	R/W
b6 to b4	DVSQ[2:0]	Device State Flag	b6 b4 0 0 0: Powered state 0 0 1: Default state 0 1 0: Address state 0 1 1: Configured state 1 x x: Suspended state	R
b7	VBSTS	VBUS Input Status Flag	0: USB0_VBUS pin is low. 1: USB0_VBUS pin is high.	R
b8	BRDY	Buffer Ready Interrupt Status Flag	0: BRDY interrupts are not generated. 1: BRDY interrupts are generated.	R
b9	NRDY	Buffer Not Ready Interrupt Status Flag	0: NRDY interrupts are not generated. 1: NRDY interrupts are generated.	R
b10	BEMP	Buffer Empty Interrupt Status Flag	0: BEMP interrupts are not generated. 1: BEMP interrupts are generated.	R
b11	CTRT	Control Transfer Stage Transition Interrupt Status Flag*5	0: Control transfer stage transition interrupts are not generated. 1: Control transfer stage transition interrupts are generated.	R/W*4
b12	DVST	Device State Transition Interrupt Status Flag*5	0: Device state transition interrupts are not generated. 1: Device state transition interrupts are generated.	R/W*4
b13	SOFR	Frame Number Refresh Interrupt Status Flag	0: SOF interrupts are not generated. 1: SOF interrupts are generated.	R/W*4
b14	RESM	Resume Interrupt Status Flag*5,*6	0: Resume interrupts are not generated. 1: Resume interrupts are generated.	R/W*4
b15	VBINT	VBUS Interrupt Status Flag*6	0: VBUS interrupts are not generated. 1: VBUS interrupts are generated.	R/W*4

x: Don't care

Note 1. The value is 0 when the MCU is reset and 1 after a USB bus reset.

Note 2. The value is 1 when the USB0_VBUS pin is high and 0 when the USB0_VBUS pin is low.

Note 3. The value is 000b when the MCU is reset and 001b after a USB bus reset.

Note 4. To clear the VBINT, RESM, SOFR, DVST, CTRT, or VALID flag, write 0 only to the flags to be cleared; write 1 to the other flags. Do not write 0 to the status flags indicating 0.

Note 5. The status of the RESM, DVST, and CTRT flags are changed only when the function controller is selected. Set the corresponding interrupt enable bits to 0 (disabled) when the host controller is selected.

Note 6. A change in the status indicated by the VBINT and RESM flags can be detected even while the clock supply is stopped (the SCCK bit = 0), and the interrupts are output when the corresponding interrupt enable bits are enabled. Clearing the status through software should be done after enabling the clock supply.

CTSQ[2:0] Flags (Control Transfer Stage Flag)

When the host controller is selected, the read value is invalid.

VALID Flag (USB Request Reception Flag)

When the host controller is selected, the read value is invalid.

DVSQ[2:0] Flags (Device State Flag)

The DVSQ[2:0] flags are initialized by a USB bus reset.

When the host controller is selected, the read value is invalid.

BRDY Flag (Buffer Ready Interrupt Status Flag)

Indicates the BRDY interrupt status.

The USB sets the BRDY flag to 1 when at least one PIPEnBRDY flag (n = 0 to 9) is set to 1 among the PIPEnBRDY flags. These bits correspond to the BRDYENB.PIPEnBRDYE bits (n = 0 to 9) to which 1 has been set, when the USB detects the BRDY interrupt status in at least one pipe among the pipes for which the BRDY interrupt output is enabled by software.

For the conditions for PIPEnBRDY status assertion, refer to [section 40.3.3.1, BRDY Interrupt](#).

The USB sets the BRDY flag to 0 when 0 is written by software to all the PIPEnBRDY flags corresponding to the PIPEnBRDYE bits that have been set to 1.

The BRDY flag cannot be set to 0 even if 0 is written to this bit by software.

NRDY Flag (Buffer Not Ready Interrupt Status Flag)

The USB sets the NRDY flag to 1 when at least one PIPEnNRDY flag (n = 0 to 9) is set to 1 among the PIPEnNRDY flags corresponding to the PIPEnNRDYE bits (n = 0 to 9) to which 1 has been set (when the USB detects the NRDY interrupt status in at least one pipe among the pipes for which software enables the NRDY interrupt output).

For the conditions for PIPEnNRDY status assertion, refer to [section 40.3.3.2, NRDY Interrupt](#).

The USB sets the NRDY flag to 0 when 0 is written by software to all the PIPEnNRDY flags corresponding to the PIPEnNRDYE bits that have been set to 1.

The NRDY flag cannot be set to 0 even if 0 is written to this bit by software.

BEMP Flag (Buffer Empty Interrupt Status Flag)

The USB sets the BEMP flag to 1 when at least one PIPEnBEMP flag (n = 0 to 9) is set to 1 among the PIPEnBEMP flags corresponding to the PIPEnBEMPE bits (n = 0 to 9) to which 1 has been set (when the USB detects the BEMP interrupt status in at least one pipe among the pipes for which the BEMP interrupt output is enabled by software).

For the conditions for PIPEnBEMP status assertion, refer to [section 40.3.3.3, BEMP Interrupt](#).

The USB sets the BEMP flag to 0 when 0 is written by software to all the PIPEnBEMP flags corresponding to the PIPEnBEMPE bits that have been set to 1.

The BEMP flag cannot be set to 0 even if 0 is written to this bit by software.

CTRT Flag (Control Transfer Stage Transition Interrupt Status Flag)

When the function controller is selected, the USB updates the value of the CTSQ[2:0] flags and sets the CTRT flag to 1 on detecting a change in the control transfer stage.

When a control transfer stage transition interrupt is generated, clear the status before the USB detects the next control transfer stage transition.

When the host controller is selected, the read value is invalid.

DVST Flag (Device State Transition Interrupt Status Flag)

When the function controller is selected, the USB updates the DVSQ[2:0] value and sets the DVST flag to 1 on detecting a change in the device state.

When a device state transition interrupt is generated, clear the status before the USB detects the next device state transition.

When the host controller is selected, the read value is invalid.

SOFR Flag (Frame Number Refresh Interrupt Status Flag)

(1) When the host controller is selected

The USB sets the SOFR flag to 1 on updating the frame number when the DVSTCTR0.UACT bit has been set to 1 by software. (A frame number refresh interrupt is detected every 1 ms.)

(2) When the function controller is selected

The USB sets the SOFR flag to 1 on updating the frame number. (A frame number refresh interrupt is detected every 1 ms.)

The USB can detect an SOFR interrupt through the internal interpolation function even when a damaged SOF packet is received from the USB host.

RESM Flag (Resume Interrupt Status Flag)

When the function controller is selected, the USB sets the RESM flag to 1 on detecting the falling edge of the signal on the USB0_DP pin in the suspended state (DVSQ[2:0] = 1xxb).

When the host controller is selected, the read value is invalid.

VBINT Flag (VBUS Interrupt Status Flag)

The USB sets the VBINT flag to 1 on detecting a level change (high to low or low to high) in the USB0_VBUS pin input value. The USB sets the VBSTS flag to indicate the USB0_VBUS pin input value. When the VBUS interrupt is generated, use software to repeat reading the VBSTS flag until the same value is read three or more times, and eliminate chattering.

40.2.14 Interrupt Status Register 1 (INTSTS1)

Address(es): USB0.INTSTS1 000A 0042h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	OVRCR	BCHG	—	DTCH	ATTCH	—	—	—	—	EOFERR	SIGN	SACK	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	SACK	Setup Transaction Normal Response Interrupt Status Flag	0: SACK interrupts are not generated. 1: SACK interrupts are generated.	R/W *1
b5	SIGN	Setup Transaction Error Interrupt Status Flag	0: SIGN interrupts are not generated. 1: SIGN interrupts are generated.	R/W *1
b6	EOFERR	EOF Error Detection Interrupt Status Flag	0: EOFERR interrupts are not generated. 1: EOFERR interrupts are generated.	R/W *1
b10 to b7	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b11	ATTCH	ATTCH Interrupt Status Flag	0: ATTCH interrupts are not generated. 1: ATTCH interrupts are generated.	R/W *1
b12	DTCH	USB Disconnection Detection Interrupt Status Flag	0: DTCH interrupts are not generated. 1: DTCH interrupts are generated.	R/W *1
b13	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b14	BCHG	USB Bus Change Interrupt Status Flag *2	0: BCHG interrupts are not generated. 1: BCHG interrupts are generated.	R/W *1
b15	OVRCCR	Overcurrent Input Change Interrupt Status Flag*2	0: OVRCCR interrupts are not generated. 1: OVRCCR interrupts are generated.	R/W *1

Note 1. To clear the status indicated by the flags in the INTSTS1 register, write 0 only to the flags to be cleared; write 1 to the other flags.

Note 2. A change in the status indicated by the OVRCCR or BCHG flag can be detected even while the clock supply is stopped (while the SYSCFG.SCKE bit = 0), and the interrupt is output when the corresponding interrupt enable bit is enabled. Clearing the status through software should be done after setting the SYSCFG.SCKE bit to 1.

No interrupts other than those indicated by the BCHG and OVRCCR flags can be detected while the clock supply is stopped (while the SYSCFG.SCKE bit = 0).

The INTSTS1 register is used to confirm the status of each interrupt when the host controller is selected.

The various status change interrupts indicated by the flags in the INTSTS1 register should be enabled only when the host controller is selected.

SACK Flag (Setup Transaction Normal Response Interrupt Status Flag)

Indicates the status of the setup transaction normal response interrupt when the host controller is selected.

The USB detects the SACK interrupt when ACK response is returned from the peripheral device during the setup transactions issued by the USB, and sets the SACK flag to 1. Here, if the corresponding interrupt enable bit has been set to 1 by software, the USB generates the SACK interrupt.

When the function controller is selected, the read value is invalid.

SIGN Flag (Setup Transaction Error Interrupt Status Flag)

Indicates the status of the setup transaction error interrupt when the host controller is selected.

The USB detects the SIGN interrupt when ACK response is not returned from the peripheral device three consecutive times during the setup transactions issued by this module, and sets the SIGN flag to 1. Here, if the corresponding interrupt enable bit has been set to 1 by software, the USB generates the SIGN interrupt.

Specifically, the USB detects the SIGN interrupt when any of the following response conditions occur for three consecutive setup transactions.

- Timeout is detected by the USB when the peripheral device has returned no response.
- A damaged ACK packet is received.
- A handshake other than ACK (NAK, NYET, or STALL) is received.

When the function controller is selected, the read value is invalid.

EOFERR Flag (EOF Error Detection Interrupt Status Flag)

Indicates the status of the EOFERR interrupt when the host controller is selected.

The USB detects the EOFERR interrupt on detecting that communication is not completed at the EOF2 timing prescribed by USB Specification 2.0, and sets the EOFERR flag to 1. Here, if the corresponding interrupt enable bit has been set to 1 by software, the USB generates the EOFERR interrupt.

After detecting the EOFERR interrupt, the USB controls hardware as described below (irrespective of the setting of the corresponding interrupt enable bit). All pipes in which communications are currently carried out for the USB port should be terminated by software and perform re-enumeration of the USB port.

- Modifies the DVSTCTR0.UACT bit for the port in which an EOFERR interrupt has been detected to 0.
- Puts the port in which an EOFERR interrupt has been generated into the idle state.

When the function controller is selected, the read value is invalid.

ATTCH Flag (ATTCH Interrupt Status Flag)

Indicates the status of the ATTCH interrupt when the host controller is selected.

The USB detects the ATTCH interrupt on detecting J-state or K-state of the full-speed or low-speed signal level for 2.5 μ s, and sets the ATTCH flag to 1. Here, if the corresponding interrupt enable bit has been set to 1 by software, the USB generates the interrupt.

Specifically, the USB detects the ATTCH interrupt on any of the following conditions.

- K-state, SE0, or SE1 changes to J-state, and J-state continues for 2.5 μ s.
- J-state, SE0, or SE1 changes to K-state, and K-state continues for 2.5 μ s.

When the function controller is selected, the read value is invalid.

DTCH Flag (USB Disconnection Detection Interrupt Status Flag)

Indicates the status of the USB disconnection detection interrupt when the host controller is selected.

The USB detects the DTCH interrupt on detecting USB bus disconnection, and sets the DTCH flag to 1. Here, if the corresponding interrupt enable bit has been set to 1 by software, the USB generates the interrupt.

The USB detects bus disconnection based on USB Specification 2.0.

After detecting the DTCH interrupt, the USB controls hardware as described below (irrespective of the setting of the corresponding interrupt enable bit). All the pipes in which communications are currently carried out for the USB port should be terminated by software and make a transition to the wait state for bus connection to the USB port (wait state for ATTCH interrupt generation).

- Modifies the DVSTCTR0.UACT bit for the port in which a DTCH interrupt has been detected to 0.
- Puts the port in which a DTCH interrupt has been generated into the idle state.

When the function controller is selected, the read value is invalid.

BCHG Flag (USB Bus Change Interrupt Status Flag)

Indicates the status of the USB bus change interrupt.

The USB detects the BCHG interrupt when a change in the full-speed or low-speed signal level occurs on the USB port (a change from J-state, K-state, or SE0 to J-state, K-state, or SE0), and sets the BCHG flag to 1. Here, if the corresponding interrupt enable bit has been set to 1 by software, the USB generates the interrupt.

The USB sets the LNST[1:0] flags to indicate the current input state of the USB port. When the BCHG interrupt is generated, use software to repeat reading the LNST[1:0] flags until the same value is read three or more times, and eliminate chattering.

A change in the USB bus state can be detected even while the internal clock supply is stopped.

When the function controller is selected, the read value is invalid.

OVRCCR Flag (Overcurrent Input Change Interrupt Status Flag)

Indicates the status of the USB0_OVRCURA and USB0_OVRCURB input pin change interrupt.

The USB detects the OVRCCR interrupt when a change (high to low or low to high) occurs in at least one of the input values to the USB0_OVRCURA and USB0_OVRCURB pins, and sets the OVRCCR flag to 1. Here, if the corresponding interrupt enable bit has been set to 1 by software, the USB generates the interrupt.

40.2.15 BRDY Interrupt Status Register (BRDYSTS)

Address(es): USB0.BRDYSTS 000A 0046h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	PIPE9B RDY	PIPE8B RDY	PIPE7B RDY	PIPE6B RDY	PIPE5B RDY	PIPE4B RDY	PIPE3B RDY	PIPE2B RDY	PIPE1B RDY	PIPE0B RDY
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	PIPE0BRDY	BRDY Interrupt Status Flag for Pipe 0*2	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b1	PIPE1BRDY	BRDY Interrupt Status Flag for Pipe 1*2	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b2	PIPE2BRDY	BRDY Interrupt Status Flag for Pipe 2*2	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b3	PIPE3BRDY	BRDY Interrupt Status Flag for Pipe 3*2	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b4	PIPE4BRDY	BRDY Interrupt Status Flag for Pipe 4*2	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b5	PIPE5BRDY	BRDY Interrupt Status Flag for Pipe 5*2	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b6	PIPE6BRDY	BRDY Interrupt Status Flag for Pipe 6*2	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b7	PIPE7BRDY	BRDY Interrupt Status Flag for Pipe 7*2	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b8	PIPE8BRDY	BRDY Interrupt Status Flag for Pipe 8*2	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b9	PIPE9BRDY	BRDY Interrupt Status Flag for Pipe 9*2	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b15 to b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. When the SOFCFG.BRDYM bit is set to 0, to clear the status indicated by the flags in the BRDYSTS register, write 0 only to the flags to be cleared; write 1 to the other flags.

Note 2. When the SOFCFG.BRDYM bit is set to 0, clearing BRDY Interrupts should be done before accessing the FIFO.

40.2.16 NRDY Interrupt Status Register (NRDYSTS)

Address(es): USB0.NRDYSTS 000A 0048h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	PIPE9NRDY	PIPE8NRDY	PIPE7NRDY	PIPE6NRDY	PIPE5NRDY	PIPE4NRDY	PIPE3NRDY	PIPE2NRDY	PIPE1NRDY	PIPE0NRDY
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	PIPE0NRDY	NRDY Interrupt Status Flag for Pipe 0	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b1	PIPE1NRDY	NRDY Interrupt Status Flag for Pipe 1	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b2	PIPE2NRDY	NRDY Interrupt Status Flag for Pipe 2	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b3	PIPE3NRDY	NRDY Interrupt Status Flag for Pipe 3	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b4	PIPE4NRDY	NRDY Interrupt Status Flag for Pipe 4	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b5	PIPE5NRDY	NRDY Interrupt Status Flag for Pipe 5	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b6	PIPE6NRDY	NRDY Interrupt Status Flag for Pipe 6	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b7	PIPE7NRDY	NRDY Interrupt Status Flag for Pipe 7	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b8	PIPE8NRDY	NRDY Interrupt Status Flag for Pipe 8	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b9	PIPE9NRDY	NRDY Interrupt Status Flag for Pipe 9	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b15 to b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. To clear the status indicated by the flags in the NRDYSTS register, write 0 only to the flags to be cleared; write 1 to the other flags.

40.2.17 BEMP Interrupt Status Register (BEMPSTS)

Address(es): USB0.BEMPSTS 000A 004Ah

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	PIPE9B EMP	PIPE8B EMP	PIPE7B EMP	PIPE6B EMP	PIPE5B EMP	PIPE4B EMP	PIPE3B EMP	PIPE2B EMP	PIPE1B EMP	PIPE0B EMP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	PIPE0BEMP	BEMP Interrupt Status Flag for Pipe 0	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b1	PIPE1BEMP	BEMP Interrupt Status Flag for Pipe 1	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b2	PIPE2BEMP	BEMP Interrupt Status Flag for Pipe 2	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b3	PIPE3BEMP	BEMP Interrupt Status Flag for Pipe 3	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b4	PIPE4BEMP	BEMP Interrupt Status Flag for Pipe 4	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b5	PIPE5BEMP	BEMP Interrupt Status Flag for Pipe 5	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b6	PIPE6BEMP	BEMP Interrupt Status Flag for Pipe 6	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b7	PIPE7BEMP	BEMP Interrupt Status Flag for Pipe 7	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b8	PIPE8BEMP	BEMP Interrupt Status Flag for Pipe 8	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b9	PIPE9BEMP	BEMP Interrupt Status Flag for Pipe 9	0: Interrupts are not generated. 1: Interrupts are generated.	R/W *1
b15 to b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. To clear the status indicated by the flags in the BEMPSTS register, write 0 only to the flags to be cleared; write 1 to the other flags.

40.2.18 Frame Number Register (FRMNUM)

Address(es): USB0.FRNUM 000A 004Ch

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	OVRN	CRCE	—	—	—						FRNM[10:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b10 to b0	FRNM[10:0]	Frame Number Flag	Latest frame number	R
b13 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b14	CRCE	Receive Data Error Flag	0: No error 1: An error occurred	R/W*1
b15	OVRN	Overflow/Underflow Detection Status Flag	0: No error 1: An error occurred	R/W*1

Note 1. To clear the status, write 0 only to the bits to be cleared; write 1 to the other bits.

FRNM[10:0] Flags (Frame Number Flag)

These bits indicate the latest frame number for the USB after the issuing of an SOF packet every 1 ms or writing to the FRNM[10:0] flags at the SOF packet reception.

CRCE Flag (Receive Data Error Flag)

Indicates whether a CRC error or bit stuffing error has been detected in the pipe during isochronous transfer.

The CRCE flag can be set to 0 by writing 0 to the CRCE flag by software. Here, 1 should be written to the other bits in FRMNUM.

On detecting a CRC error, the USB generates the internal NRDY interrupt request.

OVRN Flag (Overflow/Underflow Detection Status Flag)

Indicates whether an overflow/underflow error has been detected in the pipe during isochronous transfer.

The OVRN flag can be set to 0 by writing 0 to the OVRN flag by software. Here, 1 should be written to the other bits in FRMNUM.

(1) When the host controller is selected

The USB sets the OVRN flag to 1 on any of the following conditions.

- For the isochronous transfer pipe in the transmitting direction, the time to issue an OUT token comes before all the transmit data has been written to the FIFO buffer.
- For the isochronous transfer pipe in the receiving direction, the time to issue an IN token comes when no FIFO buffer planes are empty.

(2) When the function controller is selected

The USB sets the OVRN flag to 1 on any of the following conditions.

- For the isochronous transfer pipe in the transmitting direction, the IN token is received before all the transmit data has been written to the FIFO buffer.
- For the isochronous transfer pipe in the receiving direction, the OUT token is received when no FIFO buffer planes are empty.

40.2.19 Device State Change Register (DVCHGR)

Address(es): USB0.DVCHGR 000A 004Eh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	DVCHG	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b14 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15	DVCHG	Device State Change	0: Disables the writing to the USBADDR.STSRECOV[3:0] bits and USBADDR.USBADDR[6:0] bits. 1: Enables the writing to the USBADDR.STSRECOV[3:0] bits and USBADDR.USBADDR[6:0] bits.	R/W

For details, refer to section 40.3.1.5, Release from Deep Software Standby Mode Due to USB Suspend/Resume Interrupts.

40.2.20 USB Address Register (USBADDR)

Address(es): USB0.USBADDR 000A 0050h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	STSRECOV[3:0]				—	USBADDR[6:0]						
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Value after reset:

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b6 to b0	USBADDR[6:0]	USB Address	When the function controller is selected, these bits indicate the USB address assigned by the host when the SET_ADDRESS request is successfully processed.	R/W
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b11 to b8	STSRECOV[3:0]	Status Recovery	• Recovery when the function controller is selected b11b8 1 0 0 1: Return to the full-speed state (bits DVSTCTR0.RHST[2:0] = 010b), bits INTSTS0.DVSQ[2:0] = 001b (Default state) 1 0 1 0: Return to the full-speed state (bits DVSTCTR0.RHST[2:0] = 010b), bits INTSTS0.DVSQ[2:0] = 010b (Address state) 1 0 1 1: Return to the full-speed state (bits DVSTCTR0.RHST[2:0] = 010b), bits INTSTS0.DVSQ[2:0] = 011b (Configured state) Settings other than above are prohibited. • Recovery when the host controller is selected b11b8 1 0 0 0: Return to the full-speed state (bits DVSTCTR0.RHST[2:0] = 010b) Settings other than above are prohibited. <td>R/W</td>	R/W
b15 to b12	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

USBADDR[6:0] Bits (USB Address)

On detecting the USB bus reset, the USB sets the USBADDR[6:0] bits to 00h.

The writing to these bits is enabled while the DVCHGR.DVCHG bits are set to 1. On returning from the USB power shut-off, the operation can resume to the USB address before the shut-off by the software.

When the host controller is selected, the USBADDR[6:0] bits are invalid.

The USBADDR[6:0] bits are initialized when a USB bus reset is detected.

STSRECOV[3:0] Bits (Status Recovery)

These bits are used to resume the state of the internal sequencer on returning from the USB power shut-off. For details, refer to section 40.3.1.5, Release from Deep Software Standby Mode Due to USB Suspend/Resume Interrupts.

The writing to the STSRECOV[3:0] bits is enabled while the DVCHGR.DVCHG bit is set to 1.

40.2.21 USB Request Type Register (USBREQ)

Address(es): USB0.USBREQ 000A 0054h



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	BMREQUESTTYPE[7:0]	Request Type	These bits store the USB request bmRequestType value.	R/W *1
b15 to b8	BREQUEST[7:0]	Request	These bits store the USB request bRequest value.	R/W *1

Note 1. When the function controller is selected, these bits can only be read from, and writing to these bits is invalid. When the host controller is selected, these bits can be read from and written to.

The USBREQ register stores setup requests for control transfers.

When the function controller is selected, the values of bRequest and bmRequestType that have been received are stored.

When the host controller is selected, the values of bRequest and bmRequestType to be transmitted are set.

The USBREQ register is initialized by a USB bus reset.

BMREQUESTTYPE[7:0] Bits (Request Type)

These bits hold the value of the bmRequestType field of a USB request.

- When the host controller is selected:
Set these bits to the value of the USB request data in setup transactions for transmission. Do not modify the value of the BMREQUESTTYPE[7:0] bits while the DCPCTR.SUREQ bit is 1.
- When the function controller is selected:
These bits indicate the value of the USB request data in setup transactions for reception. Writing to the bits has no effect.

BREQUEST[7:0] Bits (Request)

These bits store bRequest value of the USB request.

- When the host controller is selected:
Set these bits to the value of the USB request data in setup transactions for transmission. Do not modify the value of the BREQUEST[7:0] bits while the DCPCTR.SUREQ bit is 1.
- When the function controller is selected:
These bits indicate the value of the USB request data in setup transactions for reception. Writing to the bits has no effect.

40.2.22 USB Request Value Register (USBVAL)

Address(es): USB0.USBVAL 000A 0056h



When the function controller is selected, the value of wValue that has been received is stored in the USBVAL register.

When the host controller is selected, the value of wValue to be transmitted is set.

The USBVAL register is initialized by a USB bus reset.

- When the host controller is selected:
Set these bits to the value of the wValue field in USB requests of setup transactions for transmission. Do not modify the value of the USBVAL register while the DCPCTR.SUREQ bit is 1.
- When the function controller is selected:
These bits indicate the value of the wValue field in USB requests received in setup transactions for reception. Writing to the USBVAL register has no effect.

40.2.23 USB Request Index Register (USBINDX)

Address(es): USB0.USBINDX 000A 0058h



The USBINDX register stores setup requests for control transfers.

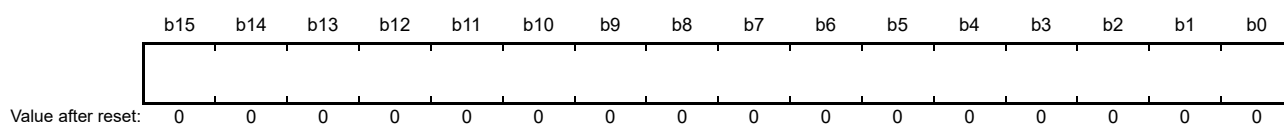
When the function controller is selected, the value of wIndex that has been received is stored. When the host controller is selected, the value of wIndex to be transmitted is set.

The USBINDX register is initialized by a USB bus reset.

- When the host controller is selected:
Set these bits to the value of the wIndex field in USB requests of setup transactions for transmission. Do not modify the value of the USBINDX register while the DCPCTR.SUREQ bit is 1.
- When the function controller is selected:
These bits indicate the value of the wIndex field in USB requests received in setup transactions for reception. Writing to the USBINDX register has no effect.

40.2.24 USB Request Length Register (USBLENG)

Address(es): USB0.USBLENG 000A 005Ah



The USBLENG register stores setup requests for control transfers.

When the function controller is selected, the value of wLength that has been received is stored. When the host controller is selected, the value of wLength to be transmitted is set.

The USBLENG register is initialized by a USB bus reset.

- When the host controller is selected:
Set these bits to the value of the wLength field in USB requests of setup transactions for transmission. Do not modify the value of the USBLENG register while the DCPCTR.SUREQ bit is 1.
- When the function controller is selected:
These bits indicate the value of the wLength field in USB requests received in setup transactions for reception. Writing to the USBLENG register has no effect.

40.2.25 DCP Configuration Register (DCPCFG)

Address(es): USB0.DCPCFG 000A 005Ch

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	SHTNA K	—	—	DIR	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	DIR	Transfer Direction*1	0: Data receiving direction 1: Data transmitting direction	R/W
b6, b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7	SHTNAK	Pipe Disabled at End of Transfer*1	0: Pipe continued at the end of transfer 1: Pipe disabled at the end of transfer	R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Modify this bit while DCPCTR.PID[1:0] bits are 00b (NAK). Before modifying this bit after modifying the DCPCTR.PID[1:0] bits for the DCP from 01b (BUF) to 00b (NAK), check that the DCPCTR.PBUSY flag is 0. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software.

DIR Bit (Transfer Direction)

When the host controller is selected, the DIR bit sets the transfer direction of the data stage and status stage.

When the function controller is selected, the DIR bit should be set to 0.

SHTNAK Bit (Pipe Disabled at End of Transfer)

The SHTNAK bit specifies whether to modify PID[1:0] bits to 00b (NAK) upon the end of transfer when the selected pipe is in the receiving direction.

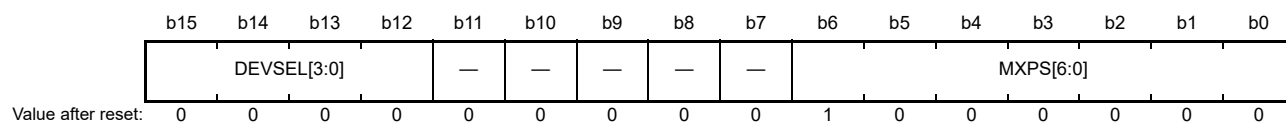
The SHTNAK bit is valid when the selected pipe is in the receiving direction.

When the SHTNAK bit is set to 1, the USB modifies the DCPCTR.PID[1:0] bits for the DCP to 00b (NAK) on determining the end of the transfer. The USB determines that the transfer has ended on the following condition.

- A short packet (including a zero-length packet) is successfully received.

40.2.26 DCP Maximum Packet Size Register (DCPMAXP)

Address(es): USB0.DCPMAXP 000A 005Eh



Bit	Symbol	Bit Name	Description	R/W
b6 to b0	MXPS[6:0]	Maximum Packet Size*1	These bits set the maximum amount of data (maximum packet size) in payloads for the DCP.	R/W
b11 to b7	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15 to b12	DEVSEL[3:0]	Device Select*2	b15 b12 0 0 0 0: Address 0000 0 0 0 1: Address 0001 0 0 1 0: Address 0010 0 0 1 1: Address 0011 0 1 0 0: Address 0100 0 1 0 1: Address 0101 Settings other than above are prohibited.	R/W

Note 1. Modify the MXPS[6:0] bits while PID[1:0] bits are 00b (NAK). Before modifying these bits after modifying the DCPCTR.PID[1:0] bits for the DCP from 01b (BUF) to 00b (NAK), check that the DCPCTR.PBUSY flag is 0. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software. After modifying the MXPS[6:0] bits and the DCP has been set to the CURPIPE[3:0] bits in a port select register, clear the buffer by setting the BCLR bit the port control register to 1.

Note 2. Modify the DEVSEL[3:0] bits while PID[1:0] bits are 00b (NAK) and the DCPCTR.SUREQ bit is 0. To modify these bits after modifying the DCPCTR.PID[1:0] bits for the DCP from 01b (BUF) to 00b (NAK), check that the DCPCTR.PBUSY flag is 0. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software.

MXPS[6:0] Bits (Maximum Packet Size)

The MXPS[6:0] bits specify the maximum amount of data (maximum packet size) in payloads for the DCP. The initial value of the bits is 40h (64 bytes).

Ensure that the setting of the MXPS[6:0] bits is in compliance with USB Specification 2.0.

Do not write to the FIFO buffer or set PID[1:0] = 01b (BUF) while the setting of the MXPS[6:0] bits is 0.

DEVSEL[3:0] Bits (Device Select)

When the host controller is selected, these bits specify the address of the peripheral device which is the communication target during control transfer.

The DEVSEL[3:0] bits should be set after setting the address to the DEVADDn (n = 0 to 5) register corresponding to the value to be set in the DEVSEL[3:0] bits. For example, before setting the DEVSEL[3:0] bits to 0010b, the address should be set to DEVADD2.

When the function controller is selected, the DEVSEL[3:0] bits should be set to 0000b.

40.2.27 DCP Control Register (DCPCTR)

Address(es): USB0.DCPCTR 000A 0060h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
BSTS	SUREQ	—	—	SUREQ CLR	—	—	SQCLR	SQSET	SQMON	PBUSY	—	—	CCPL	PID[1:0]	
0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b1, b0	PID[1:0]	Response PID	b1 b0 0 0: NAK response 0 1: BUF response (depending on the buffer state) 1 0: STALL response 1 1: STALL response	R/W
b2	CCPL	Control Transfer End Enable	0: Invalid 1: Completion of control transfer is enabled.	R/W
b4, b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b5	PBUSY	Pipe Busy Flag	0: DCP is not used for the transaction. 1: DCP is used for the transaction.	R
b6	SQMON	Sequence Toggle Bit Monitor Flag	0: DATA0 1: DATA1	R
b7	SQSET	Sequence Toggle Bit Set*2	0: Invalid 1: Specifies DATA1.	R/W*1
b8	SQCLR	Sequence Toggle Bit Clear*2	0: Invalid 1: Specifies DATA0.	R/W*1
b10, b9	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b11	SUREQCLR	SUREQ Bit Clear	0: Invalid 1: Clears the SUREQ bit to 0.	R/W
b13, b12	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b14	SUREQ	Setup Token Transmission	0: Invalid 1: Transmits the setup packet.	R/W
b15	BSTS	Buffer Status Flag	0: Buffer access is disabled. 1: Buffer access is enabled.	R

Note 1. Only 0 can be read.

Note 2. Write 1 to the SQSET and SQCLR bits while PID[1:0] bits are 00b (NAK). Before modifying these bits after modifying the PID[1:0] bits for the DCP from 01b (BUF) to 00b (NAK), check that the PBUSY flag is 0. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software.

PID[1:0] Bits (Response PID)

The PID[1:0] bits control the response type of the USB during control transfer.

(1) When the host controller is selected

Modify the setting of the PID[1:0] bits from 00b (NAK) to 01b (BUF) using the following procedure.

- When the transmitting direction is set
Write all the transmit data to the FIFO buffer while the DVSTCTR0.UACT bit is 1 and PID[1:0] bits are 00b (NAK), and then write 01b (BUF response). After PID[1:0] have been set to 01b (BUF), the USB executes the OUT transaction.
- When the receiving direction is set
Check that the FIFO buffer is empty (or empty the buffer) while the DVSTCTR0.UACT bit is 1 and PID[1:0] bits are 00b (NAK), and then set PID[1:0] bits to 01b (BUF). After PID[1:0] bits have been set to 01b (BUF), the USB executes the IN transaction.

The USB modifies the setting of the PID[1:0] bits as follows.

- The USB sets PID[1:0] bits to 11b (STALL) on receiving the data of a size exceeding the maximum packet size when the PID[1:0] bits has been set to 01b (BUF) by software.
- The USB sets PID[1:0] bits to 00b (NAK) on detecting a receive error, such as a CRC error, three consecutive times.
- The USB also sets PID[1:0] bits to 11b (STALL) on receiving the STALL handshake.

(2) When the function controller is selected

The USB modifies the setting of the PID[1:0] bits as follows.

- The USB modifies the PID[1:0] bits to 00b (NAK) on receiving the setup packet. Here, the USB sets the INTSTS0.VALID flag to 1. The setting of the PID[1:0] bits cannot be modified until the VALID flag is set to 0 by software.
- The USB sets PID[1:0] bits to 11b (STALL) on receiving the data of a size exceeding the maximum packet size when the PID[1:0] bits have been set to 01b (BUF) by software.
- The USB sets PID[1:0] bits to 1xb (STALL) on detecting the control transfer sequence error.
- The USB sets PID[1:0] bits to 00b (NAK) on detecting the USB bus reset.

The USB does not check the setting of the PID[1:0] bits while the SET_ADDRESS request is processed.

The PID[1:0] bits are initialized by a USB bus reset.

CCPL Bit (Control Transfer End Enable)

When the function controller is selected, setting the CCPL bit to 1 enables the status stage of the control transfer to be completed.

When the CCPL bit is set to 1 by software while the corresponding PID[1:0] bits are set to 01b (BUF), the USB completes the control transfer status stage.

During control read transfer, the USB transmits the ACK handshake in response to the OUT transaction from the USB host, and transmits the zero-length packet in response to the IN transaction from the USB host during control write or no-data control transfer. However, on detecting the SET_ADDRESS request, the USB operates in auto response mode from the setup stage up to the status stage completion irrespective of the setting of the CCPL bit.

The USB modifies the CCPL bit from 1 to 0 on receiving a new setup packet.

1 cannot be written to the CCPL bit by software while the INTSTS0.VALID flag is 1.

The CCPL bit is initialized by a USB bus reset.

When the host controller is selected, be sure to write 0 to the CCPL bit.

PBUSY Flag (Pipe Busy Flag)

The PBUSY flag indicates whether DCP is used or not for the transaction when USB changes the PID[1:0] bits from 01b (BUF) to 00b (NAK).

The USB modifies the PBUSY flag from 0 to 1 upon start of the USB transaction for the relevant pipe, and modifies the PBUSY flag from 1 to 0 upon completion of one transaction.

Reading the PBUSY flag after the PID[1:0] bits have been set to 00b (NAK) by software allows checking whether modification of the pipe settings is possible.

For details, refer to section 40.3.4.1, Pipe Control Register Switching Procedures.

SQMON Flag (Sequence Toggle Bit Monitor Flag)

The SQMON flag indicates the expected value of the sequence toggle bit for the next transaction during the DCP transfer.

The USB allows the SQMON flag to toggle upon normal completion of the transaction. However, the SQMON flag is not allowed to toggle when a data PID mismatch occurs during the transfer in the receiving direction.

When the function controller is selected, the USB sets the SQMON flag to 1 (specifies DATA1 as the expected value) upon successful reception of the setup packet.

When the function controller is selected, the USB does not reference the SQMON flag during the IN/OUT transaction of the status stage, and does not allow the SQMON flag to toggle upon normal completion.

SQSET Bit (Sequence Toggle Bit Set)

The SQSET bit specifies DATA1 as the expected value of the sequence toggle bit for the next transaction during the DCP transfer.

Do not set the SQCLR and SQSET bits to 1 simultaneously.

SQCLR Bit (Sequence Toggle Bit Clear)

The SQCLR bit specifies DATA0 as the expected value of the sequence toggle bit for the next transaction during the DCP transfer. The SQCLR bit indicates 0.

Do not set the SQCLR and SQSET bits to 1 simultaneously.

SUREQCLR Bit (SUREQ Bit Clear)

When the host controller is selected, setting the SUREQCLR bit to 1 clears the SUREQ bit to 0. The SUREQCLR bit indicates 0.

Set the SUREQCLR bit to 1 through software when communication has stopped with the SUREQ bit being 1 during the setup transaction. However, for normal setup transactions, the USB automatically clears the SUREQ bit to 0 upon completion of the transaction; therefore, clearing the SUREQ bit through software is not necessary.

Controlling the SUREQ bit through the SUREQCLR bit must be done while the DVSTCTR0.UACT bit is 0 and thus communication is halted or while no transfer is being performed with bus disconnection detected.

When the function controller is selected, be sure to write 0 to the SUREQCLR bit.

SUREQ Bit (Setup Token Transmission)

The USB transmits the setup packet by setting the SUREQ bit to 1 when the host controller is selected.

After completing the setup transaction process, the USB generates either the SACK or SIGN interrupt and sets the SUREQ bit to 0.

The USB also sets the SUREQ bit to 0 when software sets the SUREQCLR bit to 1.

Before setting the SUREQ bit to 1, set the DCPMAXP.DEVSEL[3:0] bits, registers USBREQ, USBVAL, USBINDX, and USBLENG appropriately to transmit the desired USB request in the setup transaction. Before setting this bit to 1, check that the PID[1:0] bits for the DCP are set to 00b (NAK). After setting the SUREQ bit to 1, do not modify the DCPMAXP.DEVSEL[3:0] bits, registers USBREQ, USBVAL, USBINDX, or USBLENG until the setup transaction is completed (the SUREQ bit = 1).

Write 1 to the SUREQ bit only when transmitting the setup token; for other purposes, write 0.

When the function controller is selected, be sure to write 0 to the SUREQ bit.

BSTS Flag (Buffer Status Flag)

Indicates whether DCP FIFO buffer access is enabled or disabled.

The meaning of the BSTS flag depends on the setting of ISEL bit in the port select register as shown below.

- When the ISEL bit = 0, the BSTS flag indicates whether the received data can be read from the buffer.
- When the ISEL bit = 1, the BSTS flag indicates whether the data to be transmitted can be written to the buffer.

40.2.28 Pipe Window Select Register (PIPESEL)

Address(es): USB0.PIPESEL 000A 0064h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	—	—	—	—	—	PIPESEL[3:0]			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	PIPESEL[3:0]	Pipe Window Select	b3 b0 0 0 0 0: No pipe selected 0 0 0 1: Pipe 1 0 0 1 0: Pipe 2 0 0 1 1: Pipe 3 0 1 0 0: Pipe 4 0 1 0 1: Pipe 5 0 1 1 0: Pipe 6 0 1 1 1: Pipe 7 1 0 0 0: Pipe 8 1 0 0 1: Pipe 9 Settings other than above are prohibited.	R/W
b15 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Pipes 1 to 9 should be set using registers PIPESEL, PIPECFG, PIPEMAXP, PIPEPERI, PIPEnCTR, PIPEnTRE, and PIPEnTRN.

After selecting the pipe using the PIPESEL register, functions of the pipe should be set using registers PIPECFG, PIPEMAXP, and PIPEPERI. PIPEnCTR, PIPEnTRE, and PIPEnTRN can be set regardless of the pipe selection in the PIPESEL register.

PIPESEL[3:0] Bits (Pipe Window Select)

The PIPESEL[3:0] bits select the pipe number corresponding to registers PIPECFG, PIPEMAXP, and PIPEPERI which data are written to or read from.

Selecting a pipe number through the PIPESEL[3:0] bits allows writing to and reading from registers PIPECFG, PIPEMAXP, and PIPEPERI which correspond to the selected pipe number.

When PIPESEL[3:0] = 0000b, 0 is read from all of the bits in registers PIPECFG, PIPEMAXP, and PIPEPERI. Writing to these bits is invalid.

40.2.29 Pipe Configuration Register (PIPECFG)

Address(es): USB0.PIPECFG 000A 0068h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	TYPE[1:0]	—	—	—	BFRE	DBLB	—	SHTNAK	—	—	DIR	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	EPNUM[3:0]	Endpoint Number*1	These bits specify the endpoint number for the selected pipe. Setting 0000b means an unused pipe.	R/W
b4	DIR	Transfer Direction*2,*3	0: Receiving direction 1: Transmitting direction	R/W
b6, b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7	SHTNAK	Pipe Disabled at End of Transfer*1	0: Pipe assignment continued at the end of transfer 1: Pipe assignment disabled at the end of transfer	R/W
b8	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b9	DBLB	Double Buffer Mode*2,*3	0: Single buffer 1: Double buffer	R/W
b10	BFRE	BRDY Interrupt Operation Specification*2,*3	0: BRDY interrupt upon transmitting or receiving data 1: BRDY interrupt upon completion of reading data	R/W
b13 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15, b14	TYPE[1:0]	Transfer Type*1	• Pipes 1 and 2 b15 b14 0 0: Pipe not used 0 1: Bulk transfer 1 0: Setting prohibited 1 1: Isochronous transfer • Pipes 3 to 5 b15 b14 0 0: Pipe not used 0 1: Bulk transfer 1 0: Setting prohibited 1 1: Setting prohibited • Pipes 6 to 9 b15 b14 0 0: Pipe not used 0 1: Setting prohibited 1 0: Interrupt transfer 1 1: Setting prohibited	R/W

Note 1. Modify the TYPE[1:0], SHTNAK, and EPNUM[3:0] bits while the PIPEnCTR.PID[1:0] bits are 00b (NAK). Before modifying these bits after modifying the PIPEnCTR.PID[1:0] bits for the selected pipe from 01b (BUF) to 00b (NAK), check that the PIPEnCTR.PBUSY flag is 0. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software.

Note 2. Modify the BFRE, DBLB, and DIR bits while the PIPEnCTR.PID[1:0] bits are 00b (NAK) and before the pipe is selected by the CURPIPE[3:0] bits in the port select register. Before modifying these bits after modifying the PIPEnCTR.PID[1:0] bits for the selected pipe from 01b (BUF) to 00b (NAK), check that the PIPEnCTR.PBUSY flag is 0. However, if the USB changes the PID[1:0] bits to 00b (NAK), checking the PIPEnCTR.PBUSY flag through software is not necessary.

Note 3. To modify the BFRE, DBLB, and DIR bits after completing USB communication using the selected pipe, write 1 and then 0 to the PIPEnCTR.ACLRM bit continuously through software to clear the FIFO buffer assigned to the selected pipe while the PID[1:0] bits and CURPIPE[3:0] bits are in the state described in the above note 2.

The PIPECFG register specifies the transfer type, buffer memory access direction, and endpoint numbers for pipes 1 to 9. It also selects single or double buffer mode, and whether to continue or disable pipe operation at the end of transfer.

EPNUM[3:0] Bits (Endpoint Number)

The EPNUM[3:0] bits specify the endpoint number for the selected pipe.

Setting 0000b means an unused pipe.

Do not make the settings such that the combination of the settings of the DIR and EPNUM[3:0] bits should be the same for two or more pipes (EPNUM[3:0] bits = 0000b can be set for all of the pipes).

DIR Bit (Transfer Direction)

The DIR bit specifies the transfer direction for the selected pipe.

When the DIR bit has been set to 0 by software, the USB uses the selected pipe in the receiving direction, and when software has set the DIR bit to 1, the USB uses the selected pipe in the transmitting direction.

SHTNAK Bit (Pipe Disabled at End of Transfer)

The SHTNAK bit specifies whether to modify the PID[1:0] bits to 00b (NAK) upon the end of transfer when the selected pipe is in the receiving direction.

The SHTNAK bit is valid when the selected pipe is pipe 1 to 5 in the receiving direction.

When the SHTNAK bit has been set to 1 by software for the selected pipe in the receiving direction, the USB modifies the PIPEnCTR.PID[1:0] bits corresponding to the selected pipe to 00b (NAK) on determining the end of the transfer. The USB determines that the transfer has ended on any of the following conditions.

- A short packet (including a zero-length packet) is successfully received.
- The transaction counter is used and the number of packets specified by the counter are successfully received.

DBLB Bit (Double Buffer Mode)

The DBLB bit selects either single or double buffer mode for the FIFO buffer used by the selected pipe.

The DBLB bit is valid when the selected pipe is pipe 1 to 5.

BFRE Bit (BRDY Interrupt Operation Specification)

The BFRE bit specifies the BRDY interrupt generation timing from the USB to the CPU with respect to the selected pipe.

When the BFRE bit has been set to 1 by software and the selected pipe is in the receiving direction, the USB detects the transfer completion and generates the BRDY interrupt on having read the relevant packet.

When the BRDY interrupt is generated with the above conditions, 1 should be written to the BCLR bit in the port control register by software. The FIFO buffer assigned to the selected pipe is not enabled for reception until 1 is written to the BCLR bit.

When the BFRE bit has been set to 1 by software and the selected pipe is in the transmitting direction, the USB does not generate the BRDY interrupt.

For details, refer to section 40.3.3.1, BRDY Interrupt.

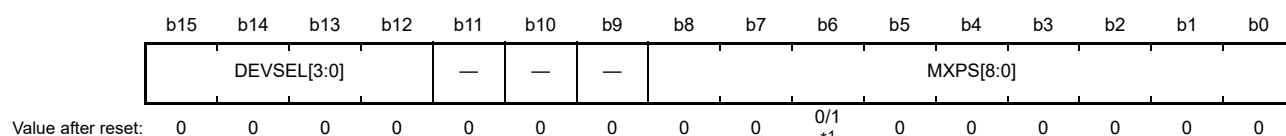
TYPE[1:0] Bits (Transfer Type)

The TYPE[1:0] bits select the transfer type for the pipe selected by the PIPESEL.PIPESEL[3:0] bits (selected pipe).

Before setting the PID[1:0] bits to 01b (BUF) for the selected pipe (before starting USB communication using the selected pipe), set the TYPE[1:0] bits to a value other than 00b.

40.2.30 Pipe Maximum Packet Size Register (PIPEMAXP)

Address(es): USB0.PIPEMAXP 000A 006Ch



Bit	Symbol	Bit Name	Description	R/W
b8 to b0	MXPS[8:0]	Maximum Packet Size*2	- Pipes 1 and 2: 1 byte (001h) to 256 bytes (100h) - Pipes 3 to 5: 8 bytes (008h), 16 bytes (010h), 32 bytes (020h), 64 bytes (040h) - Pipes 6 to 9: 1 byte (001h) to 64 bytes (040h)	R/W
b11 to b9	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15 to b12	DEVSEL[3:0]	Device Select*3	b3 b0 0 0 0 0: Address 0000 0 0 0 1: Address 0001 0 0 1 0: Address 0010 0 0 1 1: Address 0011 0 1 0 0: Address 0100 0 1 0 1: Address 0101 Settings other than above are prohibited. R/W	R/W

Note 1. The value of these bits is 0000h when no pipe is selected with the PIPESEL.PIPESEL[3:0] bits and 0040h when a pipe is selected.

Note 2. Modify the MXPS[8:0] bits while the PIPEnCTR.PID[1:0] bits are 00b (NAK) and before the pipe is selected by the CURPIPE[3:0] bits in the port select register. Before modifying these bits after modifying the PIPEnCTR.PID[1:0] bits for the selected pipe from 01b (BUF) to 00b (NAK), check that the PIPEnCTR.PBUSY flag is 0. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software.

Note 3. Modify the DEVSEL[3:0] bits while the PIPEnCTR.PID[1:0] bits are 00b (NAK). To modify these bits after modifying the PIPEnCTR.PID[1:0] bits for the selected pipe from 01b (BUF) to 00b (NAK), check that the PIPEnCTR.PBUSY flag is 0. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software.

The PIPEMAXP register specifies the maximum packet size for pipes 1 to 9.

MXPS[8:0] Bits (Maximum Packet Size)

The MXPS[8:0] bits specify the maximum data payload (maximum packet size) for the selected pipe.

These bits should be set to the appropriate value for each transfer type based on USB Specification 2.0. Note that the maximum value of pipes 1 and 2 is 256. While MXPS[8:0] = 000h, do not write to the FIFO buffer or do not set the PID[1:0] bits to 01b (BUF).

DEVSEL[3:0] Bits (Device Select)

When the host controller is selected, these bits specify the USB device address of the peripheral device which is the communication target.

The DEVSEL[3:0] bits should be set after setting the address to the DEVADDn (n = 0 to 5) register corresponding to the value to be set in the DEVSEL[3:0] bits. For example, before setting the DEVSEL[3:0] bits to 0010b, the address should be set to DEVADD2.

When the function controller is selected, the DEVSEL[3:0] bits should be set to 0000b.

40.2.31 Pipe Cycle Control Register (PIPEPERI)

Address(es): USB0.PIPEPERI 000A 006Eh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	IFIS	—	—	—	—	—	—	—	—	—	IITV[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	IITV[2:0]	Interval Error Detection Interval *1	Specify the interval error detection timing for the selected pipe in terms of frames, which is expressed as nth power of 2.	R/W
b11 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b12	IFIS	Isochronous IN Buffer Flush	0: The buffer is not flushed. 1: The buffer is flushed.	R/W
b15 to b13	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Modify the IITV[2:0] bits while the PIPEnCTR.PID[1:0] bits are 00b (NAK). To modify these bits after modifying the PIPEnCTR.PID[1:0] bits for the selected pipe from 01b (BUF) to 00b (NAK), check that the PIPEnCTR.PBUSY flag is 0. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software.

The PIPEPERI register selects whether the buffer is flushed or not when an interval error occurred during isochronous IN transfer, and sets the interval error detection interval for pipes 1 to 9.

IITV[2:0] Bits (Interval Error Detection Interval)

Before modifying the IITV[2:0] bits after USB communication has been completed with the IITV[2:0] bits set to a certain value, set the PID[1:0] bits to 00b (NAK) and then set the PIPEnCTR.ACLRM bit to 1 to initialize the interval timer.

The IITV[2:0] bits are invalid for pipes 3 to 5; set the IITV[2:0] bits to 000b for pipes 3 to 5.

IFIS Bit (Isochronous IN Buffer Flush)

Specifies whether to flush the buffer when the pipe selected by the PIPESEL.PIPESEL[3:0] bits (selected pipe) is used for isochronous IN transfers.

When the function controller is selected and the selected pipe is for isochronous IN transfers, the USB automatically clears the FIFO buffer when the USB fails to receive the IN token from the USB host within the interval set by the IITV[2:0] bits in terms of frames.

In double buffer mode (the PIPECFG.DBLB bit = 1), the USB only clears the data in the plane used earlier.

The USB clears the FIFO buffer on receiving the SOF packet immediately after the frame in which the USB has expected to receive the IN token. Even if the SOF packet is damaged, the USB also clears the FIFO buffer at the right timing to receive the SOF packet by using the internal interpolation function.

When the host controller is selected, set the IITV[2:0] bits to 000b.

When the selected pipe is not for isochronous transfer, set the IITV[2:0] bits to 000b.

40.2.32 Pipe n Control Registers (PIPEnCTR) (n = 1 to 9)

- PIPEnCTR (n = 1 to 5)

Address(es): USB0.PIPE1CTR 000A 0070h, USB0.PIPE2CTR 000A 0072h, USB0.PIPE3CTR 000A 0074h, USB0.PIPE4CTR 000A 0076h, USB0.PIPE5CTR 000A 0078h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	BSTS	INBUFM	—	—	—	ATREPM	ACLRM	SQCLR	SQSET	SQMON	PBUSY	—	—	—	PID[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	PID[1:0]	Response PID	b1 b0 0 0: NAK response 0 1: BUF response (depending on the buffer state) 1 0: STALL response 1 1: STALL response	R/W
b4 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b5	PBUSY	Pipe Busy Flag	0: The relevant pipe is not used for the transaction. 1: The relevant pipe is used for the transaction.	R
b6	SQMON	Sequence Toggle Bit Confirmation	0: DATA0 1: DATA1	R
b7	SQSET	Sequence Toggle Bit Set* ²	0: Write disabled 1: Specifies DATA1.	R/W* ¹
b8	SQCLR	Sequence Toggle Bit Clear* ²	0: Write disabled 1: Specifies DATA0.	R/W* ¹
b9	ACLRM	Auto Buffer Clear Mode* ³	0: Disabled 1: Enabled (all buffers are initialized)	R/W
b10	ATREPM	Auto Response Mode* ²	0: Auto response is disabled. 1: Auto response is enabled.	R/W
b13 to b11	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b14	INBUFM	Transmit Buffer Monitor Flag	0: There is no data to be transmitted in the buffer memory. 1: There is data to be transmitted in the buffer memory.	R
b15	BSTS	Buffer Status Flag	0: Buffer access by the CPU is disabled. 1: Buffer access by the CPU is enabled.	R

Note 1. Only 0 can be read.

Note 2. Modify the ATREPM bit or write 1 to the SQCLR or SQSET bit while the PID[1:0] bits are 00b (NAK). Before modifying these bits after modifying the PID[1:0] bits for the selected pipe from 01b (BUF) to 00b (NAK), check that the PBUSY flag is 0. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software.

Note 3. Modify the ACLRM bit while PID[1:0] bits are 00b (NAK) and before the pipe is selected by the CURPIPE[3:0] bits in the port select register. Before modifying this bit after modifying the PID[1:0] bits for the selected pipe from 01b (BUF) to 00b (NAK), check that the PBUSY flag is 0. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software.

The PIPEnCTR register can be set regardless of the pipe selection in the PIPESEL register.

PID[1:0] Bits (Response PID)

The PID[1:0] bits specify the response type for the next transaction of the relevant pipe.

The default setting of the PID[1:0] bits are 00b (NAK). Modify the setting of the PID[1:0] bits to 01b (BUF) to use the relevant pipe for USB transfer. Table 40.6 and Table 40.7 show the basic operation (operation when there are no errors in the transmitted and received packets) of the USB depending on the PID[1:0] bit setting.

After modifying the setting of the PID[1:0] bits through software from 01b (BUF) to 00b (NAK) during USB communication using the relevant pipe, check that the PBUSY flag is 1 to see if USB transfer using the relevant pipe has actually entered the NAK state. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software.

The USB modifies the setting of the PID[1:0] bits in the following cases.

- The USB sets the PID[1:0] bits to 00b (NAK) on recognizing the completion of the transfer when the relevant pipe is in the receiving direction and the PIPECFG.SHTNAK bit for the selected pipe has been set to 1 by software.
- The USB sets the PID[1:0] bits to 11b (STALL) on receiving a data packet with a payload exceeding the maximum packet size of the relevant pipe.
- The USB sets the PID[1:0] bits to 00b (NAK) on detecting a USB bus reset when the function controller is selected.
- The USB sets the PID[1:0] bits to 00b (NAK) on detecting a receive error, such as a CRC error, three consecutive times when the host controller is selected.
- The USB sets the PID[1:0] bits to 11b (STALL) on receiving the STALL handshake when the host controller is selected.

To specify each response type, set the PID[1:0] bits as follows.

- To make a transition from NAK (00b) to STALL, set 10b.
- To make a transition from BUF (01b) to STALL, set 11b.
- To make a transition from STALL (11b) to NAK, set 10b and then 00b.
- To make a transition from STALL (11b) to BUF, set 10b, 00b, and then 01b.
- To make a transition from STALL (10b) to BUF, set 00b and then 01b.

PBUSY Flag (Pipe Busy Flag)

The PBUSY flag indicates whether the relevant pipe is being currently used or not for the transaction.

The USB modifies the PBUSY flag from 0 to 1 upon start of the USB transaction for the relevant pipe, and modifies the PBUSY flag from 1 to 0 upon completion of one transaction.

Reading the PBUSY flag after the PID[1:0] bits have been set to 00b (NAK) by software allows checking whether modification of the pipe settings is possible.

For details, refer to section 40.3.4.1, Pipe Control Register Switching Procedures.

SQMON Bit (Sequence Toggle Bit Confirmation)

The SQMON flag indicates the expected value of the sequence toggle bit for the next transaction of the relevant pipe.

When the relevant pipe is not for the isochronous transfer, the USB allows the SQMON flag to toggle upon normal completion of the transaction. However, the SQMON flag is not allowed to toggle when a data PID mismatch occurs during the transfer in the receiving direction.

SQSET Bit (Sequence Toggle Bit Set)

The SQSET bit should be set to 1 to set DATA1 as the expected value of the sequence toggle bit for the next transaction of the relevant pipe.

Setting the SQSET bit to 1 through software allows the USB to set DATA1 as the expected value of the sequence toggle bit of the relevant pipe. The USB sets the SQSET bit to 0.

SQCLR Bit (Sequence Toggle Bit Clear)

The SQCLR bit should be set to 1 to clear the expected value (to set DATA0 as the expected value) of the sequence toggle bit for the next transaction of the relevant pipe.

Setting the SQCLR bit to 1 through software allows the USB to set DATA0 as the expected value of the sequence toggle bit of the relevant pipe. The USB sets the SQCLR bit to 0.

ACLRM Bit (Auto Buffer Clear Mode)

Enables or disables auto buffer clear mode for the relevant pipe.

To delete the information in the FIFO buffer assigned to the relevant pipe completely, write 1 and then 0 to the ACLRM bit continuously.

Table 40.8 shows the information cleared by writing 1 and 0 to the ACLRM bit continuously and the cases in which clearing the information is necessary.

ATREPM Bit (Auto Response Mode)

Enables or disables auto response mode for the relevant pipe.

When the function controller is selected and the relevant pipe is for bulk transfer, the ATREPM bit can be set to 1.

When the ATREPM bit is set to 1, the USB responds to the token from the USB host as described below.

(1) When the relevant pipe is for bulk IN transfer (the PIPECFG.TYPE[1:0] bits = 01b and the PIPECFG.DIR bit = 1)
When the ATREPM bit = 1 and PID[1:0] = 01b (BUF), the USB transmits a zero-length packet in response to the IN token.

The USB updates (allows toggling of) the sequence toggle bit (data PID) each time the USB receives ACK from the USB host (in a single transaction, IN token is received, zero-length packet is transmitted, and then ACK is received.).

In this case, the USB does not generate the BRDY or BEMP interrupt.

(2) When the relevant pipe is for bulk OUT transfer (the PIPECFG.TYPE[1:0] bits = 01b and the PIPECFG.DIR bit = 0)
When the ATREPM bit = 1 and PID[1:0] = 01b (BUF), the USB returns NAK in response to the OUT token and generates the NRDY interrupt.

For USB communication in auto response mode, set the ATREPM bit to 1 while the FIFO buffer is empty. Do not write to the FIFO buffer during USB communication in auto response mode.

When the relevant pipe is for isochronous transfer, be sure to set the ATREPM bit to 0.

When the host controller is selected, be sure to set the ATREPM bit to 0.

INBUFM Flag (Transmit Buffer Monitor Flag)

Indicates the relevant FIFO buffer status when the relevant pipe is in the transmitting direction.

When the relevant pipe is in the transmitting direction (the PIPECFG.DIR bit = 1), the USB sets the INBUFM flag to 1 when the CPU or DMAC/DTC completes writing data to at least one FIFO buffer plane.

The USB sets the INBUFM flag to 0 when the USB completes transmitting the data from the FIFO buffer plane to which all the data has been written. In double buffer mode (the PIPECFG.DBLB bit = 1), the USB sets the INBUFM flag to 0 when the USB completes transmitting the data from the two FIFO buffer planes before the CPU or DMAC/DTC completes writing data to one FIFO buffer plane.

The INBUFM flag indicates the same value as the BSTS flag when the relevant pipe is in the receiving direction (the PIPECFG.DIR bit = 0).

BSTS Flag (Buffer Status Flag)

Indicates the FIFO buffer status for the relevant pipe.

The meaning of the BSTS flag depends on the settings of the PIPECFG.DIR bit, PIPECFG.BFRE bit, and DnFIFOSEL.DCLRM bits as shown in Table 40.9.

Table 40.6 Operation of USB depending on PID[1:0] Bits Setting (When Host Controller is Selected)

PID[1:0] Bits	Transfer Type	Transfer Direction (DIR Bit)	Operation of USB
00b (NAK)	Operation does not depend on the setting.	Operation does not depend on the setting.	Does not issue tokens.
01b (BUF)	Bulk or interrupt	Operation does not depend on the setting.	Issues tokens while the DVSTCTR0.UACT bit is 1 and the FIFO buffer corresponding to the relevant pipe is ready for transmission and reception. Does not issue tokens while the DVSTCTR0.UACT bit is 0 or the FIFO buffer corresponding to the relevant pipe is not ready for transmission or reception.
	Isochronous	Operation does not depend on the setting.	Issues tokens irrespective of the status of the FIFO buffer corresponding to the relevant pipe.
10b (STALL) or 11b (STALL)	Operation does not depend on the setting.	Operation does not depend on the setting.	Does not issue tokens.

Table 40.7 Operation of USB depending on PID[1:0] Bits Setting (When Function Controller is Selected)

PID[1:0] Bits	Transfer Type	Transfer Direction (DIR Bit)	Operation of USB
00b (NAK)	Bulk or interrupt	Operation does not depend on the setting.	Returns NAK in response to the token from the USB host.
	Isochronous	Operation does not depend on the setting.	Returns nothing in response to the token from the USB host.
01b (BUF)	Bulk	Receiving direction (DIR bit = 0)	Receives data and returns ACK in response to the OUT token from the USB host if the FIFO buffer corresponding to the relevant pipe is ready for reception.
	Interrupt	Receiving direction (DIR bit = 0)	Receives data and returns ACK in response to the OUT token from the USB host if the FIFO buffer corresponding to the relevant pipe is ready for reception.
	Bulk or interrupt	Transmitting direction (DIR bit = 1)	Transmits data in response to the token from the USB host if the corresponding FIFO buffer is ready for transmission. Returns NAK if not ready.
	Isochronous	Receiving direction (DIR bit = 0)	Receives data in response to the OUT token from the USB host if the FIFO buffer corresponding to the relevant pipe is ready for reception. Discards data if not ready.
	Isochronous	Transmitting direction (DIR bit = 1)	Transmits data in response to the token from the USB host if the corresponding FIFO buffer is ready for transmission. Transmits the zero-length packet if not ready.
10b (STALL) or 11b (STALL)	Bulk or interrupt	Operation does not depend on the setting.	Returns STALL in response to the token from the USB host.
	Isochronous	Operation does not depend on the setting.	Returns nothing in response to the token from the USB host.

Table 40.8 Information Cleared by USB by Setting ACLRM = 1

No.	Information Cleared by ACLRM Bit Manipulation	Cases in which Clearing Information is Necessary
1	All the information in the FIFO buffer assigned to the relevant pipe (both FIFO buffer planes are cleared when double buffer mode is selected)	When the pipe is to be initialized
2	The interval count value when the relevant pipe is for isochronous transfer	When the interval count value is to be reset
3	Internal flags concerning the PIPECFG.BFRE bit	When the PIPECFG.BFRE setting is modified
4	FIFO buffer toggle control	When the PIPECFG.DBLB setting is modified
5	Internal flags concerning the transaction count	When the transaction count function is forcibly terminated

Table 40.9 Operation of BSTS Flag

DIR Bit	BFRE Bit	DCLRM Bit	BSTS Flag Function
0	0	0	The received data can be read from the FIFO buffer. The received data has been completely read from the FIFO buffer.
		1	Setting prohibited
	1	0	The received data can be read from the FIFO buffer. The BCLR bit in the port control register has been set to 1 by software after the received data has been completely read from the FIFO buffer.
		1	The received data can be read from the FIFO buffer. The received data has been completely read from the FIFO buffer.
		0	The transmit data can be written to the FIFO buffer. The transmit data has been completely written to the FIFO buffer.
		1	Setting prohibited
1	0	0	The transmit data can be written to the FIFO buffer. The transmit data has been completely written to the FIFO buffer.
		1	Setting prohibited
	1	0	Setting prohibited
		1	Setting prohibited

- PIPEnCTR (n = 6 to 9)

Address(es): USB0.PIPE6CTR 000A 007Ah, USB0.PIPE7CTR 000A 007Ch, USB0.PIPE8CTR 000A 007Eh, USB0.PIPE9CTR 000A 0080h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	BSTS	—	—	—	—	—	ACLRM	SQCLR	SQSET	SQMON	PBUSY	—	—	—	PID[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b1, b0	PID[1:0]	Response PID	b1 b0 0 0: NAK response 0 1: BUF response (depending on the buffer state) 1 0: STALL response 1 1: STALL response	R/W
b4 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b5	PBUSY	Pipe Busy Flag	0: The relevant pipe is not used at the USB bus. 1: The relevant pipe is used at the USB bus.	R
b6	SQMON	Sequence Toggle Bit Confirmation Flag	0: DATA0 1: DATA1	R
b7	SQSET	Sequence Toggle Bit Set*2	0: Invalid 1: Specifies DATA1.	R/W *1
b8	SQCLR	Sequence Toggle Bit Clear*2	0: Invalid 1: Specifies DATA0.	R/W *1
b9	ACLRM	Auto Buffer Clear Mode*2,*3	0: Auto buffer clear mode is disabled. 1: Auto buffer clear mode is enabled (all buffers are initialized)	R/W
b14 to b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15	BSTS	Buffer Status Flag	0: Buffer access is disabled. 1: Buffer access is enabled.	R

Note 1. Only 0 can be read. Only 1 can be written.

Note 2. Write 1 to the SQCLR or SQSET bit while the PID[1:0] bits are 00b (NAK). Before modifying these bits after modifying the PID[1:0] bits for the selected pipe from 01b (BUF) to 00b (NAK), check that the PBUSY flag is 0. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software.

Note 3. Modify the ACLRM bit while the PID[1:0] bits are 00b (NAK) and before the pipe is selected by the CURPIPE[3:0] bits in the port select register. Before modifying this bit after modifying the PID[1:0] bits for the selected pipe from 01b (BUF) to 00b (NAK), check that the PBUSY flag is 0. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software.

PID[1:0] Bits (Response PID)

The PID[1:0] bits specify the response type for the next transaction of the relevant pipe.

The default setting of the PID[1:0] bits are 00b (NAK). Modify the setting of the PID[1:0] bits to 01b (BUF) to use the relevant pipe for USB transfer. Table 40.6 and Table 40.7 show the basic operation (operation when there are no errors in the transmitted and received packets) of the USB depending on the setting of the PID[1:0] bits.

After modifying the setting of the PID[1:0] bits through software from 01b (BUF) to 00b (NAK) during USB communication using the relevant pipe, check that the PBUSY flag is 1 to see if USB transfer using the relevant pipe has actually entered the NAK state. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software.

The USB modifies the setting of the PID[1:0] bits in the following cases.

- The USB sets the PID[1:0] bits to 11b (STALL) on receiving a data packet with a payload exceeding the maximum packet size of the relevant pipe.
- The USB sets the PID[1:0] bits to 00b (NAK) on detecting a USB bus reset when the function controller is selected.
- The USB sets the PID[1:0] bits to 00b (NAK) on detecting a receive error, such as a CRC error, three consecutive times when the host controller is selected.
- The USB sets the PID[1:0] bits to 11b (STALL) on receiving the STALL handshake when the host controller is selected.

To specify each response type, set the PID[1:0] bits as follows.

- To make a transition from NAK (00b) to STALL, set 10b.
- To make a transition from BUF (01b) to STALL, set 11b.
- To make a transition from STALL (11b) to NAK, set 10b and then 00b.
- To make a transition from STALL (11b) to BUF, set 10b, 00b, and then 01b.
- To make a transition from STALL (10b) to BUF, set 00b and then 01b.

PBUSY Flag (Pipe Busy Flag)

The PBUSY flag indicates whether the relevant pipe is being currently used or not for the transaction.

The USB modifies the PBUSY flag from 0 to 1 upon start of the USB transaction for the relevant pipe, and modifies the PBUSY flag from 1 to 0 upon completion of one transaction.

Reading the PBUSY flag after the PID[1:0] bits has been set to 00b (NAK) by software allows checking whether modification of the pipe settings is possible.

SQMON Flag (Sequence Toggle Bit Confirmation Flag)

The SQMON flag indicates the expected value of the sequence toggle bit for the next transaction of the relevant pipe.

The USB allows the SQMON flag to toggle upon normal completion of the transaction. However, the SQMON flag is not allowed to toggle when a data PID mismatch occurs during the transfer in the receiving direction.

SQSET Bit (Sequence Toggle Bit Set)

The SQSET bit should be set to 1 to set DATA1 as the expected value of the sequence toggle bit for the next transaction of the relevant pipe.

Setting the SQSET bit through software allows the USB to set DATA1 as the expected value of the sequence toggle bit of the relevant pipe. The USB sets the SQSET bit to 0.

SQCLR Bit (Sequence Toggle Bit Clear)

The SQCLR bit should be set to 1 to clear the expected value (to set DATA0 as the expected value) of the sequence toggle bit for the next transaction of the relevant pipe.

Setting the SQCLR bit to 1 through software allows the USB to set DATA0 as the expected value of the sequence toggle bit of the relevant pipe. The USB sets the SQCLR bit to 0.

ACLRM Bit (Auto Buffer Clear Mode)

Enables or disables auto buffer clear mode for the relevant pipe.

To delete the information in the FIFO buffer assigned to the relevant pipe completely, write 1 and then 0 to the ACLRM bit continuously.

Table 40.10 shows the information cleared by writing 1 and 0 to the ACLRM bit continuously and the cases in which clearing the information is necessary.

BSTS Flag (Buffer Status Flag)

Indicates the FIFO buffer status for the relevant pipe.

The meaning of the BSTS flag depends on the settings of the PIPECFG.DIR bit, PIPECFG.BFRE bit, and DnFIFOSEL.DCLRM bits as shown in Table 40.9.

Table 40.10 Information Cleared by USB by Setting the ACLRM Bit = 1

No.	Information Cleared by ACLRM Bit Manipulation	Cases in which Clearing Information is Necessary
1	All the information in the FIFO buffer assigned to the selected pipe	When the pipe is to be initialized
2	The interval count value when the selected pipe is for interrupt transfer and the host controller is selected	When the interval count value is to be reset
3	Internal flags concerning the PIPECFG.BFRE bit	When the PIPECFG.BFRE setting is modified
4	Internal flags concerning the transaction count	When the transaction count function is forcibly terminated

40.2.33 Pipe n Transaction Counter Enable Register (PIPEnTRE) (n = 1 to 5)

Address(es): USB0.PIPE1TRE 000A 0090h, USB0.PIPE2TRE 000A 0094h, USB0.PIPE3TRE 000A 0098h, USB0.PIPE4TRE 000A 009Ch, USB0.PIPE5TRE 000A 00A0h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	TRENB	TRCLR	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b8	TRCLR	Transaction Counter Clear	0: Invalid 1: The current counter value is cleared.	R/W
b9	TRENB	Transaction Counter Enable	0: Transaction counter is disabled. 1: Transaction counter is enabled.	R/W
b15 to b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note: Modify each bit in the PIPEnTRE register while the PID[1:0] bits are 00b (NAK). Before modifying these bits after modifying the PIPEnCTR.PID[1:0] bits for the selected pipe from 01b (BUF) to 00b (NAK), check that the PIPEnCTR.PBUSY flag is 0. However, if the USB changes the PID[1:0] bits to 00b (NAK), the PBUSY flag does not need to be checked by software.

TRCLR Bit (Transaction Counter Clear)

Clears the current value of the transaction counter corresponding to the relevant pipe and then sets the TRCLR bit to 0.

TRENB Bit (Transaction Counter Enable)

Enables or disables the transaction counter.

For the pipe in the receiving direction, setting the TRENB bit to 1 after setting the total number of the packets to be received in the PIPEnTRN register through software allows the USB to control hardware as described below on having received the number of packets equal to the setting of the PIPEnTRN register.

- While the PIPECFG.SHTNAK bit is 1, the USB modifies the PID[1:0] bits to 00b (NAK) for the corresponding pipe on having received the number of packets equal to the setting of the PIPEnTRN register.
- While the PIPECFG.BFRE bit is 1, the USB asserts the BRDY interrupt on having received the number of packets equal to the setting of the PIPEnTRN register and then reading the last received data.

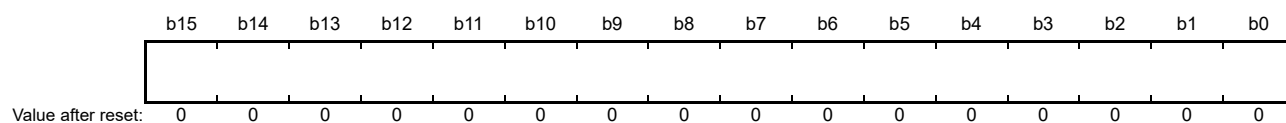
For the pipe in the transmitting direction, set the TRENB bit to 0.

When the transaction counter is not used, set the TRENB bit to 0.

When the transaction counter is used, set the PIPEnTRN register before setting the TRENB bit to 1. Set the TRENB bit to 1 before receiving the first packet to be counted by the transaction counter.

40.2.34 Pipe n Transaction Counter Register (PIPE_nTRN) (n = 1 to 5)

Address(es): USB0.PIPE1TRN 000A 0092h, USB0.PIPE2TRN 000A 0096h, USB0.PIPE3TRN 000A 009Ah, USB0.PIPE4TRN 000A 009Eh, USB0.PIPE5TRN 000A 00A2h



The PIPE_nTRN register retains the setting by a USB bus reset.

- When written to:
This register specifies the total of packets (number of transactions) to be received in corresponding pipe.
- When read from:
This register indicates the specified number of transactions if the PIPE_nTRE.TRENB bit is 0.
This register indicates the number of currently counted transactions if the PIPE_nTRE.TRENB bit is 1.

The USB increments the value of the PIPE_nTRN register by one when all of the following conditions are satisfied on receiving the packet.

- The PIPE_nTRE.TRENB bit = 1
- (PIPE_nTRN set value $\neq$ current counter value + 1) on receiving the packet.
- The payload of the received packet agrees with the setting of the PIPEMAXP.MXPS[8:0] bits.

The USB sets the value of the PIPE_nTRN register to 0000h when any of the following conditions are satisfied.

(1) All of the following conditions are satisfied.

- The PIPE_nTRE.TRENB bit = 1
- (PIPE_nTRN set value = current counter value + 1) on receiving the packet.
- The payload of the received packet agrees with the setting of the PIPEMAXP.MXPS[8:0] bits.

(2) All of the following conditions are satisfied.

- The PIPE_nTRE.TRENB bit = 1
- The USB has received a short packet.

(3) All of the following conditions are satisfied.

- The PIPE_nTRE.TRENB bit = 1
- The PIPE_nTRE.TRCLR bit has been set to 1 by software.

For the pipe in the transmitting direction, set the PIPE_nTRN register to 0000h.

When the transaction counter is not used, set the PIPE_nTRN register to 0000h.

Setting the number of transactions to be transferred to the PIPE_nTRN register is only enabled when the PIPE_nTRE.TRENB bit is 0. To modify the number of transactions to be transferred, set the TRCLR bit to 1 (to clear the current counter value) before setting the PIPE_nTRE.TRENB bit to 1.

40.2.35 Device Address n Configuration Register (DEVADDn) (n = 0 to 5)

Address(es): USB0.DEVADD0 000A 00D0h, USB0.DEVADD1 000A 00D2h, USB0.DEVADD2 000A 00D4h, USB0.DEVADD3 000A 00D6h, USB0.DEVADD4 000A 00D8h, USB0.DEVADD5 000A 00DAh

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	USBSPD[1:0]	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b5 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7, b6	USBSPD[1:0]	Transfer Speed of Communication Target Device	b7 b6 0 0: DEVADDn is not used 0 1: Low-speed 1 0: Full-speed 1 1: Setting prohibited	R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The DEVADDn register specifies the transfer speed of the peripheral device which is the communication target for pipes 0 to 9.

When the host controller is selected, the bits in the DEVADDn register should be set before starting communication using each pipe.

The bits in the DEVADDn register should be modified while no valid pipes are using the settings of the bits. Valid pipes refer to the pipes satisfying both of the following conditions:

- The DEVADDn register is selected by the DEVSEL[3:0] bits.
- The PID[1:0] bits are set to 01b (BUF) for the selected pipe or the selected pipe is the DCP with the DCPCTR.SUREQ bit set to 1.

USBSPD[1:0] Bits (Transfer Speed of Communication Target Device)

The USBSPD[1:0] bits specify the USB transfer speed of the communication target peripheral device.

Set these bits to 10b when a full-speed device is connected via the HUB.

When the host controller is selected, the USB refers to the setting of the USBSPD[1:0] bits to generate packets.

When the function controller is selected, set these bits to 00b.

40.2.36 PHY Cross Point Adjustment Register (PHYSLEW)

Address(es): USB0.PHYSLEW 000A 00F0h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	SLEWFO1	SLEWFO0	SLEWRO1	SLEWRO0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0

Bit	Symbol	Bit Name	Description	R/W
b0	SLEWRO0	Driver Cross Point Adjustment 00	Set this bit to 1.	R/W
b1	SLEWRO1	Driver Cross Point Adjustment 01	Set this bit to 0.	R/W
b2	SLEWFO0	Driver Cross Point Adjustment 00	Set this bit to 1.	R/W
b3	SLEWFO1	Driver Cross Point Adjustment 01	Set this bit to 0.	R/W
b31 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The PHYSLEW register adjusts the cross point of the driver.

Set the value (00000005h) to the register before starting the USB operation.

40.2.37 Deep Standby USB Transceiver Control/Pin Monitoring Register (DPUSR0R)

Address(es): USB.DPUSR0R 000A 0400h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	DVBST S0	—	DOVCB 0	DOVCA 0	—	—	DM0	DP0
Value after reset:	0	0	0	0	0	0	0	0	x	0	x	x	0	0	x	x

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	FIXPH Y0	DRPD0	—	RPUE0	SRPC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

x: Undefined

Bit	Symbol	Bit Name	Description	R/W
b0	SRPC0	USB0 Single End Receiver Control	0: Input through the D+ and D– inputs is disabled. 1: Input through the D+ and D– inputs is enabled.	R/W
b1	RPUE0*1	USB0 D+ Pull-Up Resistor Control	0: Disables D+ pull-up resistor. 1: Enables D+ pull-up resistor.	R/W
b2	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b3	DRPD0*1	USB0 D+/D– Pull-Down Resistor Control	0: Disables D+/D– pull-down resistor. 1: Enables D+/D– pull-down resistor.	R/W
b4	FIXPHY0	USB0 Transceiver Output Fix	0: The outputs are fixed in normal mode and on return from deep software standby mode. 1: The outputs are fixed on transitions to deep software standby mode.	R/W
b15 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b16	DP0	USB0 D+ Input Flag	Indicates the D+ input signal of the USB0.	R
b17	DM0	USB0 D– Input Flag	Indicates the D– input signal of the USB0.	R
b19, b18	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b20	DOVCA0	USB0 OVRCURA Input Flag	Indicates the OVRCURA input signal of the USB0.	R
b21	DOVCB0	USB0 OVRCURB Input Flag	Indicates the OVRCURB input signal of the USB0.	R
b22	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b23	DVBSTS0	USB0 VBUS Input Flag	Indicates the VBUS input signal of the USB0.	R
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Use this bit during operation in deep standby mode. For details, refer to section 40.3.1.5, Release from Deep Software Standby Mode Due to USB Suspend/Resume Interrupts.

SRPC0 Bit (USB0 Single End Receiver Control)

The SRPC0 bit controls the D+ and D– inputs of the USB0 transceiver.
This bit is only valid when the FIXPHY0 bit is 1.

FIXPHY0 Bit (USB0 Transceiver Output Fix)

The FIXPHY0 bit keeps the outputs of the USB0 transceiver disabled.

40.2.38 Deep Standby USB Suspend/Resume Interrupt Register (DPUSR1R)

Address(es): USB.DPUSR1R 000A 0404h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	DVBINT0	—	DOVRCRB0	DOVRCRA0	—	—	DMINT0	DPINT0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	DVBSE0	—	DOVRCRBE0	DOVRCRAE0	—	—	DMINTE0	DPINTE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DPINTE0	USB0 D+ Interrupt Enable/Clear	0: Recovery from deep software standby mode is disabled. 1: Recovery from deep software standby mode is enabled.	R/W
b1	DMINTE0	USB0 D– Interrupt Enable/Clear	0: Recovery from deep software standby mode is disabled. 1: Recovery from deep software standby mode is enabled.	R/W
b3, b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	DOVRCRAE0	USB0 OVRCURA Interrupt Enable/Clear	0: Recovery from deep software standby mode is disabled. 1: Recovery from deep software standby mode is enabled.	R/W
b5	DOVRCRBE0	USB0 OVRCURB Interrupt Enable/Clear	0: Recovery from deep software standby mode is disabled. 1: Recovery from deep software standby mode is enabled.	R/W
b6	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b7	DVBSE0	USB0 VBUS Interrupt Enable/Clear	0: Recovery from deep software standby mode is disabled. 1: Recovery from deep software standby mode is enabled.	R/W
b15 to b8	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b16	DPINT0	USB0 D+ Interrupt Source Recovery Flag	0: The system has not returned from deep software standby mode. 1: The system has returned from deep software standby mode.	R
b17	DMINT0	USB0 D– Interrupt Source Recovery Flag	0: The system has not returned from deep software standby mode. 1: The system has returned from deep software standby mode.	R
b19, b18	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b20	DOVRCRA0	USB0 OVRCURA Interrupt Source Recovery Flag	0: The system has not returned from deep software standby mode. 1: The system has returned from deep software standby mode.	R
b21	DOVRCRB0	USB0 OVRCURB Interrupt Source Recovery Flag	0: The system has not returned from deep software standby mode. 1: The system has returned from deep software standby mode.	R
b22	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b23	DVBINT0	USB0 VBUS Interrupt Source Recovery Flag	0: The system has not returned from deep software standby mode. 1: The system has returned from deep software standby mode.	R
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

DPINTE0 Bit (USB0 D+ Interrupt Enable/Clear)

This bit enables or disables triggering of recovery from deep software standby mode by the D+ input of the USB0. Writing 0 to the DPINTE0 bit while the DPINT0 flag is 1 sets the DPINT0 flag to 0.

DMINTE0 Bit (USB0 D– Interrupt Enable/Clear)

This bit enables or disables triggering of recovery from deep software standby mode by the D– input of the USB0. Writing 0 to the DMINTE0 bit while the DMINT0 flag is 1 sets the DMINT0 flag to 0.

DOVRCRAE0 Bit (USB0 OVRCURA Interrupt Enable/Clear)

This bit enables or disables triggering of recovery from deep software standby mode by the OVRCURA input of the USB0. Writing 0 to the DOVRCRAE0 bit while the DOVRCRA0 flag is 1 sets the DOVRCRA0 flag to 0.

DOVRCRBE0 Bit (USB0 OVRCURB Interrupt Enable/Clear)

This bit enables or disables triggering of recovery from deep software standby mode by the OVRCURB input of the USB0. Writing 0 to the DOVRCRBE0 bit while the DOVRCRB0 flag is 1 sets the DOVRCRB0 flag to 0.

DVBSE0 Bit (USB0 VBUS Interrupt Enable/Clear)

This bit enables or disables triggering of recovery from deep software standby mode by the VBUS input of the USB0. Writing 0 to the DVBSE0 bit while the DVBINT0 flag is 1 sets the DVBINT0 flag to 0.

DPINT0 Flag (USB0 D+ Interrupt Source Recovery Flag)

This bit indicates that the system has returned from deep software standby mode due to the D+ input of the USB0. Recovery from deep software standby mode due to D+ input is only enabled when the DPINTE0 bit is 1. Writing 0 to the DPINTE0 bit while the DPINT0 flag is 1 sets the DPINT0 flag to 0.

DMINT0 Flag (USB0 D– Interrupt Source Recovery Flag)

This bit indicates that the system has returned from deep software standby mode due to the D– input of the USB0. Recovery from deep software standby mode due to D– input is only enabled when the DMINTE0 bit is 1. Writing 0 to the DPINTE0 bit while the DMINT0 flag is 1 sets the DMINT0 flag to 0.

DOVRCRA0 Flag (USB0 OVRCURA Interrupt Source Recovery Flag)

This bit indicates that the system has returned from deep software standby mode due to the OVRCURA input of the USB0. Recovery from deep software standby mode due to OVRCURA input is only enabled when the DOVRCRAE0 bit is 1. Writing 0 to the DOVRCRAE0 bit while the DOVRCRA0 flag is 1 sets the DOVRCRA0 flag to 0.

DOVRCRB0 Flag (USB0 OVRCURB Interrupt Source Recovery Flag)

This bit indicates that the system has returned from deep software standby mode due to the OVRCURB input of the USB0. Recovery from deep software standby mode due to OVRCURB input is only enabled when the DOVRCRBE0 bit is 1. Writing 0 to the DOVRCRBE0 bit while the DOVRCRB0 flag is 1 sets the DOVRCRB0 flag to 0.

DVBINT0 Flag (USB0 VBUS Interrupt Source Recovery Flag)

This indicates that the system has returned from deep software standby mode due to the VBUS input of the USB0. Recovery from deep software standby mode due to VBUS input is only enabled when the DVBSE0 bit is 1. Writing 0 to the DVBSE0 bit while the DVBINT0 flag is 1 sets the DVBINT0 flag to 0.

40.3 Operation

40.3.1 System Control

This section describes the register settings that are necessary for initialization of this module and power consumption control.

40.3.1.1 Setting Data to the USB Related Register

Setting the SYSCFG.USBFE bit to 1 after starting the clock supply to the USB (SYSCFG.SCKE bit = 1) enables and starts USB operation.

40.3.1.2 Controller Function Selection

For the USB, the host or function controller can be selected using the SYSCFG.DCFM bit. Note that the DCFM bit should be modified in the initial settings immediately after a reset is released or when pull-up resistor of the D+line and pull-down resistors of the D+ and D– lines are disabled (the SYSCFG.DPRPU bit = 0 and DRPD bit = 0).

40.3.1.3 Controlling USB Data Bus Resistors

The USB has pull-up and pull-down resistors for the D+ and D– lines. Pull up or pull down these lines by setting the SYSCFG.DPRPU and DRPD bits.

When the function controller is selected, confirm that connection to the USB host is made, then set the DPRPU bit to 1 and pull up the D+ line (during full-speed).

When the DPRPU bit is set to 0 during communication with the PC, the USB module disables the pull-up resistor of the USB data line, thus notifying the USB host of disconnection.

When the host controller is selected, set the DRPD bit and pull down the D+ and D– lines.

Table 40.11 USB Data Bus Resistor Control

SYSCFG register		D–	D+	Function
DRPD bit	DPRPU bit			
0	0	Open	Open	Not in use
0	1	Open	Pull-up	When operating as the function controller (in full-speed)
1	0	Pull-down	Pull-down	When operating as the host controller
1	1	—	—	Setting prohibited

40.3.1.4 Example of USB External Connection Circuit

Figure 40.2 shows an example of connection between OTG device and USB connector in the self-powered state.

The USB controls the signals for enabling a pull-up resistor for the D+ signal and pull-down resistors for the D+ and D- signals. These signals can be pulled up or down using the SYSCFG.DPRPU and DRPD bits.

When the function controller is selected and the DPRPU bit is set to 0 during communication with the host controller, the pull-up resistor of the USB data line is disabled, making it possible to notify the USB host of the device disconnection.

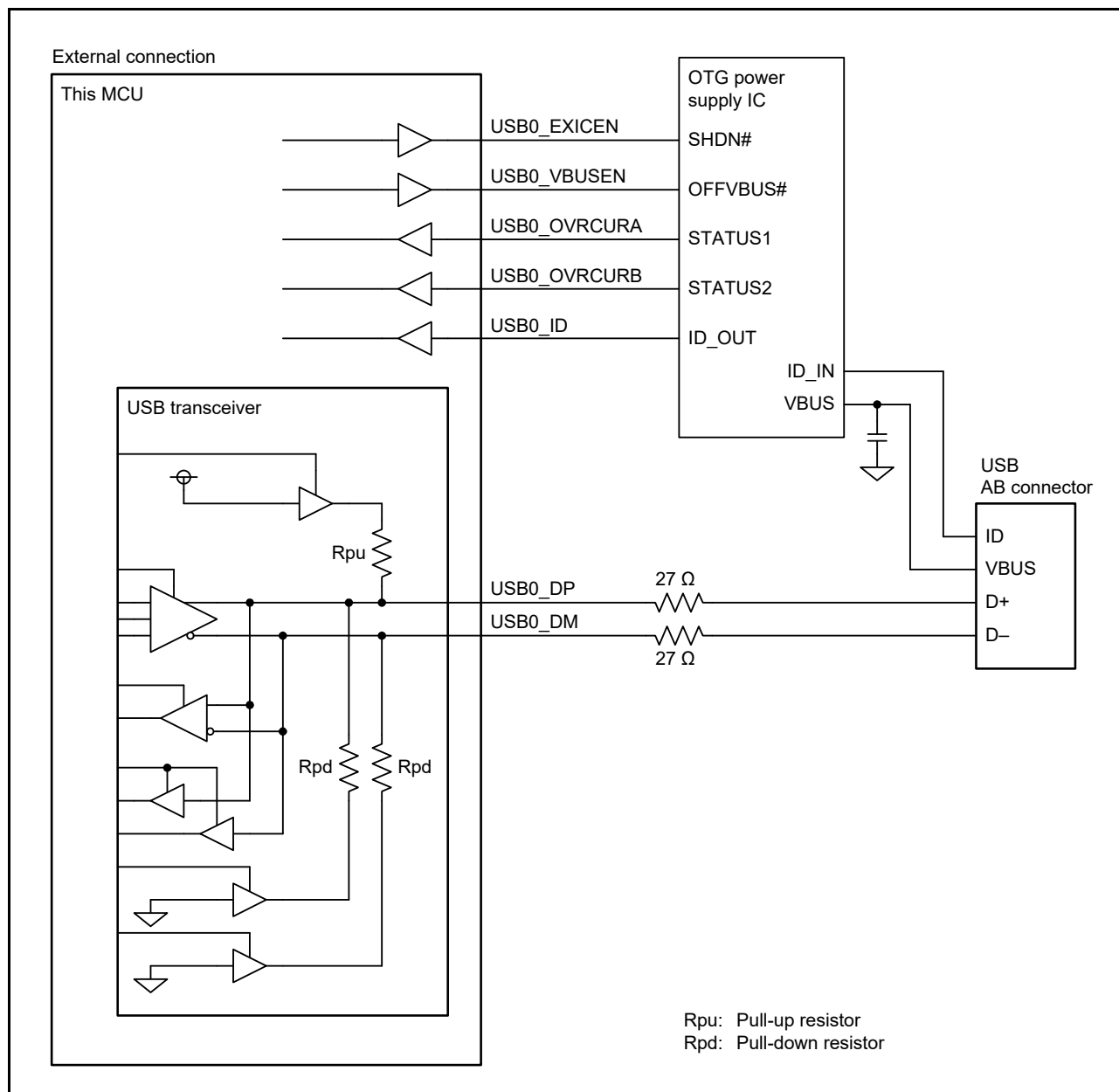


Figure 40.2 Example of Connection between Self-Powered OTG Device and USB Connector

Figure 40.3 shows an example of connection between function controller and USB connector in self-powered state.

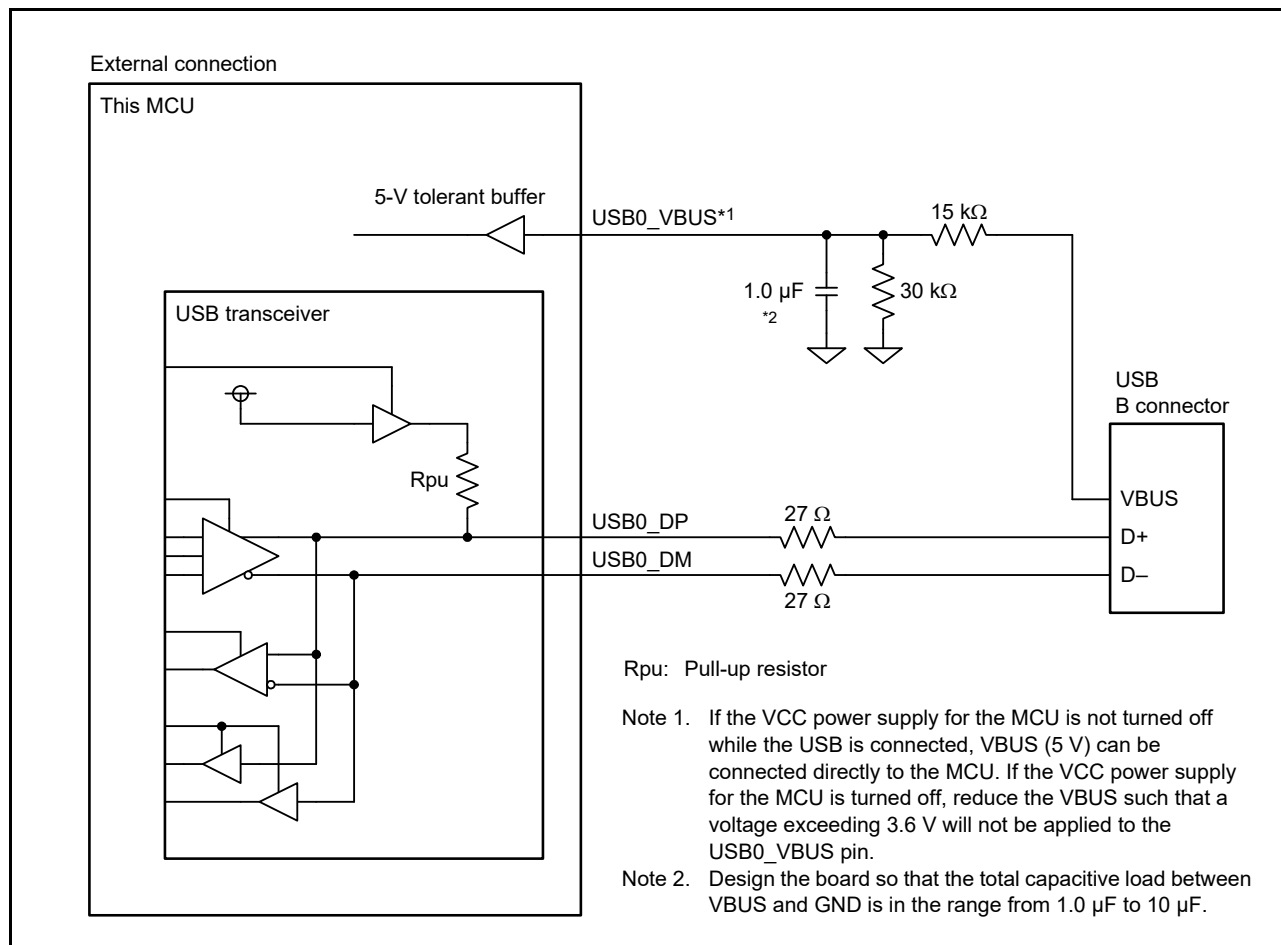


Figure 40.3 Example of Connection between Self-Powered Function and USB Connector

Figure 40.4 shows an example of connection between host controller and USB connector.

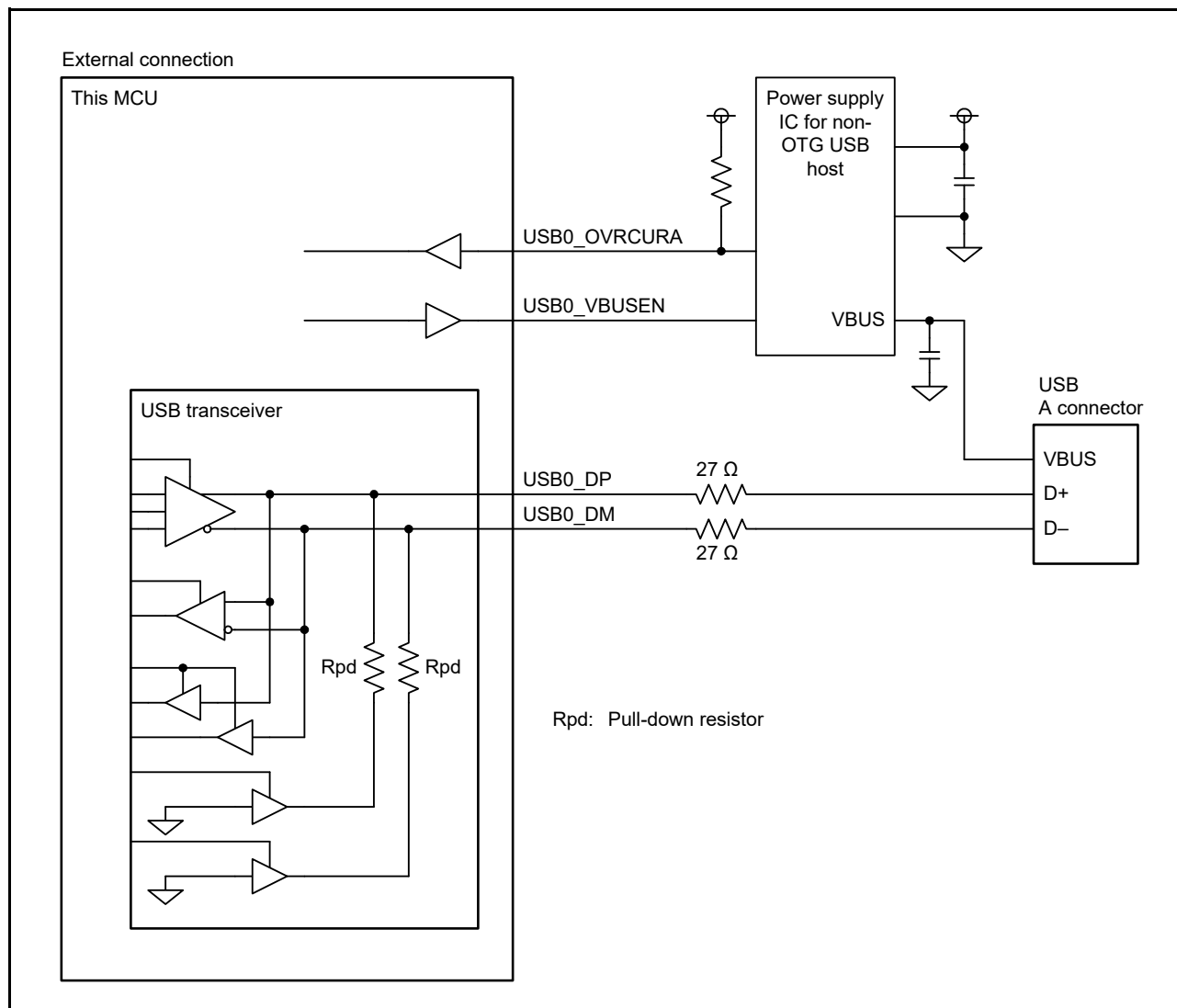


Figure 40.4 Example of Connection between Host and USB Connector

Figure 40.5 shows an example of connection between function controller and USB connector in bus powered state.

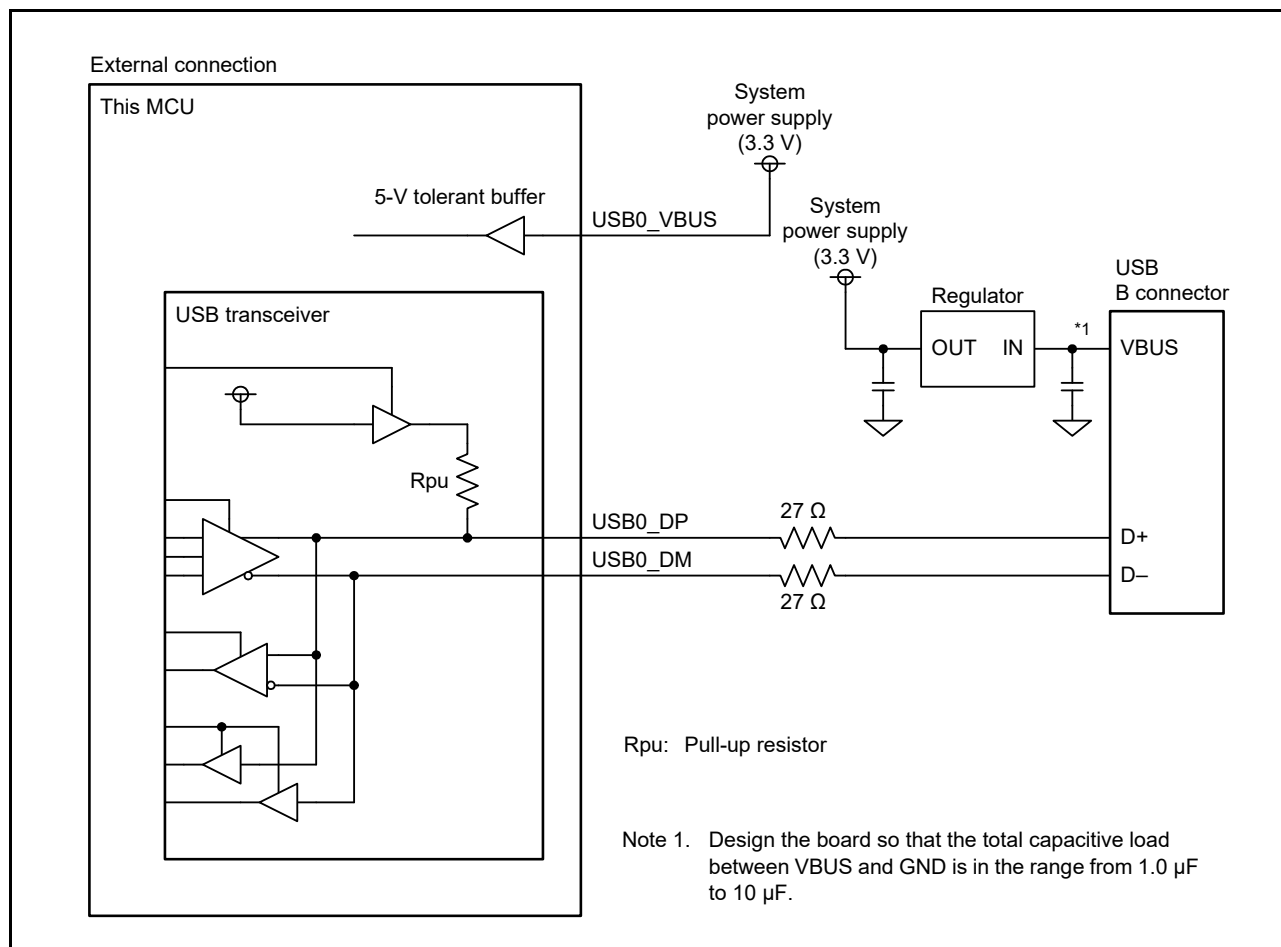


Figure 40.5 Example of Connection between Bus-Powered Function and USB Connector

The examples of external circuits given in this section are simplified circuits, and their operation in every system is not guaranteed.

40.3.1.5 Release from Deep Software Standby Mode Due to USB Suspend/Resume Interrupts

Deep software standby mode can be canceled by a USB0 suspend/resume interrupt.

A USB0 suspend/resume interrupt is detected at the USB resume detecting unit. The USB resume detecting unit controls and monitors the I/O pins for USB0 to detect USB0 suspend/resume interrupts.

Figure 40.6 shows a schematic diagram of connection between the USB resume detecting unit and the I/O pins for USB0.

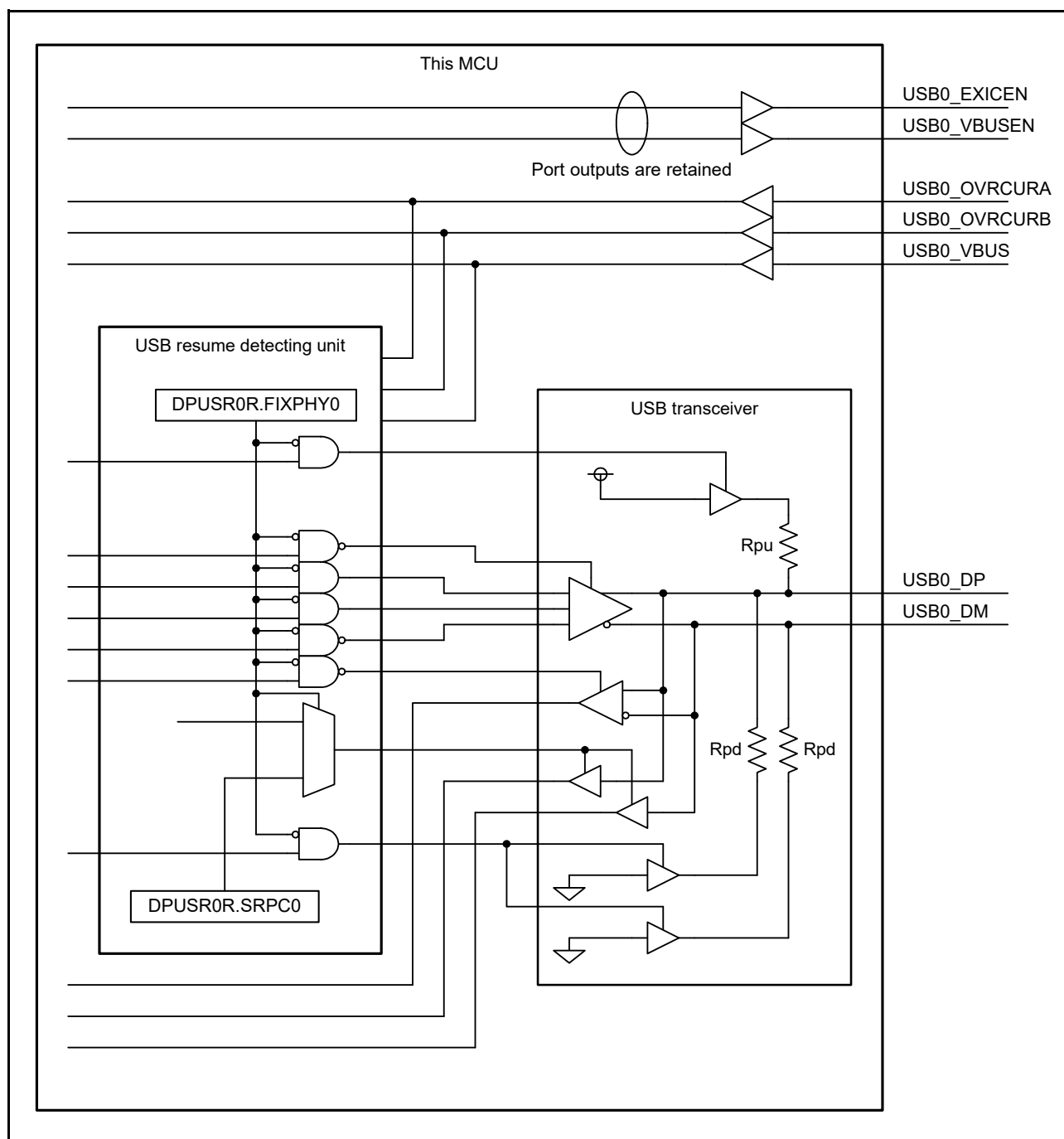


Figure 40.6 Overview of Connection between USB Resume Detecting Unit and USB0 I/O Pins

Table 40.12 shows the USB0 suspend/resume interrupt sources and their corresponding I/O pins.

Table 40.12 USB0 Suspend/Resume Interrupt Sources and Corresponding I/O Pins

USB Operating Mode	Source	Pin Name
Function/OTG	Resume	USB0_DP
Host/OTG	Connection/Disconnection	USB0_DP/USB0_DM
Function	Connection/Disconnection	USB0_VBUS
Host	Overcurrent detection	USB0_OVRCURA
OTG	Overcurrent detection	USB0_OVRCURA/USB0_OVRCURB

Figure 40.7 shows a flowchart for setting the USB when entering deep software standby mode while the host controller or function controller is selected. Figure 40.8 shows a flowchart for setting the USB0 when canceling deep software standby mode while the host controller is selected. Figure 40.9 shows a flowchart for setting the USB0 when canceling deep software standby mode while the function controller is selected.

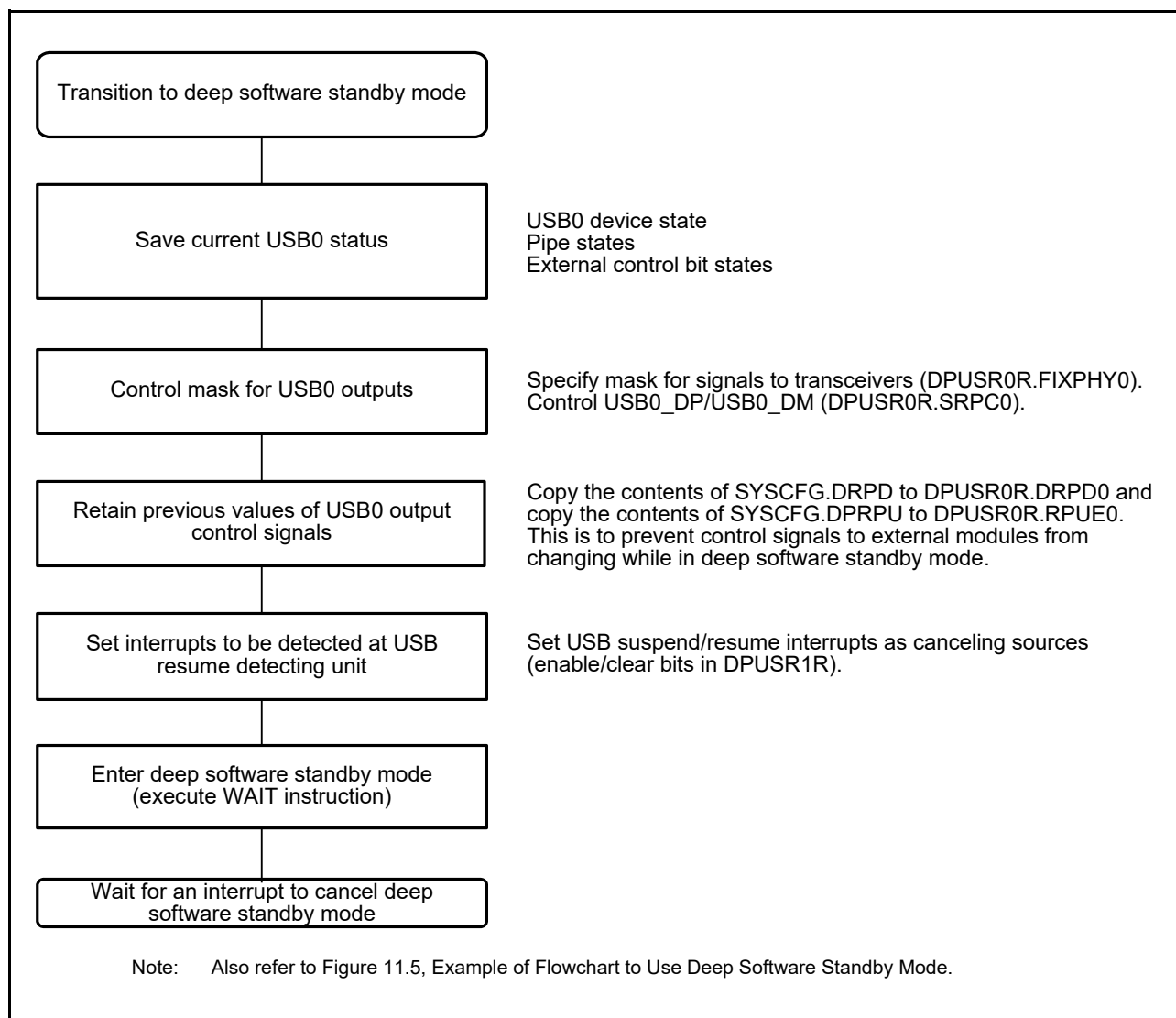


Figure 40.7 USB Setting Flowchart for Transition to Deep Software Standby Mode as Host/Function Controller

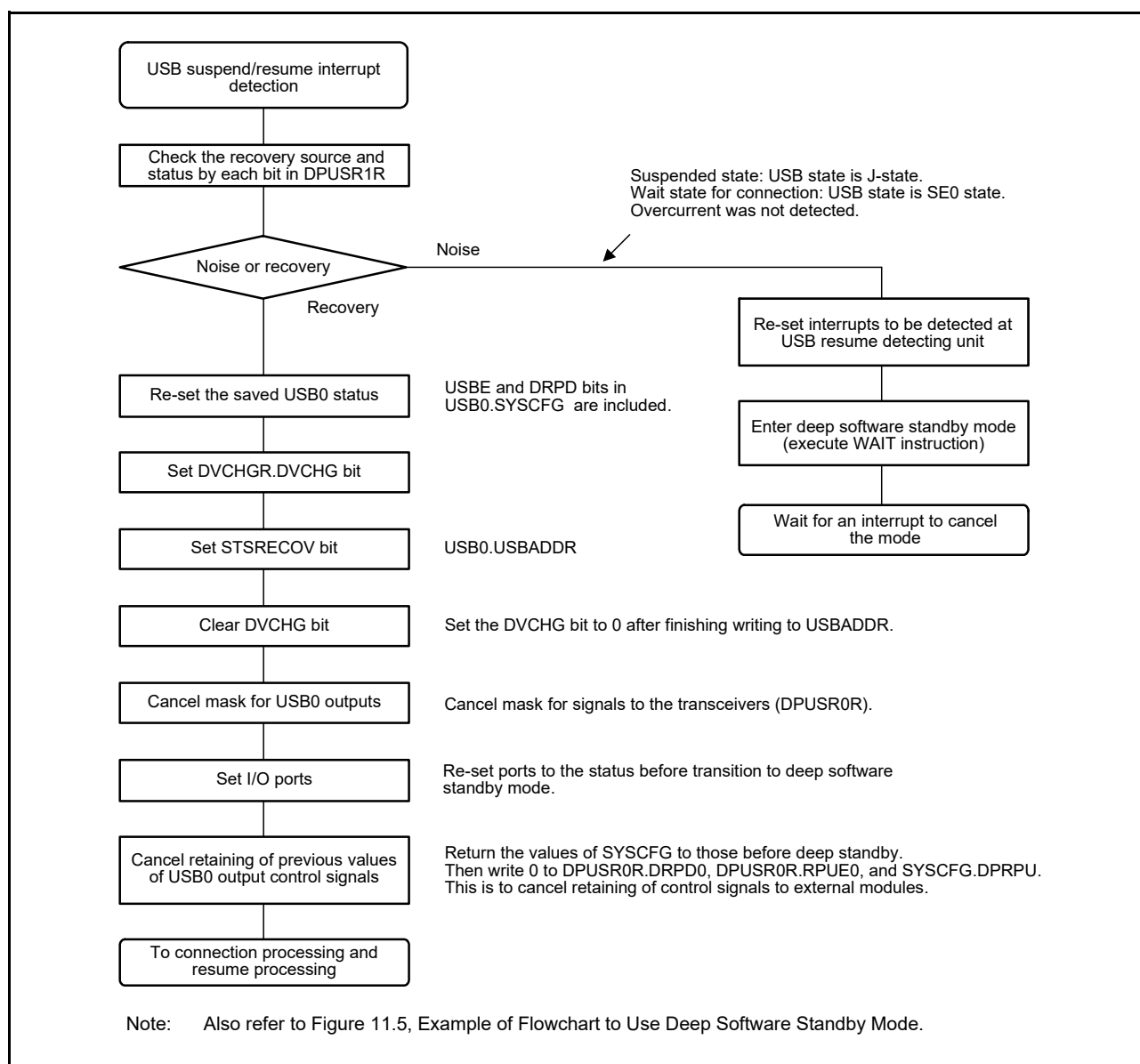


Figure 40.8 USB Setting Flowchart for Canceling Deep Software Standby Mode as Host Controller

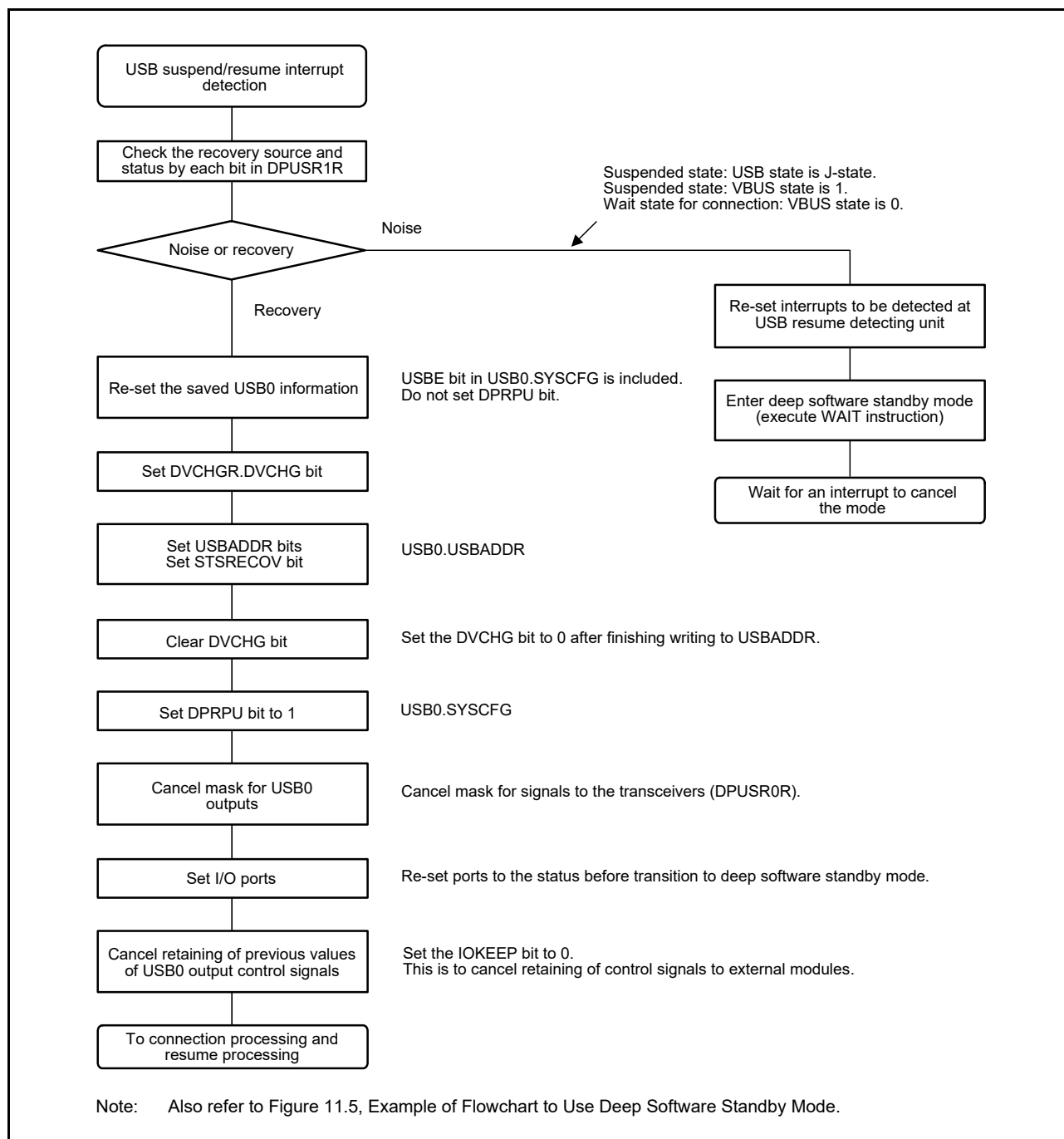


Figure 40.9 USB Setting Flowchart for Canceling Deep Software Standby Mode as Function Controller

40.3.2 Interrupt Sources

Table 40.13 lists the interrupt sources in the USB.

When an interrupt generation condition is satisfied and the interrupt output is enabled using the corresponding interrupt enable register, a USB interrupt request is issued the Interrupt Controller (ICU) and an USB interrupt will be generated.

Table 40.13 Interrupt Sources

Bit to be Set	Name	Interrupt Source	Function That Generates the Interrupt	Status Flag
VBINT	VBUS interrupt	When a change in the state of the USB0_VBUS input pin has been detected (low to high or high to low)	Host/function ^{*1}	INTSTS0.VBSTS
RESM	Resume interrupt	When a change in the state of the USB bus has been detected in the suspended state (J-state to K-state or J-state to SE0)	Function	—
SOFR	Frame number update interrupt	[Host controller is selected] - When an SOF packet with a different frame number has been transmitted [Function controller is selected] - When an SOF packet with a different frame number has been received	Host/function	—
DVST	Device state transition interrupt	- When a device state transition has been detected (any of the following conditions) - A USB bus reset detected - Suspended state detected - SET_ADDRESS request received - SET_CONFIGURATION request received	Function	INTSTS0.DVSQ[2:0]
CTRT	Control transfer stage transition interrupt	- When a stage transition has been detected in control transfer (any of the following conditions) - Setup stage completed - Control write transfer status stage transition - Control read transfer status stage transition - Control transfer completed - A control transfer sequence error occurred	Function	INTSTS0.CTSQ[2:0]
BEMP	Buffer empty interrupt	- When transmission of all data in the buffer memory has been completed and the buffer has become empty - When a packet larger than the maximum packet size has been received	Host/function	BEMPSTS.PIPEnBEMP
NRDY	Buffer not ready interrupt	[Host controller is selected] - When STALL has been received from the peripheral device for the issued token - When a response has not been received correctly from the peripheral device for the issued token (no response was returned three consecutive times or a packet reception error occurred three consecutive times) - When an overrun/underrun occurred during isochronous transfer [Function controller is selected] - When NAK has been returned for an IN or OUT token while the PID[1:0] bits are 01b (BUF) - When a CRC error or a bit stuffing error occurred during data reception in isochronous transfer - When an overrun/underrun occurred during data reception in isochronous transfer	Host/function	NRDYSTS.PIPEnNRDY
BRDY	Buffer ready interrupt	When the buffer has become ready (reading or writing is enabled)	Host/function	BRDYSTS.PIPEnBRDY
OVRCR	Overcurrent input change interrupt	When a change in the state of the USB0_OVRCURA or USB0_OVRCURB input pin has been detected (low to high or high to low)	Host	INTSTS1.OVRCR
BCHG	Bus change interrupt	When a change of USB bus state has been detected	Host/function	SYSSTS0.LNST[1:0]
DTCH	Disconnection detection during full-speed operation	When disconnection of a peripheral device has been detected in full-speed operation	Host	DVSTCTR0.RHST[2:0]
ATTCH	Device connection detection	When J-state or K-state is detected on the USB port for 2.5 μs. Used for checking whether a peripheral device is connected.	Host	—
EOFERR	EOF error detection	When an EOF error of a peripheral device has been detected	Host	—
SACK	Normal setup operation	When the normal response (ACK) for the setup transaction has been received	Host	—
SIGN	Setup error	When a setup transaction error (no response or ACK packet corruption) was detected three consecutive times	Host	—

Note 1. Though this interrupt can be generated while the host function is selected, it is not usually used with the host function.

Figure 40.10 shows the circuits related to the interrupts in the USB.

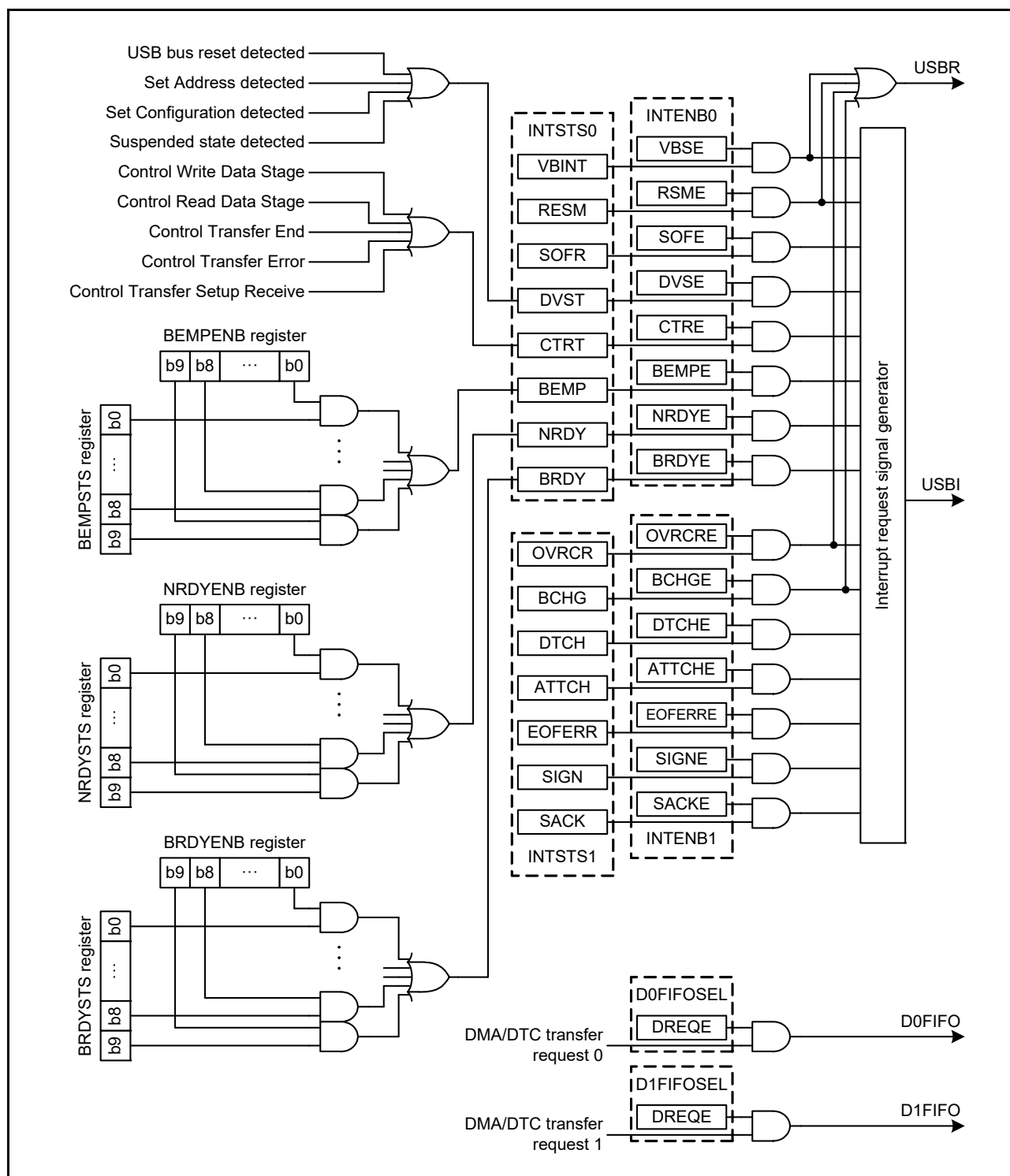


Figure 40.10 Circuits Related to Interrupts in USB

Table 40.14 shows the interrupts generated in the USB0.

Table 40.14 USB Interrupts

Interrupt Name	Interrupt Status Flag	DTC Activation	DMAC Activation
D0FIFO	DMA/DTC transfer request 0	Possible	Possible
D1FIFO	DMA/DTC transfer request 1	Possible	Possible
USBI	VBUS interrupt, resume interrupt, frame number update interrupt, device state transition interrupt, control transfer stage transition interrupt, buffer empty interrupt, buffer not ready interrupt, buffer ready interrupt, overcurrent input change interrupt, bus change interrupt, disconnection detection during full-speed operation, device connection detection, EOF error detection, normal setup operation, and setup error	Not possible	Not possible
USBR	VBUS interrupt, resume interrupt, overcurrent input change interrupt, and bus change interrupt	Not possible	Not possible

40.3.3 Interrupt Descriptions

40.3.3.1 BRDY Interrupt

The BRDY interrupt is generated when either of the host controller or function controller is selected. The following shows the conditions under which the USB sets 1 to a corresponding bit in the BRDYSTS register. Under this condition, the USB generates a BRDY interrupt if software has set 1 to the BRDYENB.PIPE_nBRDYE bit that corresponds to the pipe and 1 to the INTENB0.BRDYE bit.

The conditions for generating and clearing the BRDY interrupt depend on the settings of the SOFCFG.BRDYM bit and PIPECFG.BFRE bit for each pipe as described below.

(1) When the SOFCFG.BRDYM Bit = 0 and the PIPECFG.BFRE Bit = 0

With these settings, the BRDY interrupt indicates that the FIFO port is accessible.

On any of the following conditions, the USB generates an internal BRDY interrupt request trigger and sets 1 to the BRDYSTS.PIPE_nBRDY flag corresponding to the pertinent pipe.

(a) For the pipe in the transmitting direction:

- When the DIR bit is changed from 0 to 1 by software.
- When packet transmission is completed using the pertinent pipe while write-access from the CPU to the FIFO buffer for the pertinent pipe is disabled (when the BSTS flag is read as 0).
- When one FIFO buffer is empty on completion of writing data to the other FIFO buffer in double buffer mode.
- No request trigger is generated until completion of writing data to the currently-written FIFO buffer even if transmission to the other FIFO buffer is completed.
- When the hardware flushes the buffer of the pipe for isochronous transfers.
- When 1 is written to the PIPE_nCTR.ACLRM bit, which causes the FIFO buffer to make transition from the write-disabled to write-enabled state.

No request trigger is generated for the DCP (that is, during data transmission for control transfers).

(b) For the pipe in the receiving direction:

- When packet reception is completed successfully thus enabling the FIFO buffer to be read while read-access from the CPU to the FIFO buffer for the pertinent pipe is disabled (when the BSTS flag is read as 0).
No request trigger is generated for the transaction in which data PID mismatch has occurred.
- When one FIFO buffer is read-enabled on completion of reading data from the other FIFO buffer in double buffer mode.
No request trigger is generated until completion of reading data from the currently-read FIFO buffer even if reception by the other FIFO buffer is completed.

When the function controller is selected, the BRDY interrupt is not generated in the status stage of control transfers. The BRDY interrupt status of the pertinent pipe can be set to 0 by writing 0 to the corresponding PIPEnBRDY flag through software. In this case, 1s should be written to the PIPEnBRDY flags for the other pipes. Clear the BRDY status before accessing the FIFO buffer.

(2) When the SOFCFG.BRDYM Bit = 0 and the PIPECFG.BFRE Bit = 1

With these settings, the USB generates a BRDY interrupt on completion of reading all data for a single transfer using the pipe in the receiving direction, and sets 1 to the bit in the BRDYSTS register corresponding to the pertinent pipe. On any of the following conditions, the USB determines that the last data for a single transfer has been received.

- When a short packet including a zero-length packet is received.
- When the pipe n transaction counter register (PIPenTRN) is used and the number of packets specified by the PIPenTRN register are completely received.

When the pertinent data is completely read after any of the above conditions has been satisfied, the USB determines that all data for a single transfer has been completely read.

When a zero-length packet is received while the FIFO buffer is empty, the USB module determines that all data for a single transfer has been completely read when the FRDY flag in the FIFO port control register is 1 and the DTLN[8:0] flags are 0. In this case, to start the next transfer, write 1 to the BCLR bit in the corresponding port control register through software.

With these settings, the USB does not detect a BRDY interrupt for the pipe in the transmitting direction.

The BRDY interrupt status of the pertinent pipe can be set to 0 by writing 0 to the corresponding BRDYSTS.PIPenBRDY flag through software. In this case, 1s should be written to the PIPenBRDY flags for the other pipes.

In this mode, the PIPECFG.BFRE bit setting should not be modified until all data for a single transfer has been processed. When it is necessary to modify the PIPECFG.BFRE bit before completion of processing, all FIFO buffers for the pertinent pipe should be cleared using the PIPenCTR.ACLRM bit.

(3) When the SOFCFG.BRDYM Bit = 1 and the PIPECFG.BFRE Bit = 0

With these settings, the BRDYSTS.PIPEnBRDY values are linked to the BSTS flag setting for each pipe. In other words, the BRDY interrupt status flags (PIPEnBRDY) are set to 1 or 0 by the USB depending on the FIFO buffer status.

(a) For the pipe in the transmitting direction:

The BRDY interrupt status bits are set to 1 when the FIFO buffer is ready for write access, and are set to 0 when it is not ready.

However, the BRDY interrupt is not generated even if the DCP in the transmitting direction is ready for write access.

(b) For the pipe in the receiving direction:

The BRDY interrupt status bits are set to 1 when the FIFO buffer is ready for read access, and are set to 0 when all data have been read (not ready for read access).

When a zero-length packet is received while the FIFO buffer is empty, the pertinent bit is set to 1 and the BRDY interrupt is continuously generated until BCLR = 1 is written through software.

With this setting, the PIPEnBRDY flag cannot be set to 0 through software.

When the SOFCFG.BRDYM bit is set to 1, all PIPECFG.BFRE bits (for all pipes) should be set to 0.

Figure 40.11 shows the timing of BRDY interrupt generation.

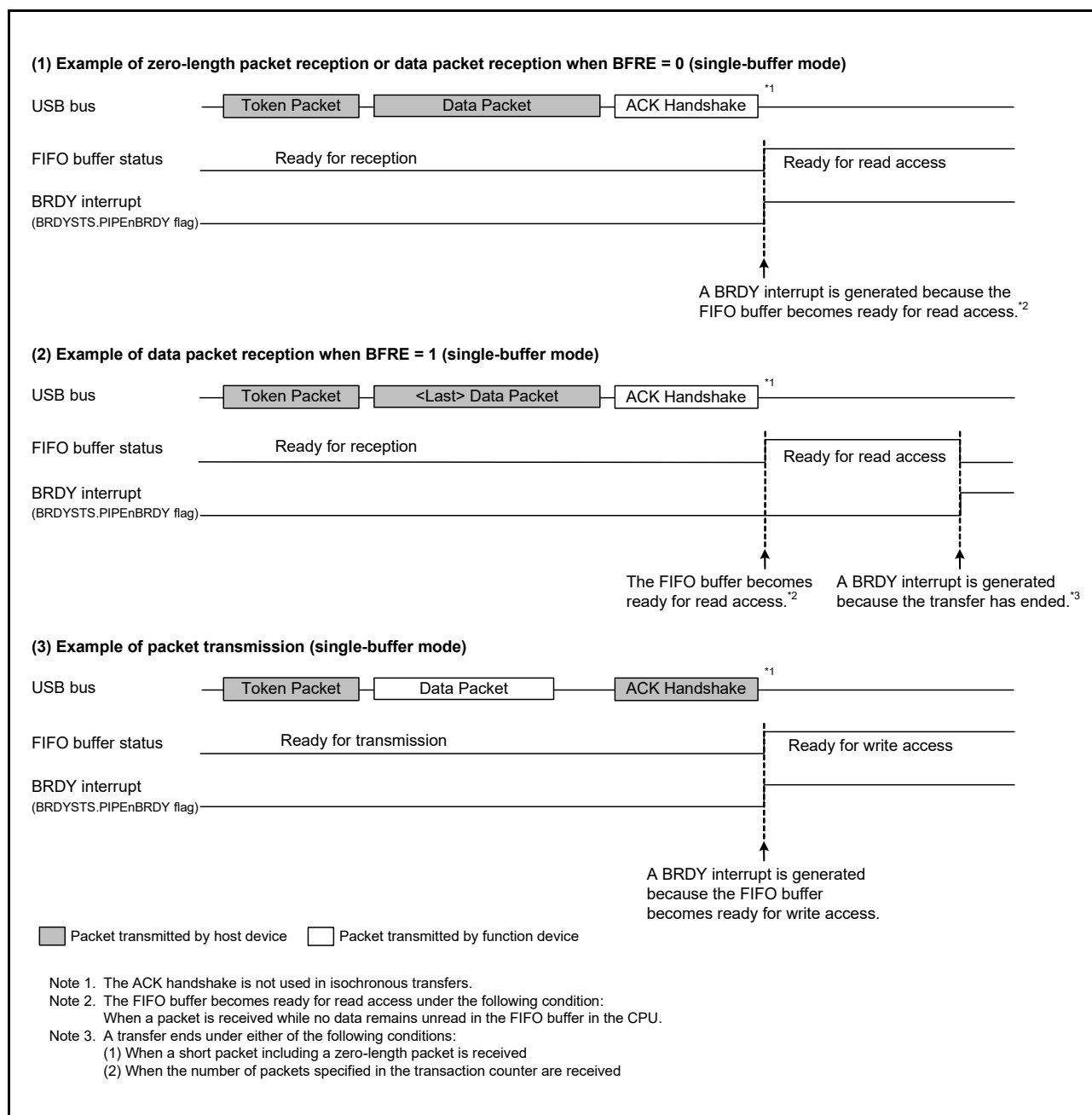


Figure 40.11 Timing of BRDY Interrupt Generation

The condition that USB clears the INTSTS0.BRDY flag depends on the SOFCFG.BRDYM bit setting. Table 40.15 shows the condition for clearing the BRDY flag.

Table 40.15 Condition for Clearing BRDY Flag

BRDYM Bit	Condition for Clearing BRDY Flag
0	The USB sets the BRDY flag to 0 when all bits in the BRDYSTS register have been set to 0 by software.
1	The USB sets the BRDY flag to 0 when the BSTS flags for all pipes have become 0.

40.3.3.2 NRDY Interrupt

On generating an internal NRDY interrupt request for the pipe whose PID[1:0] bits are set to 01b (BUF) by software, the USB sets the corresponding NRDYSTS.PIPEnNRDY flag to 1. If the corresponding bit in the NRDYENB register has been set to 1 by software, the USB sets the INTSTS0.NRDY flag to 1 and generates a USB interrupt.

The following describes the conditions on which the USB generates the internal NRDY interrupt request for a given pipe.

Note that the internal NRDY interrupt request is not generated during setup transaction execution when the host controller is selected. During setup transactions when the host controller is selected, the SACK or SIGN interrupt is detected.

The internal NRDY interrupt request is not generated during status stage execution of the control transfer when the function controller is selected.

(1) When Host Controller is Selected

(a) For the pipe in the transmitting direction:

On any of the following conditions, the USB detects an NRDY interrupt.

- For the pipe for isochronous transfers, when the time to issue an OUT token comes while there is no data to be transmitted in the FIFO buffer.

In this case, the USB transmits a zero-length packet following the OUT token and sets the bit corresponding to the NRDYSTS.PIPEnNRDY flag and the FRMNUM.OVRN flag to 1.

- During communications other than setup transactions using the pipe for the transfers other than isochronous transfers, when any combination of the following two cases occur three consecutive times: 1) no response is returned from the peripheral device (when timeout is detected before detection of the handshake packet from the peripheral device) and 2) an error is detected in the packet from the peripheral device.

In this case, the USB sets the bit corresponding to the PIPEnNRDY flag to 1 and modifies the setting of the PID[1:0] bits of the corresponding pipe to 00b (NAK).

- During communications other than setup transactions, when the STALL handshake is received from the peripheral device.

In this case, the USB sets the bit corresponding to the PIPEnNRDY flag to 1 and modifies the setting of the PID[1:0] bits of the corresponding pipe to 11b (STALL).

(b) For the pipe in the receiving direction:

- For the pipe for isochronous transfers, when the time to issue an IN token comes while there is no space available in the FIFO buffer.

In this case, the USB discards the received data for the IN token and sets the PIPEnNRDY flag corresponding to the pipe and the OVRN flag to 1.

When a packet error is detected in the received data for the IN token, the USB also sets the FRMNUM.CRCE flag to 1.

- For the pipe for the transfers other than isochronous transfers, when any combination of the following two cases occur three consecutive times: 1) no response is returned from the peripheral device for the IN token issued by the USB (when timeout is detected before detection of the DATA packet from the peripheral device) and 2) an error is detected in the packet from the peripheral device.

In this case, the USB sets the PIPEnNRDY flag corresponding to the pipe to 1 and modifies the setting of the PID[1:0] bits of the corresponding pipe to 00b (NAK).

- For the pipe for isochronous transfers, when no response is returned from the peripheral device for the IN token (when timeout is detected before detection of the DATA packet from the peripheral device) or an error is detected in the packet from the peripheral device.

In this case, the USB sets the PIPEnNRDY flag corresponding to the pipe to 1. (The setting of the PID[1:0] bits of the pipe is not modified.)

- For the pipe for isochronous transfers, when a CRC error or a bit stuffing error is detected in the received data packet.
In this case, the USB sets the PIPEnNRDY flag corresponding to the pipe and the CRCE flag to 1.
- When the STALL handshake is received.
In this case, the USB sets the PIPEnNRDY flag corresponding to the pipe to 1 and modifies the setting of the PID[1:0] bits of the corresponding pipe to 11b (STALL).

(2) When Function Controller is Selected

(a) For the pipe in the transmitting direction:

- When an IN token is received while there is no data to be transmitted in the FIFO buffer.
In this case, the USB generates a NRDY interrupt request at the reception of the IN token and sets the NRDYSTS.PIPEnNRDY flag to 1.
For the pipe for the isochronous transfers in which an interrupt is generated, the USB transmits a zero-length packet and sets the FRMNUM.OVRN flag to 1.

(b) For the pipe in the receiving direction:

- When an OUT token is received while there is no space available in the FIFO buffer.
For the pipe for the isochronous transfers in which an interrupt is generated, the USB generates a NRDY interrupt request at the reception of the OUT token and sets the PIPEnNRDY flag to 1 and OVRN flag to 1.
For the pipe for the transfers other than isochronous transfers in which an interrupt is generated, the USB generates a NRDY interrupt request when a NAK handshake is transferred after the data following the OUT token is received, and sets the PIPEnNRDY flag to 1.
However, during re-transmission (due to data PID mismatch), the NRDY interrupt request is not generated. In addition, if an error occurs in the DATA packet, the NRDY interrupt request is not generated.
- For the pipe for isochronous transfers, when a token is not received successfully within an interval frame.
In this case, the USB generates a NRDY interrupt request when SOF is received, and sets the PIPEnNRDY flag to 1.

Figure 40.12 shows the timing of NRDY interrupt generation when the function controller is selected.

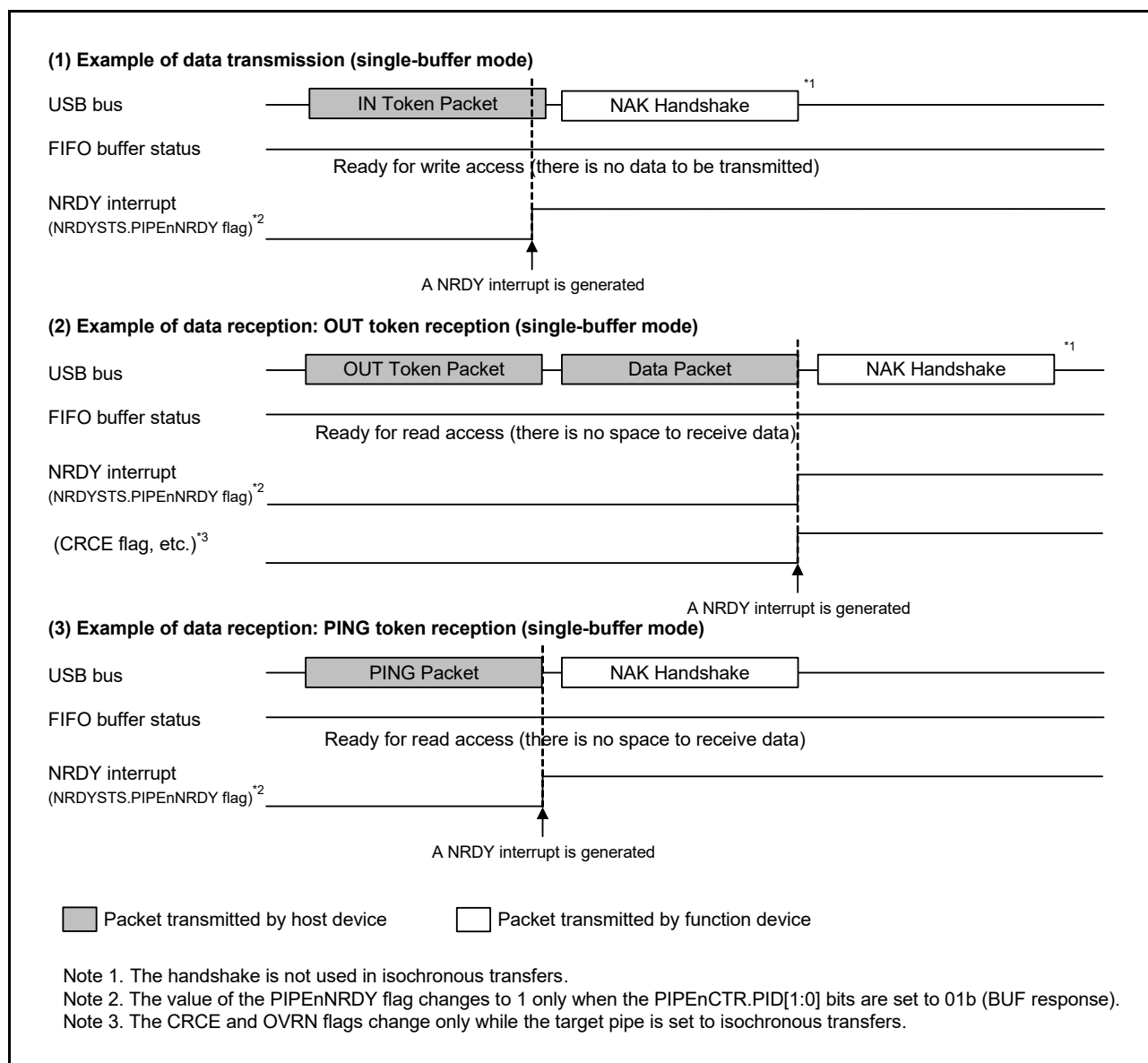


Figure 40.12 Timing of NRDY Interrupt Generation (When Function Controller is Selected)

40.3.3.3 BEMP Interrupt

On detecting a BEMP interrupt for the pipe whose PID[1:0] bits are set to 01b (BUF) by software, the USB sets the corresponding BEMPSTS.PIPEnBEMP flag to 1. If the corresponding bit in the BEMPENB register has been set to 1 by software, the USB sets the INTSTS0.BEMP flag to 1 and generates a USB interrupt.

The following describes the conditions on which the USB generates an internal BEMP interrupt request.

(1) For the pipe in the transmitting direction:

When the FIFO buffer of the corresponding pipe is empty on completion of transmission (including zero-length packet transmission).

In single buffer mode, an internal BEMP interrupt request is generated simultaneously with the BRDY interrupt for the pipe other than DCP. However, the internal BEMP interrupt request is not generated on any of the following conditions.

- When the CPU or DMAC/DTC has already started writing data to the FIFO buffer of the CPU on completion of transmitting data from one FIFO buffer in double buffer mode.
- When the buffer is cleared (emptied) by setting the PIPEnCTR.ACLRM or the BCLR bit in the port control register to 1.
- When IN transfer (zero-length packet transmission) is performed during the control transfer status stage while the function controller is selected.

(2) For the pipe in the receiving direction:

When the successfully-received data packet size exceeds the specified maximum packet size.

In this case, the USB generates a BEMP interrupt request, sets the corresponding BEMPSTS.PIPEnBEMP flag to 1, discards the received data, and modifies the setting of the PID[1:0] bits of the corresponding pipe to 11b (STALL). Here, the USB returns no response when used as the host controller, and returns STALL response when used as the function controller.

However, the internal BEMP interrupt request is not generated on any of the following conditions.

- When a CRC error or a bit stuffing error is detected in the received data.
- When a setup transaction is being performed,
Writing 0 to the BEMPSTS.PIPEnBEMP flag clears the status.
Writing 1 to the BEMPSTS.PIPEnBEMP flag has no effect.

Figure 40.13 shows the timing of BEMP interrupt generation when the function controller is selected.

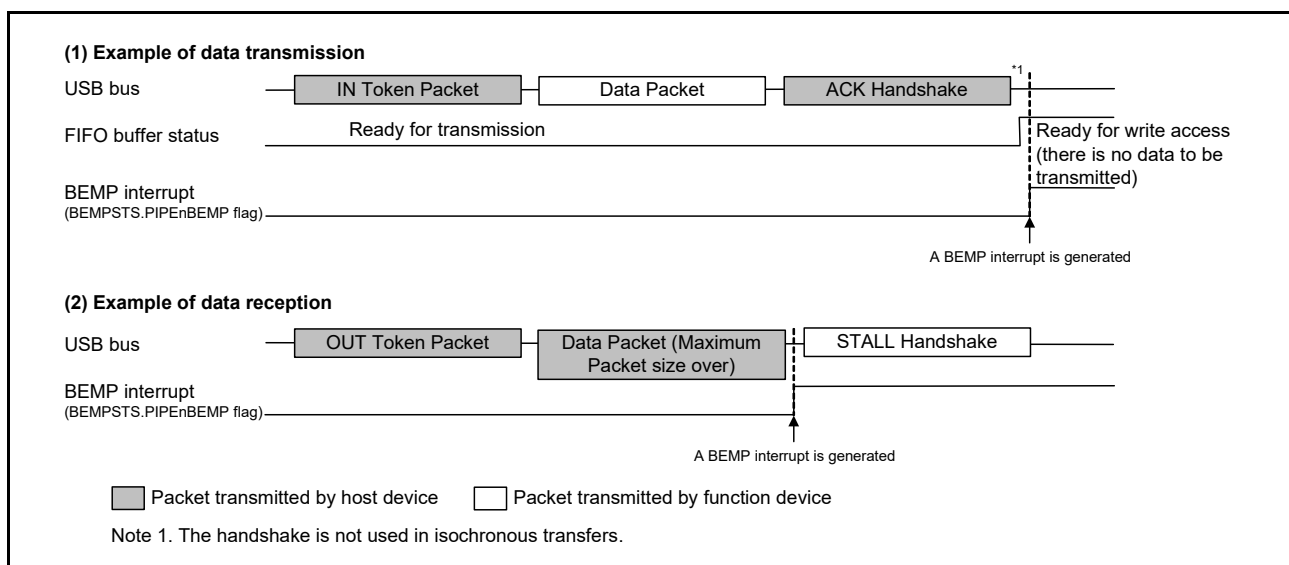


Figure 40.13 Timing of BEMP Interrupt Generation (When Function Controller is Selected)

40.3.3.4 Device State Transition Interrupt

Figure 40.14 is a diagram of device state transitions in the USB. The USB controls device state and generates device state transition interrupts. However, recovery from the suspended state (resume signal detection) is detected by means of the resume interrupt. The device state transition interrupts can be enabled or disabled individually using INTENB0. The device state to which a transition was made can be confirmed using the INTSTS0.DVSQ[2:0] flags.

When a transition is made to the default state, a device state transition interrupt is generated after a USB bus reset is detected.

Device state can be controlled only when the function controller is selected. The device state transition interrupts can also be generated only when the function controller is selected.

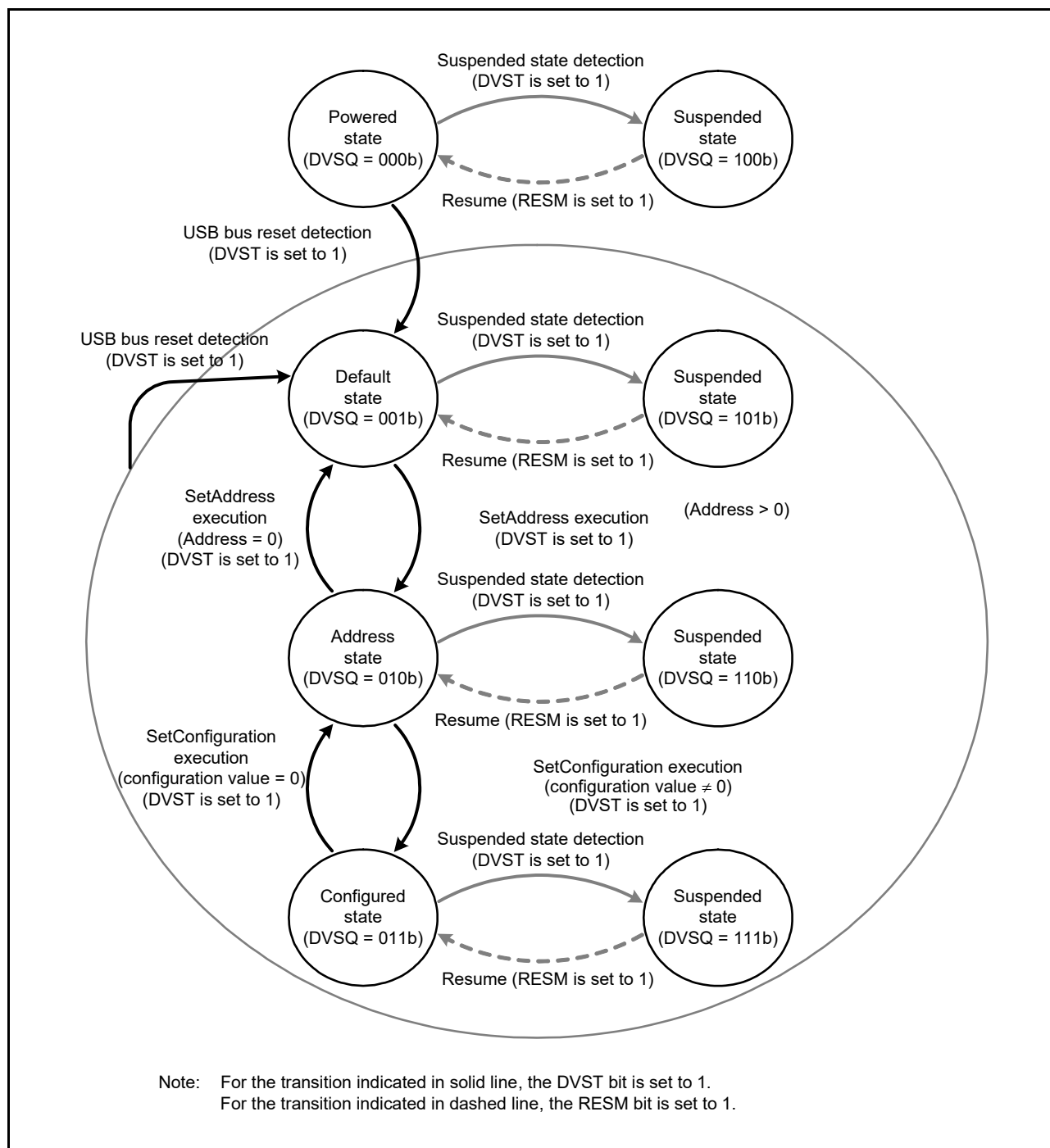


Figure 40.14 Device State Transitions

40.3.3.5 Control Transfer Stage Transition Interrupt

Figure 40.15 is a diagram of control transfer stage transitions in the USB. The USB controls the control transfer sequence and generates control transfer stage transition interrupts. The control transfer stage transition interrupts can be enabled or disabled individually using INTENB0. The transfer stage to which a transition was made can be confirmed using the INTSTS0.CTSQ[2:0] flags.

Control transfer stage transition interrupts are generated only when the function controller is selected.

The control transfer sequence errors are listed below. If an error occurs, the DCPCTR.PID[1:0] bits are set to 1xb (STALL response).

During control read transfer:

- An OUT token is received while no data has been transferred for the IN token at the data stage.
- An IN token is received at the status stage.
- A data packet with PID = DATA0 is received at the status stage.

During control write transfer:

- An IN token is received while no ACK response has been returned for the OUT token at the data stage.
- A data packet with PID = DATA0 is received for the first data packet at the data stage.
- An OUT token is received at the status stage

During no-data control transfers:

- An OUT token is received at the status stage.

At the control write transfer data stage, if the number of receive data exceeds the wLength value of the USB request, it cannot be recognized as a control transfer sequence error. At the control read transfer status stage, packets other than zero-length packets are received by an ACK response and the transfer ends normally.

When a CTRT interrupt occurs in response to a sequence error (INTSTS0.CTRT = 1), CTSQ[2:0] = 110b value is retained until the CTRT flag = 0 is written from the system (the interrupt status is cleared). Therefore, while CTSQ[2:0] = 110b is being held, the CTRT interrupt that ends the setup stage will not be generated even if a new USB request is received. (The USB retains the setup stage end, and after the interrupt status has been cleared by software, a setup stage end interrupt is generated.)

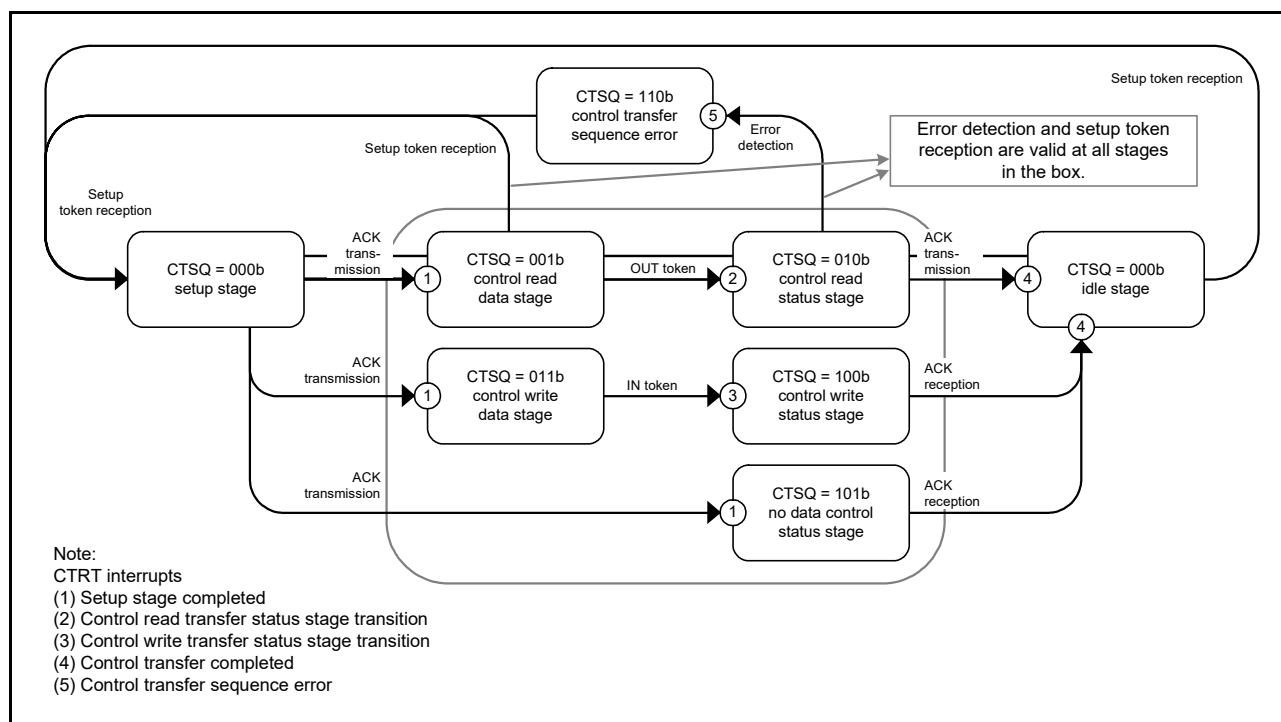


Figure 40.15 Control Transfer Stage Transitions

40.3.3.6 Frame Update Interrupt

With the host controller selected, an interrupt is generated at the timing when the frame number is updated. With the function controller selected, an SOFR interrupt is generated when the frame number is updated.

When the function controller is selected, the USB updates the frame number and generates an SOFR interrupt if it detects a new SOF packet during full-speed operation.

40.3.3.7 VBUS Interrupt

When the USB0_VBUS pin level changes, a VBUS interrupt is generated. The level of the USB0_VBUS pin can be checked with the INTSTS0.VBSTS flag. Whether the host controller is connected or disconnected can be confirmed using the VBUS interrupt. However, if the system is activated with the host controller connected, the first VBUS interrupt is not generated because there is no change in the USB0_VBUS pin level.

40.3.3.8 Resume Interrupt

When the function controller is selected, a resume interrupt is generated when the device state is the suspended state and the USB bus state has changed (from J-state to K-state, or from J-state to SE0). Recovery from the suspended state is detected by means of the resume interrupt.

When the host controller is selected, no resume interrupt is generated. Use the BCHG interrupt to detect a change in the USB bus state.

40.3.3.9 OVRCCR Interrupt

An OVRCCR interrupt is generated when the USB0_OVRCURA or USB0_OVRCURB pin level has changed. The levels of the USB0_OVRCURA and USB0_OVRCURB pins can be checked with the SYSSTS0.OVCMON[1:0] bits. The external power supply IC can check whether overcurrent has been detected using the OVRCCR interrupt.

For OTG connection, whether a change has been detected in the VBUS comparator can be checked using the OVRCCR interrupt.

40.3.3.10 BCHG Interrupt

A BCHG interrupt is generated when the USB bus state has changed. The BCHG interrupt can be used to detect whether the peripheral device is connected and can also be used to detect a remote wakeup when the host controller is selected. The BCHG interrupt is generated regardless of whether the host controller or function controller is selected.

40.3.3.11 DTCH Interrupt

A DTCH interrupt is generated when disconnection of the USB bus is detected while the host controller is selected. The USB detects bus disconnection based on USB Specification 2.0.

After detecting a DTCH interrupt, the USB controls hardware as described below (irrespective of the value set in the corresponding interrupt enable bit). All pipes in which communications are currently carried out for the pertinent port should be terminated by software and make a transition to the wait state for bus connection to the pertinent port (wait state for ATTCH interrupt generation).

- Modifies the DVSTCTR0.UACT bit for the port in which a DTCH interrupt has been detected to 0.
- Puts the port in which a DTCH interrupt has been generated into the idle state.

40.3.3.12 SACK Interrupt

A SACK interrupt is generated when an ACK response for the transmitted setup packet has been received from the peripheral device with the host controller selected. The SACK interrupt can be used to confirm that the setup transaction has been completed successfully.

40.3.3.13 SIGN Interrupt

A SIGN interrupt is generated when an ACK response for the transmitted setup packet has not been correctly received from the peripheral device three consecutive times with the host controller selected. The SIGN interrupt can be used to detect no ACK response transmitted from the peripheral device or corruption of an ACK packet.

40.3.3.14 ATTCH Interrupt

An ATTCH interrupt is generated when J-state or K-state of the full-speed signal level is detected on the USB port for 2.5 μ s with the host controller selected. To be more specific, an ATTCH interrupt is detected on any of the following conditions.

- When K-state, SE0, or SE1 changes to J-state, and J-state continues 2.5 μ s.
- When J-state, SE0, or SE1 changes to K-state, and K-state continues 2.5 μ s.

40.3.3.15 EOFERR Interrupt

An EOFERR interrupt is generated when it is detected that communication is not completed at the EOF2 timing prescribed in USB Specification 2.0.

After detecting an EOFERR interrupt, the USB controls hardware as described below (irrespective of the value set in the corresponding interrupt enable bit). All pipes in which communications are currently carried out for the pertinent port should be terminated by software and perform re-enumeration of the pertinent port.

- Modifies the DVSTCTR0.UACT bit for the port in which an EOFERR interrupt has been detected to 0.
- Puts the port in which an EOFERR interrupt has been generated into the idle state.

40.3.4 Pipe Control

Table 40.16 lists the pipe settings for the USB. With USB data transfer, data transfer is carried out using the pipe that the software has associated with the endpoint. The USB has ten pipes that are used for data transfer.

Appropriate settings should be made for each of the pipes according to the specifications of the system.

Table 40.16 Pipe Settings

Register Name	Bit Name	Setting	Remarks	
DCPCFG PIPECFG	TYPE[1:0]	Specifies the transfer type	Pipes 1 to 9:	Can be set
	BFRE	Selects the BRDY interrupt mode	Pipes 1 to 5:	Can be set
	DBLB	Selects double buffer mode	Pipes 1 to 5:	Can be set
	DIR	Selects transfer direction	IN or OUT can be set	
	EPNUM[3:0]	Endpoint number	Pipes 1 to 9:	Can be set A value other than 0000b should be set when the pipe is used.
	SHTNAK	Selects disabled state for pipe when transfer ends	Pipes 1 and 2:	Can be set (only when bulk transfer has been selected)
			Pipes 3 to 5:	Can be set
DCPMAXP PIPEMAXP	DEVSEL[3:0]	Selects a device	Referenced only when the host controller is selected.	
	MXPS[8:0]	Maximum packet size	Compliant with USB Specification 2.0.	
PIPEPERI	IFIS	Buffer flush	Pipes 1 and 2:	Can be set (only when isochronous transfer has been selected)
			Pipes 3 to 9:	Cannot be set
	IITV[2:0]	Interval counter	Pipes 1 and 2:	Can be set (only when isochronous transfer has been selected)
			Pipes 3 to 5:	Cannot be set
			Pipes 6 to 9:	Can be set (only when the host controller has been selected)
DCPCTR PIPEnCTR	BSTS	Buffer status	For the DCP, receive buffer status and transmit buffer status are switched with the ISEL bit.	
	INBUFM	IN buffer monitor	Available only for pipes 1 to 5.	
	SUREQ	SETUP request	Can be set only for the DCP. Can be controlled only when the host controller has been selected.	
	SUREQCLR	SUREQ clear	Can be set only for the DCP. Can be controlled only when the host controller has been selected.	
	ATREPM	Auto response mode	Pipes 1 to 5:	Can be set Can be set only when the function controller has been selected.
	ACLRM	Auto buffer clear	Pipes 1 to 9:	Can be set
	SQCLR	Sequence clear	Clears the data toggle bit.	
	SQSET	Sequence set	Sets the data toggle bit.	
	SQMON	Sequence monitor	Monitors the data toggle bit.	
	PBUSY	Pipe busy status		
	PID[1:0]	Response PID	Refer to section 40.3.4.6, Response PID.	
PIPEnTRE	TRENB	Transaction counter enable	Pipes 1 to 5:	Can be set
	TRCLR	Current transaction counter clear	Pipes 1 to 5:	Can be set
PIPEnTRN	—	Transaction counter	Pipes 1 to 5:	Can be set

40.3.4.1 Pipe Control Register Switching Procedures

The following bits in the pipe control registers can be modified only when USB communication is prohibited (PID[1:0] = 00b (NAK)).

The following shows the registers and bits that should not be modified when USB communication is enabled (PID[1:0] = 01b (BUF)).

- Bits in the DCPCFG and DCPMAXP registers
- The SQCLR and SQSET bits in the DCPCTR register
- Bits in registers PIPECFG, PIPEMAXP, and PIPEPERI
- The ATREPM, ACLRM, SQCLR, and SQSET bits in the PIPEnCTR register
- Bits in the PIPEnTRE and PIPEnTRN registers

In order to modify the above bits in the USB communication enabled (PID[1:0] = 01b (BUF)) state, follow the procedure shown below:

1. A request to modify bits in the pipe control register occurs.
2. Modify the PID[1:0] bits corresponding to the pipe to 00b (NAK).
3. Wait until the corresponding PBUSY flag is set to 0.
4. Modify the bits in the pipe control register.

The following bits in the pipe control registers can be modified only when the pertinent pipe information has not been set by the CURPIPE[3:0] bits in registers CFIFOSEL, D0FIFOSEL, and D1FIFOSEL.

Registers that should not be set when the CURPIPE[3:0] bits are set:

- Bits in the DCPCFG and DCPMAXP register
- Bits in registers PIPECFG, PIPEMAXP and PIPEPERI

In order to modify pipe information, the CURPIPE[3:0] bits in the port select registers should be set to a pipe other than the pipe to be modified. For the DCP, the buffer should be cleared using the BCLR bit in the port control register after the pipe information is modified.

40.3.4.2 Transfer Types

The PIPECFG.TYPE[1:0] bits are used to specify the transfer type for each pipe. The transfer types that can be set for the pipes are as follows.

- DCP: No setting is necessary (fixed at control transfer).
- Pipes 1 and 2: These should be set to bulk transfer or isochronous transfer.
- Pipes 3 to 5: These should be set to bulk transfer.
- Pipes 6 to 9: These should be set to interrupt transfer.

40.3.4.3 Endpoint Number

The PIPECFG.EPNUM[3:0] bits are used to set the endpoint number for each pipe. The DCP is fixed at endpoint 0. The other pipes can be set from endpoint 1 to endpoint 15.

- DCP: No setting is necessary (fixed at endpoint 0).
- Pipes 1 to 9: The endpoint numbers from 1 to 15 should be selected and set.
These should be set so that the combination of the PIPECFG.DIR bit and EPNUM[3:0] bits is unique.

40.3.4.4 Maximum Packet Size Setting

The DCPMAXP.MXPS[6:0] bits and the PIPEMAXP.MXPS[8:0] bits are used to specify the maximum packet size for each pipe. DCP and pipes 1 to 5 can be set to any of the maximum pipe sizes defined by USB Specification 2.0. For pipes 6 to 9, 64 bytes are the upper limit of the maximum packet size. The maximum packet size should be set before beginning the transfer (PID[1:0] = 01b (BUF)).

- DCP: Set 8, 16, 32, or 64.
- Pipes 1 to 5: Set 8, 16, 32, or 64 when using bulk transfer.
- Pipes 1 and 2: Set a value between 1 and 256 when using isochronous transfer.
- Pipes 6 to 9: Set a value between 1 and 64.

40.3.4.5 Transaction Counter (For Pipes 1 to 5 in Reading Direction)

When the specified number of transactions has been completed in the data packet receiving direction, the USB recognizes that the transfer has ended. Two transaction counters are provided: one is the PIPEnTRN register that specifies the number of transactions to be executed and the other is the current counter that internally counts the number of executed transactions. With the PIPECFG.SHTNAK bit set to 1, when the current counter value matches the specified number of transactions, the corresponding PIPEnCTR.PID[1:0] bits are set to 00b (NAK) and the subsequent transfer is disabled. The transactions can be counted again from the beginning by initializing the current counter of the transaction counter function through the PIPEnTRE.TRCLR bit. The information read from PIPEnTRN differs depending on the setting of the PIPEnTRE.TRENB bit.

- The TRENB bit = 0: The specified transaction counter value can be read.
- The TRENB bit = 1: The current counter value indicating the internally counted number of executed transactions can be read.

When operating the TRCLR bit, the following should be noted.

- If the transactions are being counted and PID[1:0] = 01b (BUF), the current counter cannot be cleared.
- If there is any data left in the buffer, the current counter cannot be cleared.

40.3.4.6 Response PID

The PID[1:0] bits in the DCPCTR and PIPEnCTR registers are used to set the response PID for each pipe. The following shows the USB operation with various response PID settings:

(1) Response PID settings when the host controller is selected:

The response PID is used to specify the execution of transactions.

- NAK setting: Using pipes is disabled. No transaction is executed.
- BUF setting: Transactions are executed based on the status of the buffer memory.
For OUT direction: If there are transmit data in the buffer memory, an OUT token is issued.
For IN direction: If there is an area to receive data in the buffer memory, an IN token is issued.
- STALL setting: Using pipes is disabled. No transaction is executed.

Note: Setup transactions for the DCP are set with the DCPCTR.SUREQ bit.

(2) Response PID settings when the function controller is selected:

The response PID is used to specify the response to transactions from the host.

- NAK setting: The NAK response is returned in response to the generated transaction.
- BUF setting: Responses are made to transactions according to the status of the buffer memory.
- STALL setting: The STALL response is returned in response to the generated transaction.

Note: For setup transactions, an ACK response is returned regardless of the PID[1:0] bits setting, and the USB request is stored in the register.

The USB may write to the PID[1:0] bits, depending on the results of the transaction as described below.

(3) When the host controller has been selected and the response PID is set by hardware:

- NAK setting: In the following cases, PID[1:0] = 00b (NAK) is set and issuing of tokens is automatically stopped:
When a transfer other than isochronous transfer has been performed and an NRDY interrupt is generated.
(For details, refer to section 40.3.3.2, NRDY Interrupt.)
- If a short packet is received when the PIPECFG.SHTNAK bit has been set to 1 for bulk transfer.
- If the transaction counting ends when the SHTNAK bit has been set to 1 for bulk transfer.
- BUF setting: There is no BUF writing by the USB.
- STALL setting: In the following cases, PID[1:0] = 1xb (STALL) is set and issuing of tokens is automatically stopped:
When STALL is received in response to the transmitted token.
When the size of the receive data packet exceeds the maximum packet size.

(4) When the function controller has been selected and the response PID is set by hardware:

- NAK setting: In the following cases, PID[1:0] = 00b (NAK) is set and NAK is returned in response to transactions:
When the SETUP token is received normally (DCP only).
If the transaction counting ends or a short packet is received when the PIPECFG.SHTNAK bit has been set to 1 for bulk transfer.
- BUF setting: There is no BUF writing by the USB.
- STALL setting: In the following cases, PID[1:0] = 1xb (STALL) is set and STALL is returned in response to transactions:
When a maximum packet size exceeded error is detected in the received data packet.
When a control transfer sequence error has been detected (DCP only).

40.3.4.7 Data PID Sequence Bit

The USB automatically toggles the sequence bit in the data PID when data is transferred successfully in the control transfer data stage, bulk transfer, and interrupt transfer. The sequence bit of the next data PID to be transmitted can be confirmed with the SQMON flag in the DCPCTR and PIPEnCTR registers. When data is transmitted, the sequence bit switches at the timing of ACK handshake reception. When data is received, the sequence bit switches at the timing of ACK handshake transmission. The DCPCTR.SQCLR bit and the PIPEnCTR.SQSET bit can be used to change the data PID sequence bit.

When the function controller has been selected and control transfer is used, the USB automatically sets the sequence bit when a stage transition is made. DATA1 is returned when the setup stage is ended. The sequence bit is not referenced and PID = DATA1 is returned in a status stage. Therefore, software settings are not required. However, when the host controller has been selected and control transfer is used, the sequence bit should be set by software at a stage transition. For the ClearFeature request transmission or reception, the data PID sequence bit should be set by software regardless of whether the host controller or function controller is selected.

40.3.4.8 Response PID = NAK Function

The USB has a function that disables pipe operation (response PID = NAK) at the timing at which the final data packet of a transaction is received (the USB automatically distinguishes this based on reception of a short packet or the transaction counter) by setting the PIPECFG.SHTNAK bit to 1.

When the double buffer mode is being used for the buffer memory, using this function enables reception of data packets in transfer units. If pipe operation has been disabled, software should set the pipe to the enabled state again (response PID = BUF).

The response PID = NAK function can be used only when bulk transfers are used.

40.3.4.9 Auto Response Mode

With the pipes for bulk transfer (pipe 1 to pipe 5), when the PIPEnCTR.ATREPM bit is set to 1, a transition is made to auto response mode. During an OUT transfer (the PIPECFG.DIR bit = 0), OUT-NAK mode is entered, and during an IN transfer (the DIR bit = 1), null auto response mode is entered.

40.3.4.10 OUT-NAK Mode

With the pipes for bulk OUT transfer, NAK is returned in response to an OUT token and an NRDY interrupt is output when the PIPEnCTR.ATREPM bit is set to 1. To make a transition from normal mode to OUT-NAK mode, OUT-NAK mode should be specified in the pipe operation disabled state (response PID = NAK) before enabling pipe operation (response PID = BUF). After pipe operation has been enabled, OUT-NAK mode becomes valid. However, if an OUT token is received immediately before pipe operation is disabled, the token data is normally received, and an ACK is returned to the host.

To make a transition from OUT-NAK mode to normal mode, OUT-NAK mode should be canceled in the pipe operation disabled state (response PID = NAK) before enabling pipe operation (response PID = BUF). In normal mode, reception of OUT data is enabled.

40.3.4.11 Null Auto Response Mode

With the pipes for bulk IN transfer, zero-length packets are continuously transmitted when the PIPEnCTR.ATREPM bit is set to 1.

To make a transition from normal mode to null auto response mode, null auto response mode should be set in the pipe operation disabled state (response PID = NAK) before enabling pipe operation (response PID = BUF). After pipe operation has been enabled, null auto response mode becomes valid. Before setting null auto response mode, the PIPEnCTR.INBUFM = 0 should be confirmed because the mode can be set only when the buffer is empty. If the INBUFM flag is 1, the buffer should be emptied with the PIPEnCTR.ACLRM bit. While a transition to null auto response mode is being made, data should not be written from the FIFO port.

To make a transition from null auto response mode to normal mode, pipe operation disabled state (response PID = NAK) should be retained for the period of zero-length packet transmission (about 10 μ s) before canceling null auto response mode. In normal mode, data can be written from the FIFO port; therefore, packet transmission to the host is enabled by enabling pipe operation (response PID = BUF).

40.3.5 FIFO Buffer Memory

40.3.5.1 FIFO Buffer Memory

The USB has FIFO buffer memory for data transfer. The memory area used for each pipe is managed by the USB. The FIFO buffer memory has two states depending on whether the access right is assigned to the system (CPU side) or the USB (SIE side).

(1) Buffer Status

Table 40.17 and Table 40.18 show the buffer status in the USB. The buffer memory status can be confirmed using the DCPCTR.BSTS flag and the PIPEnCTR.INBUFM flag. The transfer direction for the buffer memory can be specified using either the PIPECFG.DIR bit or the CFIFOSEL.ISEL bit (when DCP is selected).

The INBUFM flag is valid for pipe 0 to pipe 5 in the transmitting direction.

When a transmitting pipe uses the double buffer configuration, software can read the BSTS flag to monitor the buffer memory status on the CPU side and the INBUFM flag to monitor the buffer memory status on the SIE side. When the BEMP interrupt may not show the buffer empty status because the write access to the FIFO port by the CPU or DMAC/DTC is slow, software can use the INBUFM flag to confirm the end of transmission.

Table 40.17 Buffer Status Indicated by the BSTS Flag

ISEL or DIR	BSTS	Buffer Memory Status
0 (receiving direction)	0	There is no received data, or data is being received. Reading from the FIFO port is disabled.
0 (receiving direction)	1	There is received data, or a zero-length packet has been received. Reading from the FIFO port is allowed. Note that when a zero-length packet is received, reading is not possible and the buffer must be cleared.
1 (transmitting direction)	0	The transmission has not been completed. Writing to the FIFO port is disabled.
1 (transmitting direction)	1	The transmission has been completed. CPU write is allowed.

Table 40.18 Buffer Status Indicated by the INBUFM Flag

DIR	INBUFM	Buffer Memory Status
0 (receiving direction)	Invalid	Invalid
1 (transmitting direction)	0	The transmission has been completed. There is no waiting data to be transmitted.
1 (transmitting direction)	1	The FIFO port has written data to the buffer. There is data to be transmitted.

40.3.5.2 FIFO Buffer Clearing

Table 40.19 shows the clearing of the FIFO buffer memory by the USB. The buffer memory can be cleared using the BCLR, DnFIFOSEL.DCLRM, and PIPEnCTR.ACLRM bit in the port control register.

Either a single or double buffer configuration can be selected for pipes 1 to 5, using the PIPECFG.DBLB bit.

Table 40.19 List of Buffer Clearing Methods

FIFO Buffer Clearing Mode	Clearing Buffer Memory on CPU Side	Mode for Automatically Clearing Buffer Memory after Reading Specified Pipe Data	Auto Buffer Clear Mode for Discarding All Received Packets
Register used	CFIFOCTR DnFIFOCTR	DnFIFOSEL	PIPEnCTR
Bit used	BCLR	DCLRM	ACLRM
Clearing condition	Cleared by writing 1	1: Mode valid 0: Mode invalid	1: Mode valid 0: Mode invalid

(1) Auto Buffer Clear Mode Function

With the USB, all received data packets are discarded if the PIPEnCTR.ACLRM bit is set to 1. If a correct data packet has been received, the ACK response is returned to the host controller. The auto buffer clear mode function can be set only in the buffer memory reading direction.

If the ACLRM bit is set to 1 and then to 0, the buffer memory of the selected pipe can be cleared regardless of the access direction.

An access cycle of at least 100 ns is required for the internal hardware sequence processing time between ACLRM = 1 and ACLRM = 0.

40.3.5.3 FIFO Port Functions

Table 40.20 shows the settings for the FIFO port functions of the USB. In write access, writing data until the maximum packet size is reached automatically enables transmission of the data. To enable transmission before the maximum packet size is reached, the BVAL bit in the port control register should be set to end writing. To send a zero-length packet, the BCLR bit in the register should be used to clear the buffer and then the BVAL bit set in order to end writing.

In reading, reception of new packets is automatically enabled when all data has been read. Data cannot be read when a zero-length packet has been received (the DTLN[8:0] flags = 0), so the BCLR bit in the register should be used to clear the buffer. The length of the receive data can be confirmed using the DTLN[8:0] flags in the port control register.

Table 40.20 FIFO Port Function Settings

Register	Bit	Description
CFIFOSEL, DnFIFOSEL (n = 0, 1)	RCNT	Selects DTLN[8:0] read mode.
	REW	Buffer memory rewind (re-read, rewrite).
	DCLRM	Automatically clears receive data for a specified pipe after the data has been read (only for DnFIFO).
	DREQE	Enables DMA/DTC transfers (only for DnFIFO).
	MBW	FIFO port access bit width.
	BIGEND	Selects FIFO port endian.
	ISEL	FIFO port access direction (only for DCP).
	CURPIPE[3:0]	Selects the current pipe.
CFIFOCTR, DnFIFOCTR (n = 0, 1)	BVAL	Ends writing to the buffer memory.
	BCLR	Clears the buffer memory on the CPU side.
	DTLN[8:0]	Checks the length of receive data.

(1) FIFO Port Selection

Table 40.21 shows the pipes that can be selected with the various FIFO ports. The pipe to be accessed should be selected using the CURPIPE[3:0] bits in the port select register. After the pipe is selected, whether the written value can be correctly read from the CURPIPE[3:0] bits should be checked. (If the previous pipe number is read, it indicates that the pipe modification is being executed by the USB controller.) Then, the FRDY flag in a port control register = 1 is checked.

In addition, the bus width to be accessed should be selected using the MBW bit in the port select register. The buffer memory access direction conforms to the PIPECFG.DIR bit. Only for the DCP, the ISEL bit in the port select register determines the direction.

Table 40.21 FIFO Port Access Categorized by Pipe

Pipe	Access Method	Port that can be Used
DCP	CPU access	CFIFO port register
Pipe 1 to pipe 9	CPU access	CFIFO port register D0FIFO/D1FIFO port register
	DMAC/DTC access	D0FIFO/D1FIFO port register

(2) REW Bit

It is possible to temporarily stop access to the pipe currently being accessed, access a different pipe, and then continue processing for the current pipe again. The REW bit in the port select register is used for this processing.

If a pipe is selected through the CURPIPE[3:0] bits in the port select register with the REW bit set to 1, the pointer used for reading from and writing to the buffer memory is reset, and reading or writing can be carried out from the first byte.

If a pipe is selected with 0 set for the REW bit, data can be read and written in continuation from the previous selection, without the pointer being reset.

To access the FIFO port, the FRDY flag in the port control register = 1 should be checked after selecting a pipe.

40.3.5.4 DMA Transfers (D0FIFO and D1FIFO Ports)

(1) Overview of DMA Transfers

For pipes 1 to 9, the FIFO port can be accessed using the DMAC. When accessing the buffer for the pipe targeted for DMA transfer is enabled, a DMA transfer request is issued.

The unit of transfer to the FIFO port should be selected using the DnFIFOSEL.MBW bit and the pipe targeted for the DMA transfer should be selected using the DnFIFOSEL.CURPIPE[3:0] bits. The selected pipe should not be changed during the DMA transfer.

(2) DnFIFO Auto Clear Mode (D0FIFO and D1FIFO Port Reading Direction)

If 1 is set in the DnFIFOSEL.DCLRM bit, the USB automatically clears the buffer memory of the selected pipe when reading of data from the buffer memory has been completed.

Table 40.22 shows the packet reception and buffer memory clearing processing by software for each of the various settings. As shown in Table 40.22, the buffer clearing conditions depend on the value set in the PIPECFG.BFRE bit. Using the DnFIFOSEL.DCLRM bit eliminates the need for the buffer to be cleared by software in any situation that requires buffer clearing. This enables DMA transfers without involving software.

The DnFIFO auto clear mode can be set only in the buffer memory reading direction.

Table 40.22 Packet Reception and Buffer Memory Clearing Processing by Software

Buffer Status When Packet is Received	Register Setting			
	DCLRM = 0		DCLRM = 1	
	BFRE = 0	BFRE = 1	BFRE = 0	BFRE = 1
Buffer full	Clearing is not necessary	Clearing is not necessary	Clearing is not necessary	Clearing is not necessary
Zero-length packet reception	Clearing is necessary	Clearing is necessary	Clearing is not necessary	Clearing is not necessary
Normal short packet reception	Clearing is not necessary	Clearing is necessary	Clearing is not necessary	Clearing is not necessary
Transaction count end	Clearing is not necessary	Clearing is necessary	Clearing is not necessary	Clearing is not necessary

40.3.6 Control Transfers Using DCP

In the data stage of control transfers, data is transferred using the default control pipe (DCP).

The DCP buffer memory is a 64-byte single buffer and is a fixed area that is shared for both control reading and control writing. The buffer memory can be accessed only through the CFIFO port.

40.3.6.1 Control Transfers When the Host Controller is Selected

(1) Setup Stage

Registers USBREQ, USBVAL, USBINDX, and USBLENG are the registers that are used to transmit a USB request for setup transactions. Writing setup packet data to the registers and writing 1 to the DCPCTR.SUREQ bit transmits the specified data for setup transactions. Upon completion of the transaction, the SUREQ bit is set to 0. The above USB request registers should not be modified while SUREQ = 1.

After the attached state of the connected function device is detected, the first setup transaction for the device should be issued by using the sequence described above with the DCPMAXP.DEVSEL[3:0] bits set to 0 and the DEVADD0.USBSPPD[1:0] bits set appropriately.

After the connected function device is shifted to the Address state, setup transactions should be issued by using the sequence described above with the assigned USB address set in the DEVSEL[3:0] bits and the bits in the DEVADDn register corresponding to the specified USB address set appropriately. For example, when PIPEMAXP.DEVSEL[3:0] = 0010b, make appropriate settings in the DEVADD2 register; when PIPEMAXP.DEVSEL[3:0] = 0101b, make appropriate settings in the DEVADD5 register.

When the setup transaction data has been sent, an interrupt request is generated according to the response received from the peripheral device (SIGN or SACK flag in the INTSTS1 register), by means of which the result of the setup transactions can be confirmed.

A data packet of DATA0 (USB request) is transmitted as the data packet for a setup transaction regardless of the setting of the DCPCTR.SQMON flag.

(2) Data Stage

Data is transferred using the DCP buffer memory.

The access direction of the DCP buffer memory should be specified using the CFIFOSEL.ISEL bit. The transfer direction should be specified using the DCPCFG.DIR bit.

For the first data packet of the data stage, the data PID should be transferred as DATA1. Set data PID = DATA1 in the DCPCTR.SQSET bit and the PID[1:0] bits = 01b (BUF). Completion of data transfer is detected using the BRDY or BEMP interrupt.

For control write transfers, when the number of data bytes to be sent is an integer multiple of the maximum packet size, software should control so as to send a zero-length packet at the end.

(3) Status Stage

Zero-length packet data is transferred in the direction opposite to that in the data stage. As in the data stage, data is transferred using the DCP buffer memory. Transactions are done in the same manner as the data stage.

For the data packets of the status stage, the data PID should be set to DATA1 using the DCPCTR.SQSET bit.

For reception of a zero-length packet, the received data length should be confirmed using the CFIFOCTR.DTLN[8:0] flags after a BRDY interrupt is generated, and the buffer memory should then be cleared using the CFIFOCTR.BCLR bit.

40.3.6.2 Control Transfers When the Function Controller is Selected

(1) Setup Stage

The USB sends an ACK response for a correct setup packet targeted to the USB. The operation of the USB in the setup stage is described below.

When receiving a new setup packet, the USB sets the following bits.

- Set the INTSTS0.VALID flag to 1.
- Set the DCPCTR.PID[1:0] bits to 00b (NAK).
- Set the DCPCTR.CCPL bit to 0.

When receiving a data packet right after the setup packet, the USB stores the USB request parameters in registers USBREQ, USBVAL, USBINDX, and USBLENG.

Response processing with respect to the control transfer should be carried out after setting the VALID flag = 0. In the VALID flag = 1 state, PID[1:0] = 01b (BUF) cannot be set, and the data stage cannot be terminated.

Using the function of the VALID flag, the USB can suspend the current request processing when receiving a new USB request during a control transfer, and can send a response to the newest request.

In addition, the USB automatically detects the direction bit (bit 8 of bmRequestType) and the request data length (wLength) of the received USB request, distinguishes between control read transfer, control write transfer, and no-data control transfer, and controls stage transitions. For a wrong sequence, the sequence error of the control transfer stage transition interrupt is generated, and the software is notified of occurrence of the error. For the stage control of the USB, refer to Figure 40.15.

(2) Data Stage

Data transfers corresponding to received USB requests should be done using the DCP. Before accessing the DCP buffer memory, the access direction should be specified using the CFIFOSEL.ISEL bit.

If the transfer data is larger than the size of the DCP buffer memory, the data transfer should be carried out using the BRDY interrupt for control write transfers and the BEMP interrupt for control read transfers.

(3) Status Stage

Control transfers are terminated by setting the DCPCTR.CCPL bit to 1 while the DCPCTR.PID[1:0] bits are set to 01b (BUF).

After the above settings have been made, the USB automatically executes the status stage in accordance with the data transfer direction determined at the setup stage. The specific procedure is as follows.

- For control read transfers
A zero-length packet is received from the USB host and an ACK response is sent.
- For control write transfers and no-data control transfers
A zero-length packet is transmitted and an ACK response is received from the USB host.

(4) Control Transfer Auto Response Function

The USB automatically responds to a correct SET_ADDRESS request. If any of the following errors occurs in the SET_ADDRESS request, a response from the software is necessary.

- bmRequestType is not 00h: Any transfer other than a control write transfer
- wIndex is not 00h: Request error
- wIndex is not 00h: Any transfer other than a no-data control transfer
- wValue is larger than 7Fh: Request error
- INTSTS0.DVSQ[2:0] are 011b (Configured state): Control transfer of a device state error

For all requests other than the SET_ADDRESS request, a response is required from the corresponding software.

40.3.7 Bulk Transfers (Pipes 1 to 5)

The buffer memory usage (single/double buffer setting) can be selected for bulk transfers. The USB provides the following functions for bulk transfers.

- BRDY interrupt function (PIPECFG.BFRE bit: refer to section 40.3.3.1, (2) When the SOFCFG.BRDYM Bit = 0 and the PIPECFG.BFRE Bit = 1)
- Transaction count function
(PIPEnTRE.TRENB and TRCLR bits and PIPEnTRN register: refer to section 40.3.4.5, Transaction Counter (For Pipes 1 to 5 in Reading Direction))
- Response PID = NAK function (PIPECFG.SHTNAK bit: refer to section 40.3.4.8, Response PID = NAK Function)
- Auto response mode (PIPEnCTR.ATREPM bit: refer to section 40.3.4.9, Auto Response Mode)

40.3.8 Interrupt Transfers (Pipes 6 to 9)

When the function controller is selected, the USB carries out interrupt transfers in accordance with the timing controlled by the host controller.

When the host controller is selected, the timing of issuing a token can be specified using the interval counter.

40.3.8.1 Interval Counter during Interrupt Transfers When the Host Controller is Selected

For interrupt transfers, intervals between transactions are set in the PIPEPERI.IITV[2:0] bits. The USB controller issues interrupt transfer tokens based on the specified intervals.

(1) Counter Initialization

The interval counter is initialized when the MCU is reset or when the PIPEnCTR.ACLRM bit is set to 1. Note that the PIPEPERI.IITV[2:0] bits are not initialized when the ACLRM bit is used for initialization.

Note that the interval counter is not initialized in the following case.

- USB bus reset or USB suspended
The IITV[2:0] bits are not initialized. Setting 1 to the DVSTCTR0.UACT bit starts counting from the value before entering the USB bus reset state or USB suspended state.

(2) Operation When Transmission/Reception is Impossible at Token Issuance Timing

The USB cannot issue tokens even at token issuance timing in the following cases. In such a case, the USB attempts transactions at the subsequent interval.

- When the PID[1:0] bits are set to 00b (NAK) or 1xb (STALL).
- When the buffer memory is full at the token sending timing in the receiving (IN) direction.
- When there is no data to be sent in the buffer memory at the token sending timing in the transmitting (OUT) direction.

40.3.9 Isochronous Transfers (Pipes 1 and 2)

The USB has the following functions for isochronous transfers.

- Notification of isochronous transfer error information
- Interval counter (specified by the PIPEPERI.IITV[2:0] bits)
- Isochronous IN transfer data setup control (IDLX function)
- Isochronous IN transfer buffer flush function (specified by the PIPEPERI.IFIS bit)

40.3.9.1 Error Detection in Isochronous Transfers

The USB has a function for detecting the error information described below, so that when errors occur in isochronous transfers, they can be controlled by software. Table 40.23 and Table 40.24 show the priority in which errors are confirmed and the interrupts generated corresponding to errors.

(a) PID errors

- If the PID of the received packet is illegal.

(b) CRC errors and bit stuffing errors

- If an error occurs in the CRC of the received packet or the bit stuffing is illegal.

(c) Maximum packet size exceeded

- The data of the received packet is larger than the specified maximum packet size.

(d) Overrun and underrun errors

- When the host controller is selected
When the buffer memory is full at the token sending timing in the IN (receiving) direction.
When there is no data to be sent in the buffer memory at the token sending timing in the OUT (transmitting) direction.
- When the function controller is selected
When there is no data to be sent in the buffer memory at the token receiving timing in the IN (transmitting) direction.
When the buffer memory is full at the token receiving timing in the OUT (receiving) direction.

(e) Interval errors

An interval error is generated on any of the following conditions when the function controller is selected.

- During an isochronous IN transfer, an IN token could not be received in the interval frame.
- During an isochronous OUT transfer, an OUT token could not be received in the interval frame.

Table 40.23 Error Detection When a Token is Received

Detection Priority	Error	Generated Interrupt and Status
1	PID errors	No interrupts are generated in both cases when the host controller is selected and the function controller is selected (ignored as a corrupted packet).
2	CRC errors and bit stuffing errors	No interrupts generated in both cases when the host controller is selected and the function controller is selected (ignored as a corrupted packet).
3	Overrun and underrun errors	An NRDY interrupt is generated to set the FRMNUM.OVRN flag to 1 in both cases when the host controller is selected and function controller is selected. When the function controller is selected, a zero-length packet is transmitted in response to IN token. However, no data packets are received in response to OUT token.
4	Interval errors	An NRDY interrupt is generated when the function controller is selected. It is not generated when the host controller is selected.

Table 40.24 Error Detection When a Data Packet is Received

Detection Priority	Error	Generated Interrupt and Status
1	PID errors	No interrupts are generated (ignored as a corrupted packet).
2	CRC errors and bit stuffing errors	An NRDY interrupt is generated to set the FRMNUM.CRCE to 1 bit in both cases when the host controller is selected and the function controller is selected.
3	Maximum packet size exceeded errors	A BEMP interrupt is generated to set the PID[1:0] bits to 1xb (STALL) in both cases when the host controller is selected and the function controller is selected.

40.3.9.2 Data PID

When the function controller is selected, the USB operates as follows in response to the received PID.

IN direction

- DATA0: Sent as data packet PID
- DATA1: Not sent
- DATA2: Not sent
- mData: Not sent

OUT direction

- DATA0: Received normally as data packet PID
- DATA1: Received normally as data packet PID
- DATA2: Packets are ignored
- mData: Packets are ignored

40.3.9.3 Interval Counter

The isochronous transfer interval can be set using the PIPEPERI.IITV[2:0] bits. The interval counter enables the functions shown in Table 40.25 when the function controller is selected. When the host controller is selected, the USB generates the token issuance timing. When the host controller is selected, the interval counter operation is the same as that in the interrupt transfer.

Table 40.25 Interval Counter Function When the Function Controller is Selected

Transfer Direction	Function	Conditions for Detection
IN	Flushes transmit buffer	When an IN token cannot be successfully received in the interval frame during an isochronous IN transfer
OUT	Notifies that a token not being received	When an OUT token cannot be successfully received in the interval frame during an isochronous OUT transfer

The interval count is carried out when an SOF is received or for interpolated SOFs, so the isochronism can be maintained even if an SOF is damaged. The frame interval that can be set is the $2^{IITV[2:0]}$ frames.

(1) Counter Initialization When the Function Controller is Selected

The interval counter is initialized when the MCU is reset or when the PIPEnCTR.ACLRM bit is set to 1. Note that the PIPEPERI.IITV[2:0] bits are not initialized when the ACLRM bit is used for initialization.

After the interval counter has been initialized, counting is started under either of the following conditions 1 and 2 when a packet has been transferred successfully.

1. An SOF is received after transmission of data in response to an IN token while the PID[1:0] bits are 01b (BUF).
2. An SOF is received after reception of data of an OUT token while the PID[1:0] bits are 01b (BUF).

Note that the interval counter is not initialized under the following conditions.

- When the PID[1:0] bits are set to 00b (NAK) or 1xb (STALL)
The interval timer does not stop. The USB attempts transactions at the subsequent interval.
- When the USB bus is reset or USB is suspended
The IITV[2:0] bits are not initialized. When an SOF has been received, counting is restarted from the value prior to the reception of the SOF.

(2) Interval Counting and Transfer Control When the Host Controller is Selected

The USB controls the interval between token issuance operations based on the PIPEPERI.IITV[2:0] bit settings. Specifically, the USB issues a token for a selected pipe once every $2^{IITV[2:0]}$ frames.

The USB starts counting the token issuance interval at the frame following the frame in which the PID[1:0] bits have been set to 01b (BUF) by software.

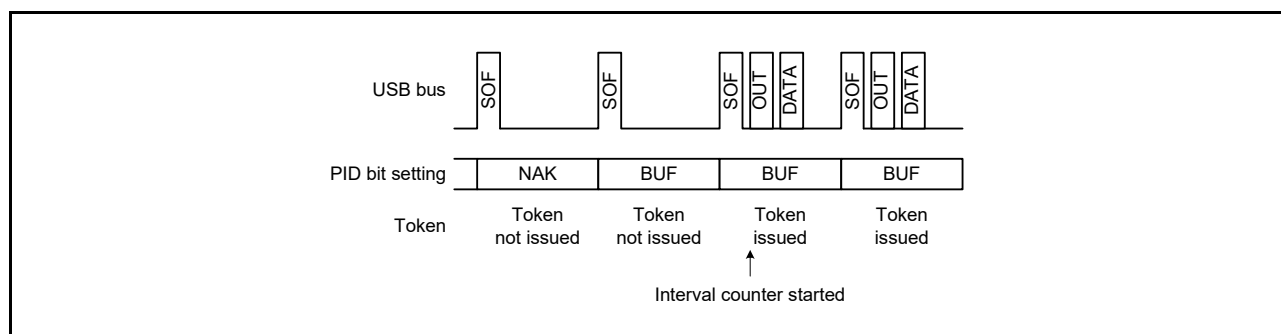


Figure 40.16 Token Issuance When IITV[2:0] = 000b

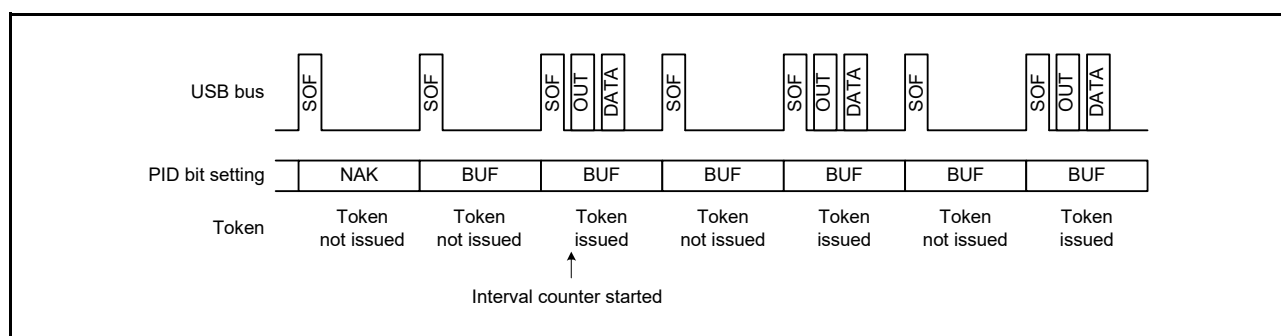


Figure 40.17 Token Issuance When IITV[2:0] = 001b

When the selected pipe is set for isochronous transfers, the USB carries out the following operation in addition to controlling the token issuance interval. The USB issues a token even when the NRDY interrupt generation condition is satisfied.

(a) When the selected pipe is for isochronous IN transfers

The USB generates an NRDY interrupt when the USB issues an IN token but does not receive a packet successfully from a peripheral device (no response or packet error).

The USB sets the FRMNUM.OVRN flag to 1 generating an NRDY interrupt when the time to issue an IN token comes while the USB cannot receive data because the FIFO buffer is full (due to the fact that the CPU or DMAC/DTC is too slow to read data from the FIFO buffer).

(b) When the selected pipe is for isochronous OUT transfers

The USB sets the OVRN flag to 1 generating an NRDY interrupt and transmitting a zero-length packet when the time to issue an OUT token comes while there is no data to be transmitted in the FIFO buffer (because the CPU or DMAC/DTC is too slow to write data to the FIFO buffer).

The token issuance interval is reset on any of the following conditions.

- When the USB is reset through a reset pin
(The IITV[2:0] bits are also set to 0).
- When the PIPEnCTR.ACLRM bit has been set to 1 by software.

(3) Interval Counting and Transfer Control When the Function Controller is Selected

(a) When the selected pipe is for isochronous OUT transfers

The USB generates an NRDY interrupt when the USB fails to receive a data packet within the interval set by the PIPEPERL.IITV[2:0] bits.

The USB also generates an NRDY interrupt when the USB fails to receive data because of a CRC error or other errors contained in the data packet or because of the FIFO buffer being full.

The NRDY interrupt is generated at the timing of SOF packet reception. Even if the SOF packet is corrupted, the internal interpolation allows the interrupt to be generated at the timing to receive the SOF packet.

However, when the IITV[2:0] bits are set to a value other than 0, the USB generates an NRDY interrupt on receiving an SOF packet for every interval after starting interval counting operation.

When the PID[1:0] bits are set to 00b (NAK) by software after starting the interval timer, the USB does not generate an NRDY interrupt on receiving an SOF packet.

The timing to start interval counting depends on the setting of IITV[2:0] bits as shown below.

- When the IITV[2:0] = 000b: The interval counting starts when software has set the PID[1:0] bits for the selected pipe to 01b (BUF).

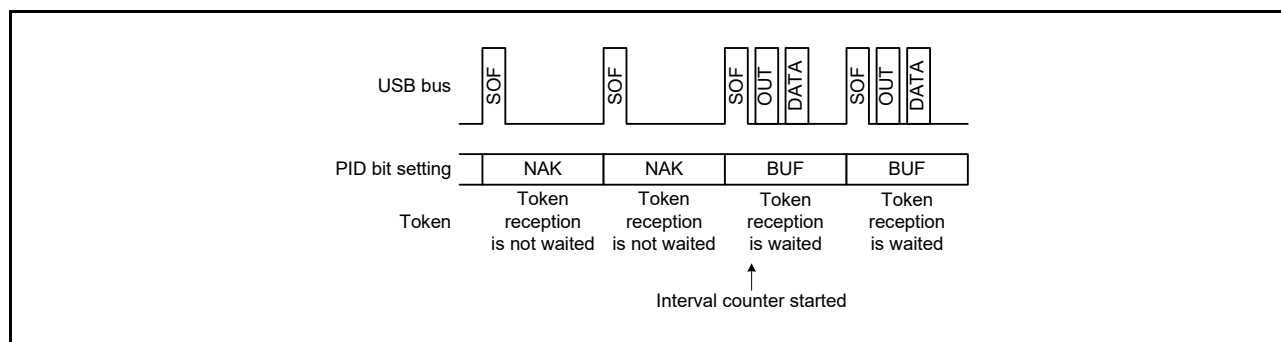


Figure 40.18 Relationship between Frames and Expected Token Reception When IITV[2:0] = 000b

- When the IITV[2:0] $\neq$ 000b: The interval counting starts on completion of successful reception of the first data packet after the PID[1:0] bits for the selected pipe have been modified to 01b (BUF).

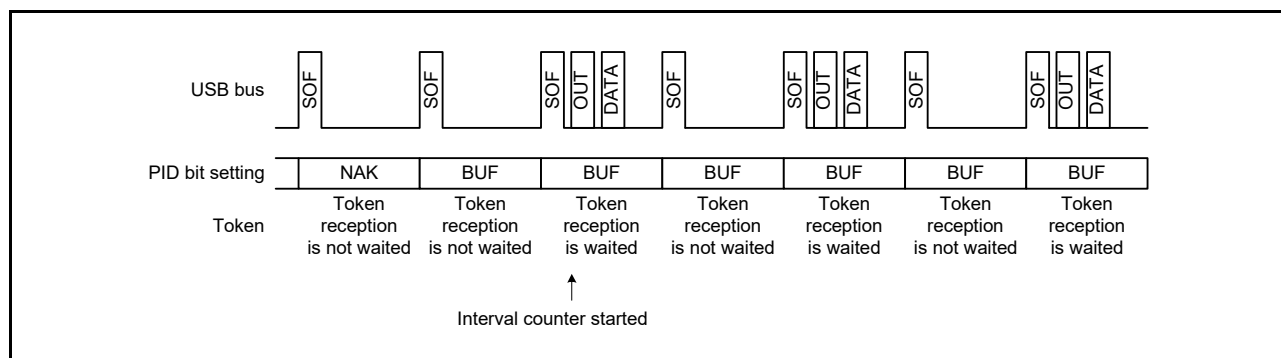


Figure 40.19 Relationship between Frames and Expected Token Reception When IITV[2:0] $\neq$ 000b

(b) When the selected pipe is for isochronous IN transfers

The PIPEPERI.IFIS bit should be 1 for this use. When IFIS = 0, the USB transmits a data packet in response to the received IN token irrespective of the setting of the PIPEPERI.IITV[2:0] bits.

When IFIS = 1, the USB clears the FIFO buffer when the USB fails to receive an IN token in the frame at the interval set by the IITV[2:0] bits while there is data to be transmitted in the FIFO buffer.

The USB also clears the FIFO buffer when the USB fails to receive an IN token successfully because of a bus error such as a CRC error contained in the IN token.

The FIFO buffer is cleared at the timing of SOF packet reception. Even if the SOF packet is corrupted, the internal interpolation allows the FIFO buffer to be cleared at the timing to receive the SOF packet.

The timing to start interval counting depends on the setting of the IITV[2:0] bits (similar to the timing during OUT transfers).

The interval is counted on any of the following conditions in function controller mode.

- When a hardware-reset is applied to the USB (here, the IITV[2:0] bits are also set to 000b).
- When the PIPEnCTR.ACLRM bit is set to 1 by software.
- When the USB detects a USB bus reset.

(4) Setup of Data to be Transmitted Using Isochronous Transfer When the Function Controller is Selected

With isochronous data transmission using the USB in the function controller, after data has been written to the buffer memory, a data packet can be transmitted with the next frame after the frame in which an SOF packet is detected. This function is called the isochronous transfer transmission data setup function, and it makes it possible to designate the frame from which transmission began.

In a double buffer configuration, even after the writing of data to both buffers has been completed, transmission will be enabled for only one buffer to which data writing was completed first. Accordingly, even if multiple IN tokens are received, only one packet of data is transmitted from a single buffer.

When an IN token is received, if the buffer memory is in the transmission enabled state, the USB transmits data as a normal response. If the buffer memory is not in the transmission enabled state, however, a zero-length packet is sent and an underrun error occurs.

Figure 40.20 shows an example of transmission using the isochronous transfer transmission data setup function with the USB when IITV[2:0] = 000b (every frame) has been set.

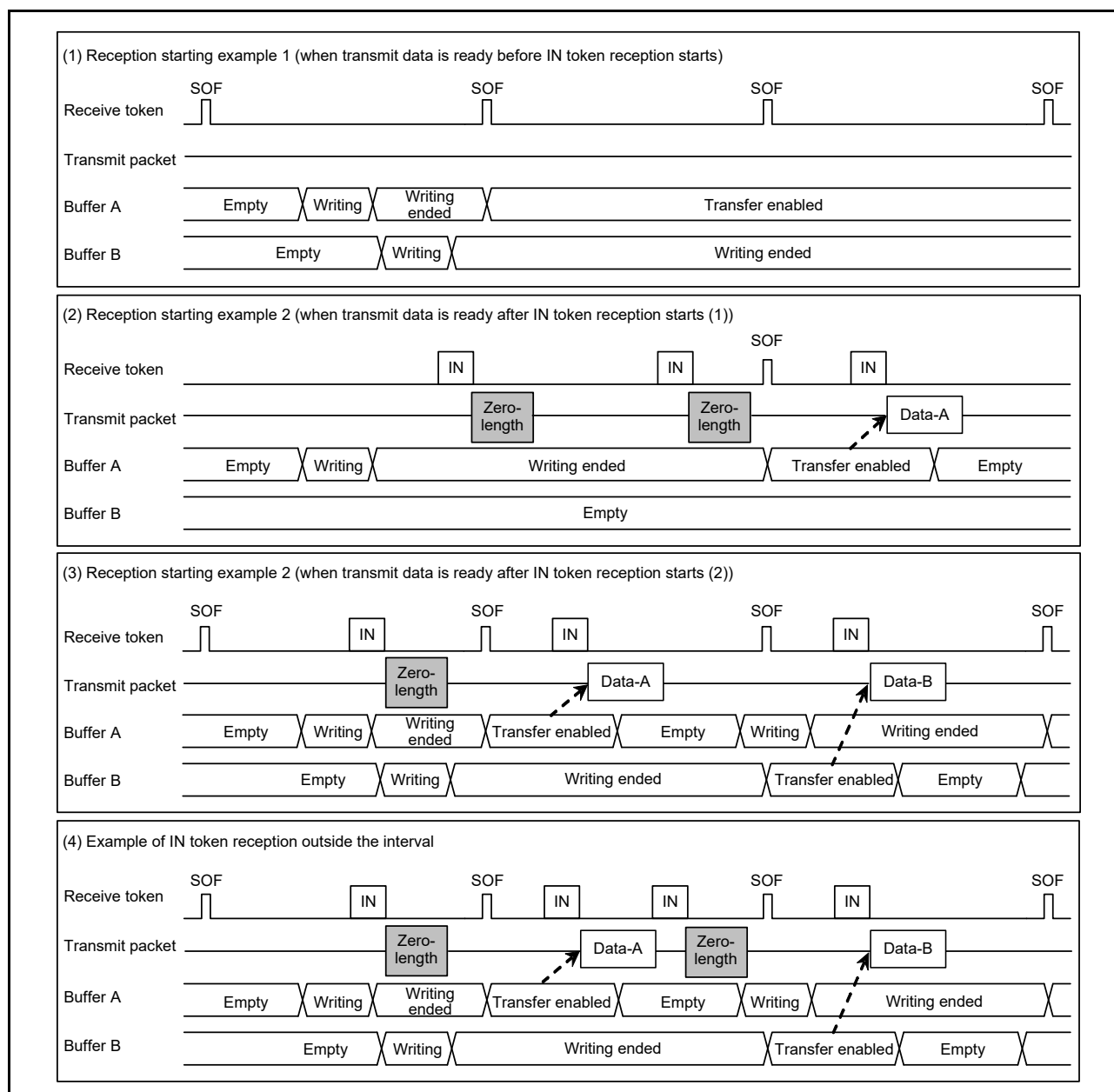


Figure 40.20 Example of Data Setup Function Operation

(5) Isochronous Transfer Transmission Buffer Flush When the Function Controller is Selected

If an SOF packet of the next frame is received without receiving an IN token in an interval frame during isochronous data transmission, the USB operates as if an IN token had been corrupted, and clears the buffer for which transmission is enabled, putting that buffer in the writing enabled state.

If a double buffer configuration is used and writing to both buffers has been completed, the buffer memory that was cleared is assumed as the data having been sent in the interval frame, and transmission is enabled for the buffer memory that is not cleared with SOF packet reception.

The timing of the buffer flush function depends on the setting of the PIPEPERI.IITV[2:0] bits.

- When the IITV[2:0] = 000b
The buffer flush operation starts from the next frame after the pipe becomes valid.
- When the IITV[2:0] ≠ 000b
The buffer flush operation is carried out after the first successful transaction.

Figure 40.21 shows an example of the buffer flush function in the USB. When an unanticipated token is received before the interval frame, the USB sends the write data or a zero-length packet as an underrun error according to the data setup state.

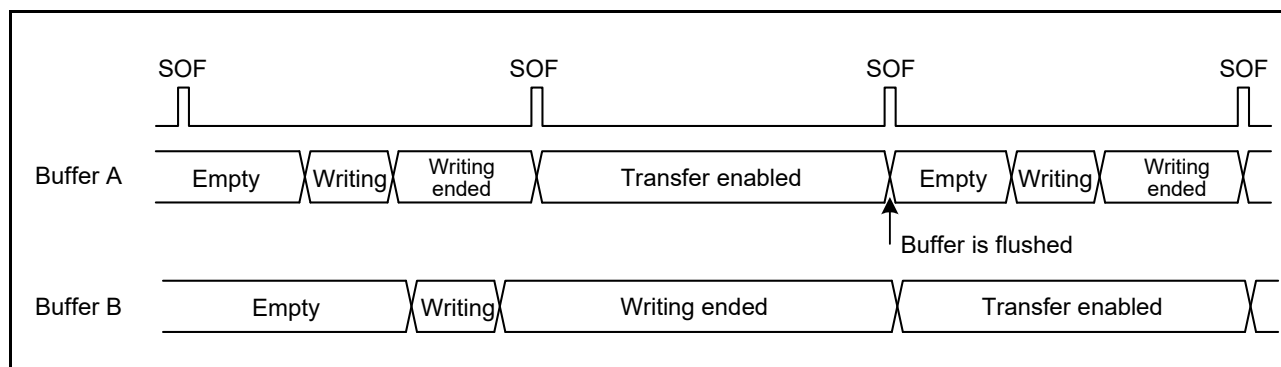


Figure 40.21 Example of Buffer Flush Operation

Figure 40.22 shows an example of interval error occurrence in the USB. There are five types of interval errors, as shown below. The interval error is generated at the timing indicated by ① in the figure, and the buffer flush function is activated.

If an interval error occurs during an IN transfers, the buffer flush function is activated; if it occurs during an OUT transfer, an NRDY interrupt is generated.

The FRMNUM.OVRN flag should be used to distinguish between NRDY interrupts such as received packet errors and overrun errors.

In response to tokens that are shaded in the figure, responses are sent according to the buffer memory status.

IN direction

- If the buffer is in the transmission enabled state, the data is transferred as a normal response.
- If the buffer is in the transmission disabled state, a zero-length packet is sent and an underrun error occurs.

OUT direction

- If the buffer is in the reception enabled state, the data is received as a normal response.
- If the buffer is in the reception disabled state, the data is discarded and an overrun error occurs.

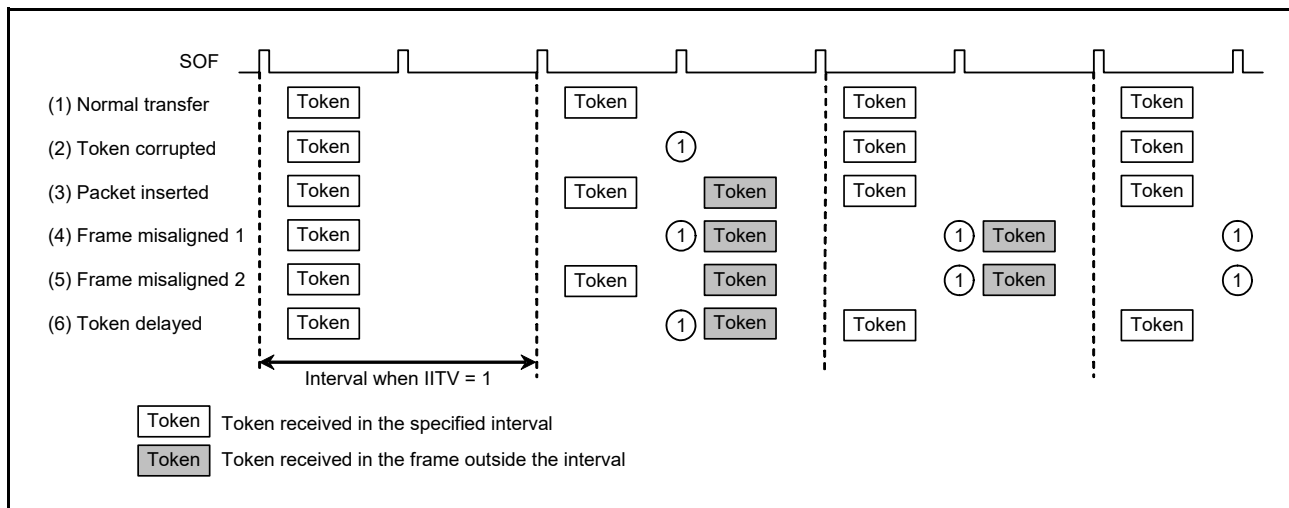


Figure 40.22 Example of Interval Error Occurrence When IITV[2:0] = 001b

40.3.10 SOF Interpolation Function

When the function controller is selected and if data could not be received at intervals of 1 ms because an SOF packet was corrupted or missing, the USB interpolates the SOF. The SOF interpolation operation begins when the USBE and SCKE bits in the SYSCFG register have been set to 1 and an SOF packet is received. The interpolation function is initialized under the following conditions.

- MCU reset
- USB bus reset
- Suspended state detected

The SOF interpolation operates as follows.

- The interpolation function is not activated until an SOF packet is received.
- After the first SOF packet is received, interpolation is carried out by counting 1 ms with an internal clock of 48 MHz.
- After the second and subsequent SOF packets are received, interpolation is carried out at the previous reception interval.
- Interpolation is not carried out in the suspended state or while a USB bus reset is being received.

The USB supports the following functions based on the SOF packet reception. These functions also operate normally with SOF interpolation, if the SOF packet was missing.

- Updating of the frame number
- SOFR interrupt timing
- Isochronous transfer interval count

If an SOF packet is missing during full-speed operation, the FRMNUM.FRNM[10:0] flags are not updated.

40.3.11 Pipe Schedule

40.3.11.1 Conditions for Generating a Transaction

When the host controller is selected and the DVSTCTR0.UACT bit has been set to 1, the USB generates a transaction under the conditions shown in Table 40.26.

Table 40.26 Conditions for Generating a Transaction

Transaction	Conditions for Generation				
	DIR	PID[1:0]	IITV[0]	Buffer State	SUREQ
Setup	—*1	—*1	—*1	—*1	1 setting
Control transfer data stage, status stage, bulk transfer	IN	BUF	Invalid	Receive area exists	—*1
	OUT	BUF	Invalid	Transmit data exists	—*1
Interrupt transfer	IN	BUF	Valid	Receive area exists	—*1
	OUT	BUF	Valid	Transmit data exists	—*1
Isochronous transfer	IN	BUF	Valid	*2	—*1
	OUT	BUF	Valid	*3	—*1

- Note 1. Symbols (—) in the table indicate that the condition is unrelated to the generating of tokens. "Valid" indicates that, for interrupt transfers and isochronous transfers, a transaction is generated only in transfer frames that are based on the interval counter. "Invalid" indicates that a transaction is generated regardless of the interval counter.
- Note 2. This indicates that a transaction is generated regardless of whether there is a receive area. If there is no receive area, however, the received data is discarded.
- Note 3. This indicates that a transaction is generated regardless of whether there is any data to be transmitted. If there is no data to be transmitted, however, a zero-length packet is transmitted.

40.3.11.2 Transfer Schedule

This section describes the transfer scheduling within a frame of the USB. After the USB sends an SOF, the transfer is carried out in the sequence described below.

- Execution of periodic transfers
A pipe is searched in the order of Pipe 1 → Pipe 2 → Pipe 6 → Pipe 7 → Pipe 8 → Pipe 9, and then, if there is a pipe for which an isochronous or interrupt transfer transaction can be generated, the transaction is generated.
- Setup transactions for control transfers
The DCP is checked, and if a setup transaction is possible, it is sent.
- Execution of bulk transfers, control transfer data stages, and control transfer status stages
A pipe is searched in the order of DCP → Pipe 1 → Pipe 2 → Pipe 3 → Pipe 4 → Pipe 5, and then, if there is a pipe for which a transaction for a bulk transfer, a control transfer data stage, or a control transfer status stage can be generated, the transaction is generated.
When a transaction is generated, processing moves to the next pipe transaction regardless of whether the response from the peripheral device is ACK or NAK. If there is time for transfer within the frame, step 3 is repeated.

40.3.11.3 Enabling USB Communication

Setting the DVSTCTR0.UACT bit to 1 initiates SOF transmission and transaction generation is enabled.
Setting the UACT bit to 0 stops SOF transmission and a suspend state is entered. If the setting of the UACT bit is changed from 1 to 0, processing stops after the next SOF is sent.

40.4 Usage Notes

40.4.1 Setting the Module-Stop Function

Operation of the USB module can be disabled or enabled by setting a bit in the module stop control register B (MSTPCRB). The USB is initially disabled after a reset. Registers in the USB only become accessible after it has been released from the module-stop state. For details, refer to **section 11, Low Power Consumption**.